Integration Bee Topics Guide for Problem Writers

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2nd Edition

I am very thankful to a lot of people. To my family for letting me pursue my dreams, my best friend for getting me into speed integration, and to the organizers of MUSA of UC Berkeley, Berkeley Math Tournament, and Stanford Math Tournament for accepting me as a problem writer for Integration Bees. This book would never exist without them.

Contents

Quality Control

Before you get crazy writing integrals for the integration bee, let's first get some things straight before you end up becoming high-school me. Every integral problem must be clean. No messy answers and no tedious process. Let me give you some examples.

A Very Long Answer

$$\int \tan^{-1}(x^2)dx$$

[UCLA Homework]

Yes, I get it, this is a cool looking integral. But the issue is, the answer is not clean at all. I mean...look at this:

$$\int \tan^{-1}(x^2)dx = x\tan^{-1}(x^2) - \int \frac{2x^2}{x^4+1}dx$$

I'm not even gonna evaluate it, I know it's a long answer, and you don't want competitors writing long answers. Imagine having to check if they're actually correct; looking at every single detail:

$$x\tan^{-1}(x^2) + \frac{1}{2\sqrt{2}}\ln\left(\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right) + \frac{1}{\sqrt{2}}\tan^{-1}(1-\sqrt{2}x) - \frac{1}{\sqrt{2}}\tan^{-1}(1+\sqrt{2}x) + C$$

Remember that not everyone has good handwriting. Yea no, we're not having this.

A Very Scary Long Process

$$\int_0^1 \sin^{-1}(x)\ln(x)dx = 2 - \frac{\pi}{2} - \ln(2)$$

Let's say you used WolframAlpha and you noticed that the indefinite integral was long. So you added bounds to make the answer shorter. You used WolframAlpha to test which bounds make the cleanest answer. You ended up getting this. Okay, yes, it's clean, but is the process clean??? Please try it out yourself before inconsiderately putting this on the "earlier rounds".

$$\int_0^1 \sin^{-1}(x)\ln(x)dx = \left[\ln(x)(x\sin^{-1}(x)+\sqrt{1-x^2})\right]_0^1 - \int_0^1 \sin^{-1}(x) + \frac{\sqrt{1-x^2}}{x}dx$$

$$= 1 - \lim_{x\to 0} x\ln(x)\sin^{-1}(x) + 1 - \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \frac{\cos^2\theta}{\sin\theta}d\theta$$

As you can see, there are some tedious issues with this problem being in the earlier rounds. Realistically, competitors will hesitate and get disgusted in the middle of the integration by parts process, there's a tedious trig-sub they need to evaluate, and an annoying limit.

A "Made in China" Integral

$$\int_0^1 (1 + x + 1 + 11 + x + 11x - x - 111x - 1010x)dx$$

[MIT Integration Bee 2015 - Round 1 Problem 6]

Can we not??? This is just straight up rude. I'm here to integrate, not do arithmetic. This is a poor quality integral. Fun to whoever wrote this, but not fun who's actually competing.

Annoying Constants

$$\int \frac{\sec^2(x)}{\sec^2(x-1)}dx$$

[My High School Integration Bee Journal]

Honestly, I don't even know how I came up with this. All I know is that there were so many annoying constants.

$$\int \frac{\sec^2(x)}{\sec^2(x-1)}dx = \cos^2(x-1)\tan(x) - \int \sin(2x-2)\tan(x)dx$$

$$\Rightarrow \int 2\sin^2(x)\cos(2) - \sin(2)\sin(x)(2\cos(x) - \sec(x))dx$$

Because of those $\sin(2)$ and $\cos(2)$ constants, I completely lost motivation to continue solving further. This is also a long answer and even with bounds, WolframAlpha would give me this:

$$\int_0^1 \frac{\sec^2(x)}{\sec^2(x-1)}dx = \cos(2) + \sin^2(1)\tan(1) - \sin(2)\ln(\cos(1))$$

Notice that I could've gotten a different correct answer. But how would you know that??? Also, this integral is cruel:

$$\int_0^1 \frac{\sec^2(x)}{\sec^2(x-1)} - \frac{\sec^2(x-1)}{\sec^2(x)}dx = 0$$

I would avoid integrals that do equal to zero. Because some competitors can mentally graph weird functions and guess zero; it makes this integral pointless and unfair for the other opponent who is actually integrating.

Caterpillar U-Substitutions

$$\int \frac{x^2e^{x^3}\cos(\ln(e^{x^3}+1))}{(e^{x^3}+1)\sqrt{\sin(\ln(e^{x^3}+1))}(\sin(\ln(e^{x^3}+1))+1)}dx$$

[My High School Integration Bee Journal]

If I was a competitor and I saw this, my eyes will be blinded from all the messiness of symbols and functions. Let $u = x^3$. Then let $w = e^u + 1$. Then let $y = \ln(w)$. Then let $t = \sin(y)$. Then let $r = \sqrt{t}$.

$$\frac{2}{3}\int \frac{dr}{r^2+1}$$

Seriously? It took me 5 u-substitutions to get here. Let's not overdo this please...

A Textbook Problem

$$\int \frac{x-1}{\sqrt{2x^2-3}}dx$$

[Caltech-Harvey Mudd Math Competition - Finals Round Problem 40]

Let's also not force competitors to bash. If you're gonna bash the competitors, you better do it with style. This integral is not one of them. Also, throwing a textbok problem for finals round is pure laziness. Let's not do this.

$$\int \frac{x-1}{\sqrt{2x^2-3}}dx = \frac{\sqrt{2x^2-3}}{2} - \frac{\ln|\sqrt{2x^2-3}+\sqrt{2}x|}{\sqrt{2}} + C$$

This literally looks like an integral table information. Terrible.

Force Graphing

$$\int_{-10}^{10} |4-|3-|2-|1-|x|||||dx$$

[Carnegie Mellon Integration Bee 2023 - Qualifying Exam]

This is very rude. I'm here to integrate, not graph and do geometry!! I get it, you want to be conceptual with integrals, but this isn't MIT. The amount of trapezoidal areas I have to add up. (Vomit emoji here.) I highly doubt anyone would mentally graph this. If so, I'm very concerned.

$$2\int_0^{10} |4-|3-|2-|1-x||||dx = 2\int_{-1}^{9} |4-|3-|2-|u||||du$$

$$= 4\int_0^1 (3-u)du + 2\int_1^9 |4-|3-|2-u|||du$$

I think I'll pass on this...

Chapter 1

Qualifying Round Topics

In this round, competitors are taking an exam trying to solve 20 integrals in one hour (usually). At least the first 5-6 integrals should be easy for competitors to integrate, then increase difficulty drastically so that we can easily choose our top 16 competitors.

1.1 Basic Integration

I think textbooks have good basic integration problems for this, but we can definitely make it more stylish or satisfying. Here are some good examples.

Textbook Problems:

1. $$\int (5x^2 - 8x + 5)dx$$

2. $$\int (e^x + x + e^{-x})dx$$

3. $$\int \frac{1+\sqrt{x}}{x}dx$$

4. $$\int 4\sin\left(\frac{x}{3}\right)dx$$

5. $$\int \frac{4}{1+x^2}dx$$

6. $$\int \frac{7}{\sqrt{1-x^2}}dx$$

Modified Problems:

1. $$\int (x-1)(x+1)(x^2+1)dx$$

2. $$\int (e^x+1)^3 dx$$

3. $$\int \frac{1}{\sqrt{x}}\left(\frac{1}{\sqrt{x}} + 1 + \sqrt{x}\right)dx$$

4. $$\int \frac{\sec(x)+\tan(x)}{\cos(x)}dx$$

5. $$\int \left(\frac{x^2+2}{x^2+1}\right)dx$$

6. $$\int \frac{1}{\sqrt{x+1}}\left(\sqrt{x+1} + \frac{1}{\sqrt{1-x}}\right)dx$$

1.2 Basic U-Substitution

Here are some good examples of basic u-substitution integrals for qualifying exam:

1. $$\int \frac{x}{x^2+1}dx$$

2. $$\int \tan(x)\sec^2(x)dx$$

3. $$\int (e^x+1)(e^x+x)dx$$

4. $$\int \sin(\sin(x)+1)\cos(x)dx$$

5. $$\int x\sqrt{x-1}\sqrt{x+1}dx$$

6. $$\int e^{x+e^x}dx$$

7. $$\int \sec^2(5x-4)dx$$

8. $$\int \frac{1}{x}\left(\ln(x)+\frac{1}{\ln(x)}\right)dx$$

9. $$\int (x^2+x+1)(2x+1)dx$$

10. $$\int (x+1)^{2024}dx$$

11. $$\int \frac{1}{\sqrt[3]{x-2024}}dx$$

12. $$\int \frac{e^x}{e^{2x}+1}dx$$

13. $$\int \frac{\cos(x)+1+e^x}{\sin(x)+x+e^x}dx$$

14. $$\int \frac{\sin(x)\cos(x)}{\sin^2(x)+1}dx$$

15. $$\int (x+1)e^{x^2+2x}dx$$

16. $$\int \frac{\sin(\ln(x))}{x}dx$$

17. $$\int \tan(x)dx$$

18. $$\int \frac{\sqrt{e^x}-\sqrt{e^{-x}}}{\sqrt{e^x}+\sqrt{e^{-x}}}dx$$

19. $$\int \left(1+\frac{1}{x}\right)\ln(xe^x)dx$$

20. $$\int \csc(1-x)\cot(1-x)dx$$

1.3 Symmetry

Symmetry Concept

For any real a, if $f(x)$ is an odd function, then

$$\int_{-a}^{a} f(x)dx = 0.$$

If $f(x)$ is an even function, then

$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx.$$

Example 1

$$\int_{-1}^{1} (x+1)(x^2+1)dx$$

$$\int_{-1}^{1} (x^3+x^2+x+1)dx = 2\int_{0}^{1} (x^2+1)dx = \frac{8}{3}$$

Example 2

$$\int_{-1}^{1} x\left(\frac{\sin^{-1}(x)}{\tan^{-1}(x)} + \frac{1}{x}\right) dx$$

The left term is an odd function, so the answer is just 2.

Example 3

$$\int_{0}^{6} \frac{(x-2)(x-3)(x-4)+1}{x^2-6x+10} dx$$

Let $u = x - 3$.

$$\int_{-3}^{3} \frac{u(u^2-1)+1}{u^2+1} du = \int_{-3}^{3} \frac{du}{u^2+1} = 2\tan^{-1}(3)$$

Here are some odd functions to give you some inspirations. All of these integrals should equal to zero.

1. $$\int_{-1}^{1} (\sqrt[7]{x} + \sqrt[5]{x} + \sqrt[3]{x})dx$$

2. $$\int_{-1}^{1} \sin^{-1}(x) + \tan^{-1}(x)dx$$

3. $$\int_{-1}^{1} \frac{1-\cos(x)}{x}dx$$

4. $$\int_{-1}^{1} \ln(x + \sqrt{x^2+1})dx$$

5. $$\int_{-1}^{1} (e^x - e^{-x})dx$$

6. $$\int_{-1}^{1} \sec(x)\tan(x)dx$$

7. $$\int_{-1}^{1} \ln\left(\frac{e^x+1}{e^{-x}+1}\right)dx$$

8. $$\int_{-1}^{1} \sin(\sin(x))dx$$

9. $$\int_{-1}^{1} \sin(x^3+x) + \sin^{-1}(x^3+x)dx$$

10. $$\int_{-1}^{1} \sqrt[3]{x + \sqrt[3]{x + \sqrt[3]{x}}}\, dx$$

11. $$\int_{-1}^{1} \sin^{-1}(\tan^{-1}(x))dx$$

12. $$\int_{-1}^{1} \frac{x+\sin(x)}{1+\cos(x)}dx$$

13. $$\int_{-1}^{1} \tan^{-1}\left(x+\frac{1}{x}\right) + \tan^{-1}\left(x-\frac{1}{x}\right)dx$$

14. $$\int_{-1}^{1} \ln\left(\frac{1+\sin(x)}{1-\sin(x)}\right)dx$$

15. $$\int_{-1}^{1} \ln\left(\frac{1+x}{1-x}\right)dx$$

16. $$\int_{-1}^{1} \frac{x^3+x}{x^4+x^2+1}dx$$

17. $$\int_{-1}^{1} \sin(x)e^{x^2}dx$$

18. $$\int_{-1}^{1} (\cos(x))^x - (\sec(x))^x dx$$

19. $$\int_{-1}^{1} f(x)dx \text{ where } f(x) = \int_0^x e^{-t^2}dt$$

20. $$\int_{-1}^{1} \left(\sqrt{1-x} - \sqrt{1+x}\right)dx$$

1.4 Basic Exponential Functions

Here are some good basic examples of exponential integrals for the qualifying exam:

1. $$\int \frac{e^x}{e^x + 2024} dx$$

2. $$\int \frac{e^x}{(e^x + 1)^2}$$

3. $$\int_0^\infty \frac{2^x + 3^x}{5^x} dx$$

4. $$\int \frac{dx}{e^x + 1}$$

5. $$\int \frac{e^x + 1}{x + e^x + 1} dx$$

6. $$\int (e^x + e^{-x})^4 dx$$

7. $$\int 2^{x-1} e^{x+1} dx$$

8. $$\int e^x \sin(e^x) dx$$

9. $$\int \frac{e^{7x} + e^{4x}}{e^{5x} + e^{2x}} dx$$

10. $$\int \frac{e^x}{\sqrt{1 - e^{2x}}} dx$$

11. $$\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx$$

12. $$\int (e^x + 1)(e^{-x} + 1) dx$$

13. $$\int \frac{dx}{e^x + e^{-x}}$$

14. $$\int \frac{e^x}{(e^x - 1)\ln(e^x - 1)} dx$$

15. $$\int \left(\frac{e^x}{\cos(e^{2x})} \right)^2 dx$$

16. $$\int e^{e^x} (e^{2x} + e^x) dx$$

17. $$\int \sqrt{e^x \sqrt{e^x}} dx$$

18. $$\int_0^\infty \frac{1 + \sqrt{1 - e^{-x}}}{e^x} dx$$

1.5 Basic Trig Integrals

Here are some good basic examples of trigonometric integrals for the qualifying exam:

1. $$\int \frac{\sin(x)+\cos(x)}{\cos(x)}dx$$

2. $$\int \sin^2(x)\cos(x)dx$$

3. $$\int \sin^2(x)\cos^3(x)dx$$

4. $$\int (\tan(x)\sec(x))^2dx$$

5. $$\int \tan(x)\sec^3(x)dx$$

6. $$\int \tan^2(x)dx$$

7. $$\int \frac{\cos(x)}{\sin^2(x)+1}dx$$

8. $$\int \frac{\csc(x)\cot(x)}{1+\csc(x)}dx$$

9. $$\int \frac{\sin(x)-\cos(x)}{\sin(x)+\cos(x)}dx$$

10. $$\int \sin^3(x)dx$$

11. $$\int \frac{\tan(x)+\cot(x)}{\sin(x)\cos(x)}dx$$

12. $$\int (\sin(x)+\cos(x))^2dx$$

13. $$\int (\tan(x)+\sec(x))^2dx$$

14. $$\int (\sin(x)+1)(\cos(x)+1)dx$$

15. $$\int (\sec(\tan(x))\sec(x))^2dx$$

16. $$\int \frac{1}{\sec(x)+\tan(x)}dx$$

17. $$\int \frac{\tan(x)+1}{\sec(x)}dx$$

18. $$\int (1+\sin(x))(1+\sec^2(x))dx$$

19. $$\int \frac{(\sec(x)-\csc(x))(\sec(x)+\csc(x))}{\tan(x)+\cot(x)}dx$$

20. $$\int \frac{\sin(x)+\cos(x)}{\sec(x)+\csc(x)}dx$$

1.6 Basic Integration by Parts

Here are some good basic examples of integration by parts integrals for the qualifying exam. Notice that there are no recursive integration by parts problems.

1. $$\int x^2 e^x dx$$

2. $$\int \ln(x) dx$$

3. $$\int x \sin(x) dx$$

4. $$\int \frac{x}{e^x} dx$$

5. $$\int x \sec^2(x) dx$$

6. $$\int x \ln(x) dx$$

7. $$\int (xe^x)^2 dx$$

8. $$\int (x + e^x)^2 dx$$

9. $$\int \frac{\ln(x)}{x^2} dx$$

10. $$\int \sin(x) \ln(\sec(x)) dx$$

11. $$\int (x+1)^2 \cos(x) dx$$

12. $$\int \tan^{-1}(x) dx$$

13. $$\int \sqrt{x} \ln(x) dx$$

14. $$\int \frac{\ln(x^2+1)}{x^2} dx$$

15. $$\int \sin^{-1}(x) dx$$

16. $$\int x \sec^2(x) \tan(x) dx$$

17. $$\int x 2^x dx$$

18. $$\int x(\cos(x) + \sin(x)) dx$$

19. $$\int \sqrt{e^x(x^2+6x+9)} dx$$

20. $$\int \frac{\ln(x)}{\sqrt{x}} dx$$

1.7 Basic Trig-Substitution

Here are some good basic examples of trig-sub integrals for the qualifying exam:

1. $$\int \frac{dx}{x\sqrt{x^2-1}}$$

2. $$\int \frac{dx}{x^2\sqrt{x^2-1}}$$

3. $$\int \frac{dx}{(1-x^2)^{\frac{3}{2}}}$$

4. $$\int \frac{dx}{(1+x^2)^{\frac{3}{2}}}$$

5. $$\int \frac{dx}{(x^2-1)^{\frac{3}{2}}}$$

6. $$\int \frac{dx}{x^2+4}$$

7. $$\int \frac{\sqrt{1-x^2}}{x^2}dx$$

8. $$\int \frac{x^2}{(1-x^2)^{\frac{3}{2}}}dx$$

9. $$\int \frac{1}{\sqrt{9-x^2}}dx$$

10. $$\int \frac{\sqrt{x^2-1}}{x}dx$$

To make an intentional trig-sub integral, start with a skeleton. For example:

$$\int \frac{d\theta}{\sec\theta+\tan\theta}$$

If I let $x=\sin\theta$, I need $dx=\cos\theta d\theta$:

$$\int \frac{\cos\theta}{(\sec\theta+\tan\theta)\cos\theta}d\theta = \int \frac{dx}{1+\sin\theta} = \int \frac{dx}{1+x}$$

Too easy, that's just basic substitution.
If I let $x=\sec\theta$, I need $dx=\sec\theta\tan\theta d\theta$:

$$\int \frac{\sec\theta\tan\theta d\theta}{(\sec\theta+\tan\theta)\sec\theta\tan\theta} = \int \frac{dx}{(x+\sqrt{x^2-1})x\sqrt{x^2-1}}$$

This looks good, let's save this!!
If I let $x=\tan\theta$, I need $dx=\sec^2\theta d\theta$: note that $\sec(\tan^{-1}(x))=\sqrt{1+x^2}$.

$$\int \frac{\sec^2\theta d\theta}{(\sec\theta+\tan\theta)\sec^2\theta} = \int \frac{dx}{(\sqrt{1+x^2}+x)(1+x^2)}$$

This also looks good as well!! Although the answer may be long, we can just add bounds to make answer cleaner. In this case, I would go from 0 to ∞ so that my skeleton goes from $\tan^{-1}(0)=0$ to $\tan^{-1}(\infty)=\frac{\pi}{2}$ (since $x=\tan\theta$).

1.8 Basic Partial Fractions

Here are some good basic examples of partial fraction integrals for the qualifying exam:

1. $$\int \frac{dx}{x(x+1)}$$

2. $$\int \frac{dx}{x^2(x+1)}$$

3. $$\int \frac{dx}{(x+1)(x+2)}$$

4. $$\int \frac{dx}{x(2x-1)}$$

5. $$\int \frac{dx}{x^2-1}$$

6. $$\int \frac{x+2}{x^2+4x+3}dx$$

7. $$\int \frac{dx}{x^3-x}$$

8. $$\int \frac{x}{(x+1)(x+2)(x+3)}dx$$

9. $$\int \frac{dx}{x^2(1+x^2)}$$

10. $$\int \frac{dx}{x(1+x)^2}$$

11. $$\int \frac{(x+1)(x+4)}{(x+2)(x+3)}dx$$

12. $$\int \frac{e^{2x}}{e^{2x}+3e^x+2}dx$$

13. $$\int \frac{x^2-x}{1+x+x^2+x^3}dx$$

14. $$\int \frac{3x}{x^2+2x-8}dx$$

15. $$\int \frac{x+5}{x^2+x-2}dx$$

16. $$\int_2^4 \frac{x-4}{x^2-6x+5}dx$$

17. $$\int_3^\infty \frac{dx}{(x-2)(x+6)}$$

1.9 Area of a Partial Circle

Area of a Partial Circle

We are familiar with the graph of a semicircle: $y = \sqrt{r^2 - x^2}$ where r is the radius of the semicircle. Then,

$$\int_{-r}^{r} \sqrt{r^2 - x^2}dx = \frac{\pi r^2}{2}$$

$$\int_{0}^{r} \sqrt{r^2 - x^2}dx = \int_{-r}^{0} \sqrt{r^2 - x^2}dx = \frac{\pi r^2}{4}$$

Basic Example

$$\int_{0}^{1} \sqrt{1 - x^2}dx$$

We see that this is a quarter of a circle of radius 1. So the integral is the area $\frac{\pi}{4}$.

Hidden Symmetry

$$\int_{-2}^{2} \left(x^3 \cos\left(\frac{x}{2}\right) + \frac{1}{2}\right) \sqrt{4 - x^2}dx$$

[Free Wi-Fi Password]

To most people this looks intimidating, but it's actually easy if you know the concept of symmetry and the partial area of a circle. Notice that the left term is an odd function:

$$\int_{-2}^{2} x^3 \cos\left(\frac{x}{2}\right) \sqrt{4 - x^2}dx = 0$$

In the right term, we can use symmetry of an even function:

$$\frac{1}{2}\int_{-2}^{2} \sqrt{4 - x^2}dx = \int_{0}^{2} \sqrt{4 - x^2}dx = \boxed{\pi}$$

Ellipse Example

$$\int_{0}^{2} \sqrt{12 - 3x^2}dx$$

[MIT Integration Bee 2013 - Qualifying Exam]

We need that x^2 to not have a coefficient other than 1.

$$\int_{0}^{2} \sqrt{3}\sqrt{4 - x^2}dx = \boxed{\sqrt{3}\pi}$$

Substitution Example

$$\int_0^3 \sqrt{6x - x^2}dx$$

Let's complete the square and perform u-sub.

$$\int_0^3 \sqrt{9 - (x-3)^2}dx = \int_{-3}^0 \sqrt{9 - u^2}du = \boxed{\frac{9\pi}{4}}$$

Exponential Example

$$\int_0^\infty \sqrt{e^{-2x} - e^{-4x}}dx$$

$$\int_0^\infty e^{-x}\sqrt{1 - e^{-2x}}dx = \int_0^1 \sqrt{1 - u^2}du = \boxed{\frac{\pi}{4}}$$

Hybrid Example

$$\int_{-1}^1 \left(\frac{2 - x^2}{\sqrt{1 - x^2}}\right)\left(1 + x^3\right)dx$$

$$\int_{-1}^1 \left(\frac{2 - x^2}{\sqrt{1 - x^2}}\right)\left(1 + x^3\right)dx = \int_{-1}^1 \left(\frac{1}{\sqrt{1 - x^2}} + \sqrt{1 - x^2}\right)\left(1 + x^3\right)dx$$

$$= \int_{-1}^1 \left(\frac{1}{\sqrt{1 - x^2}} + \sqrt{1 - x^2}\right)dx = \pi + \frac{\pi}{2} = \boxed{\frac{3\pi}{2}}$$

Hopefully this was easy to follow along. We separate it in a way where the radicals are easy to understand. Then we used symmetry to get rid of x^3. Then the rest of the integral is just inverse sine and the partial area of a circle.

1.10 Algebra Manipulation

Zero Substitution

$$\int \frac{x^2}{x^2+1}dx = \int \frac{x^2+(1-1)}{x^2+1}dx = \int 1 - \frac{1}{x^2+1}dx$$

It may not seem much, but this is actually the #1 most helpful algebra trick for speed integration!! Here, try this one. It's not always obvious sometimes.

$$\int \frac{dx}{(x+2)^2(x^2+4x+5)}$$

[JEE Main Training]
Hint: Let $1 = 5-4 = (x^2+4x+5)-(x^2+4x+4)$.

Geometric Polynomial Division

$$\int \frac{x^4}{x+1}dx = \int \frac{x^4-1+1}{x+1}dx = \int x^3-x^2+x-1+\frac{1}{x+1}dx$$

$$\int \frac{x^7}{x^2-1}dx = \int \frac{x^7-x+x}{x^2-1}dx = \int x\left(\frac{x^6-1}{x^2-1}\right)+\frac{x}{x^2-1}dx$$

Forcing an Arctangent

We mainly complete the square to give ourselves an arctan form.

$$\int \frac{dx}{x^2+x+1} = \int \frac{dx}{(x+\frac{1}{2})^2+\frac{3}{4}} = \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$$

du the Numerator

We purposely put down the derivative of the denominator on the numerator so that it's an easy u-sub. Then we just deal with the leftovers.

$$\int \frac{5x^2+3x-2}{x^3+2x^2}dx = \int \frac{3x^2+4x}{x^3+2x^2}+\frac{2x^2-x-2}{x^3+2x^2}dx$$

$$= \int \frac{3x^2+4x}{x^3+2x^2}+\frac{2x^2-(x+2)}{x^2(x+2)}dx = \int \frac{3x^2+4x}{x^3+2x^2}+\frac{2}{x+2}-\frac{1}{x^2}dx$$

Function Conjugating

We multiply top and bottom with functions to get rid of fractions or other types of simplification.

$$\int \frac{2}{1+\frac{1}{1+e^x}}dx = \int \frac{2(1+e^x)}{(e^x+1)+1}dx = \int \frac{2e^x+2}{e^x+2}dx = \int \frac{e^x}{e^x+2}+1\ dx$$

$$\int \frac{dx}{(4\cos(x)+\sin(x))^2} = \int \frac{\sec^2(x)}{((4\cos(x)+\sin(x))\sec(x))^2} = \int \frac{\sec^2(x)}{(4+\tan(x))^2}dx$$

Radical Conjugating

$$\int \frac{x}{\sqrt{x^2+1}-x}dx = \int \frac{x}{(\sqrt{x^2+1}-x)}\frac{(\sqrt{x^2+1}+x)}{(\sqrt{x^2+1}+x)}dx = \int (x\sqrt{x^2+1}+x^2)dx$$

$$\int \frac{dx}{\sqrt{x+1}-\sqrt{x-1}} = \int (\sqrt{x+1}+\sqrt{x-1})dx$$

Fake Partial Fractions

$$\int \frac{(x+3)}{(x+2)^2(x+4)^2}dx = \int \frac{x+3}{(x^2+6x+8)^2}dx$$

This is just a basic u-substitution.

$$\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}dx = \int \frac{(x+1)^2}{x^3+3x^2+3x+2}dx = \int \frac{(x+1)^2}{(x+1)^3+1}dx$$

This one is a little tricky to see, but you can simply do $u=(x+1)^3+1$.

1.11 Intermediate U-Substitution

These integrals are u-substitutions that are a little hard to see, but also not too difficult. Here are some good examples.

1. $$\int \frac{x}{1+x^4}dx$$

2. $$\int \frac{dx}{\sqrt{x}(1+x)}$$

3. $$\int x^3 e^{x^2} dx$$

4. $$\int \frac{x}{\sqrt{x-1}}dx$$

5. $$\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx$$

6. $$\int \frac{1}{x^2}e^{\frac{1}{x}}dx$$

7. $$\int \frac{dx}{\sqrt{x-x^2}}$$

8. $$\int \sqrt{x}\sin(x\sqrt{x})dx$$

9. $$\int_0^1 x^2(2x-1)^7 dx$$

10. $$\int \frac{1}{x^5}\sqrt{\frac{x^4+1}{x^4}}dx$$

11. $$\int x^2\sqrt{x-1}dx$$

12. $$\int \frac{x+5}{\sqrt{(x+2)(x+8)}}dx$$

13. $$\int \frac{e^{2x}}{1+e^{4x}}dx$$

14. $$\int_1^{\frac{3}{2}} \frac{x^2}{(x-2)^3}dx$$

15. $$\int \frac{\sqrt{\sin(x)}}{\cos^2(x)\sqrt{\cos(x)}}dx$$

16. $$\int \sec(x)\ln(\sec(x)+\tan(x))dx$$

17. $$\int \frac{dx}{\sqrt{x}(1+2\sqrt{x}+x)}$$

18. $$\int \frac{\sec^2(x)}{8+\sec^2(x)}dx$$

19. $$\int \sqrt{\frac{x}{1-x^3}}dx$$

20. $$\int \frac{\ln(x)+1}{x\ln(x)+1}dx$$

1.12 Absolute Values

Here are some good basic examples of absolute value integrals for the qualifying exam:

1. $$\int_{-2}^{2} \left|x^2 - 1\right| dx$$

2. $$\int_{0}^{5} \frac{dx}{x + |x-1|}$$

3. $$\int_{-1}^{1} |e^x - 1|\, dx$$

4. $$\int_{-1}^{1} |x - |x - |x|||dx$$

5. $$\int_{-2\pi}^{2\pi} |\sin(x)|dx$$

6. $$\int_{0}^{4} \frac{|x-1|}{|x-2| + |x-3|}dx$$

7. $$\int_{-\infty}^{\infty} e^{-|x|}dx$$

8. $$\int_{1}^{e} \frac{x+1}{|x - \ln(x)| + \ln(x)}dx$$

9. $$\int_{0}^{16} \left|\sqrt{x} - 2\right| dx$$

10. $$\int_{-2}^{2} \frac{dx}{\sqrt{|1 - x^2| + 1}}$$

WARNING EXAMPLE

$$\int_{0}^{2\pi} \sqrt{1 - \sin^2(x)}dx = \int_{0}^{2\pi} |\cos(x)|dx = 4$$

Not only this tricks competitors, but nearly everyone when it comes to square roots. Just be very careful with this, but also a nice way to give a hidden absolute value integral.

1.13 Reverse Product Rule

Reverse Product Rule Concept

$$\int [f'(x)g(x) + f(x)g'(x)]\, dx = f(x)g(x) + C$$

As simple as it looks, it's not always clear to see in most integration bee problems.

An Obvious Example

$$\int \frac{\sin(x)}{x} + \cos(x)\ln(x)dx$$

If I didn't tell you this concept, you might have struggled solving this integral. But now that I showed you this trick, you are aware that this is just a reverse product rule. So the answer is clearly $\sin(x)\ln(x) + C$.

Exponential Example

$$\int e^x(\sin(x) + \cos(x))dx$$

To most beginners, they would probably do integration by parts. But if you know the trick, you could definitely solve this in seconds. In fact, it's good to keep in mind:

$$\int e^x(f(x) + f'(x))dx = e^x f(x) + C$$

A Trig Example

$$\int \sec(x)(2x + x^2\tan(x))dx$$

$$= \int [2x]\sec(x) + x^2[\sec(x)\tan(x)]dx = \boxed{x^2\sec(x) + C}$$

Logarithm Example

$$\int (\ln^2(x) + 2\ln(x))dx$$

Natural logs are very good at hiding x^n due to the cancellation from the derivative of $\ln(x)$.

$$\int 1 \cdot \ln^2(x) + x \cdot \frac{2\ln(x)}{x} dx = x\ln^2|x| + C$$

In fact,

$$\int [f(\ln(x)) + f'(\ln(x))]\, dx = xf(\ln(x)) + C$$

Polynomial Example

$$\int_{-\frac{1}{3}}^{0} x(x^2+1)^2(3x+1)^2 + (3x+1)(x^2+1)^3 dx$$

[WolframAlpha Suggestions]

You're probably guessing that the indefinite answer is $(x^2+1)^3(3x+1)^2$. Well you're wrong lol. You must be aware of removed constants; this tricks competitors who just likes to speed guess. The indefinite answer is $\frac{1}{6}(x^2+1)^3(3x+1)^2$. So the definite integral answer should just be $\frac{1}{6}$.

1.14 Basic Series

Some Basic Series Formulas

The following properties are true if $|x| < 1$.

1. $$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

2. $$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

3. $$\sum_{n=1}^{\infty} \frac{n}{x^n} = \frac{x}{(1-x)^2}$$

4. $$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

5. $$\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x)$$

6. $$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = \tan^{-1}(x)$$

7. $$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \sin(x)$$

8. $$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \cos(x)$$

9. $$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

10. $$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Geometric Series Example

$$\int_0^1 \sqrt{1 + x^2 + x^4 + x^6 + ...}\, dx$$

$$= \int_0^1 \sqrt{\sum_{n=0}^{\infty} x^{2n}}\, dx = \int_0^1 \sqrt{\frac{1}{1-x^2}}\, dx = \frac{\pi}{2}$$

Out of Convergence Example

$$\int_0^{2024} \frac{dx}{(1+x^2)(1+x^4)(1+x^8)...}$$

$$= \int_0^1 \frac{(1+x)}{1 + x + x^2 + x^3 + x^4 + ...} dx = \int_0^1 (1-x^2) dx = \frac{2}{3}$$

Note that the integral with bounds $[1, 2024]$ is out of convergence for the geometric series. Because of this, the infinite denominator goes off to infinity and makes the integral equal to zero.

Taylor Series Example

$$\int e^{2x}\sqrt{e^{x^2}\sqrt[3]{e^{x^3}\sqrt[4]{e^{x^4}\sqrt[5]{e^{x^5}\cdots}}}}\,dx$$

[Silver]

$$=\int e^x e^{\sum_{n=1}^{\infty}\frac{x^n}{n!}}\,dx=\int e^x e^{e^x-1}dx=e^{e^x-1}+C$$

Numerical Series Example

$$\int\sqrt{2^x\sqrt{4^x\sqrt{8^x\sqrt{16^x\sqrt{...}}}}}dx$$

[Silver]

$$=\int 2^{\frac{x}{2}+\frac{2x}{4}+\frac{3x}{8}+\frac{4x}{16}+\cdots}dx=\int 2^{x\left(\sum_{n=1}^{\infty}\frac{n}{2^n}\right)}dx=\int 2^{2x}dx$$

Composite Function Series Example

$$\int\frac{e^{e^x}\sqrt[3]{e^{e^{3x}}}\sqrt[5]{e^{e^{5x}}}}{\sqrt{e^{e^{2x}}}\sqrt[4]{e^{e^{4x}}}\sqrt[6]{e^{e^{6x}}}}...dx$$

$$=\int e^{\sum_{n=1}^{\infty}\frac{-(-e^x)^n}{n}}dx=\int e^{\ln(1+e^x)}dx=x+e^x+C$$

Telescoping Series Example

$$\int\sqrt{x\sqrt[3]{x}}\cdot\sqrt[4]{\sqrt[3]{x}\sqrt[5]{x}}\cdot\sqrt[6]{\sqrt[5]{x}\sqrt[7]{x}}...dx$$

$$=\int x^{\frac{1}{2}+\frac{1}{2(3)}+\frac{1}{3(4)}+\frac{1}{4(5)}+\cdots}dx=\int x^{\sum_{n=1}^{\infty}\frac{1}{n(n+1)}}dx=\frac{x^2}{2}+C$$

Gaussian Series Example

$$\int\frac{\ln(x)}{x}+\frac{\ln(\sqrt{x})}{2x}+\frac{\ln(\sqrt[3]{x})}{3x}+\frac{\ln(\sqrt[4]{x})}{4x}...dx$$

$$=\int\frac{\ln(x)}{x}+\frac{\ln(x)}{4x}+\frac{\ln(x)}{9x}+\frac{\ln(x)}{16x}...dx$$

$$=\int\frac{\ln(x)}{x}\sum_{n=1}^{\infty}\frac{1}{n^2}dx=\frac{\pi^2\ln^2(x)}{12}+C$$

1.15 Intermediate Trig Integrals

In this section, we start utilizing trig-identities, including some uncommon ones. Here are some good examples for the qualifying exam:

1. $$\int_0^{\frac{\pi}{2}} \sin^2(x)dx$$

2. $$\int_0^{\frac{\pi}{2}} (1+\cos(x))^2dx$$

3. $$\int \sec^6(x)dx$$

4. $$\int \sin(2x)\cos(3x)dx$$

5. $$\int \cos(5x)\cos(7x)dx$$

6. $$\int (\tan^4(x)+\tan^6(x))dx$$

7. $$\int \frac{dx}{1+\sin(x)}$$

8. $$\int \sqrt{1+\cos(x)}dx$$

9. $$\int \cos^4(x)-\sin^4(x)dx$$

10. $$\int \sqrt{\csc(x)-\sin(x)}dx$$

11. $$\int \sqrt{2\tan^2(x)+2\sec(x)\tan(x)+1}\ dx$$

12. $$\int \sin(x-\sin(x))-\sin(x+\sin(x))dx$$

13. $$\int \frac{dx}{\sin^2(x)\cos^2(x)}$$

14. $$\int \frac{dx}{(\sec(x)-\tan(x))^2}$$

15. $$\int \frac{\cos^2(x)}{2\cos^2(x)-1}dx$$

16. $$\int \frac{\tan(x)}{\tan(2x)}dx$$

17. $$\int_0^{\frac{\pi}{6}} \sqrt{\frac{1}{1-\sin(x)}+\frac{1}{1+\sin(x)}}dx$$

18. $$\int \tan^3(x)dx$$

19. $$\int \frac{1+4\cot(x)}{4-\cot(x)}dx$$

20. $$\int \frac{\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx$$

1.16 Intermediate Trig-Substitution

In this section, we use trig-substitution in a nonstandard way. Remember that in trig-substitution, that whole goal is to get rid of radicals or other simplification.

Using $\sin^2(x)$ Example

$$\int \frac{\sqrt{1-x}}{1-\sqrt{x}}dx$$

[Calculus - Spivak]

Let $x = \sin^2\theta$, then $dx = \sin(2\theta)d\theta$.

$$\int \frac{\sqrt{1-x}}{1-\sqrt{x}}dx = \int \frac{\cos\theta}{1-\sin\theta}\sin(2\theta)d\theta = \int 2\sin\theta(1+\sin\theta)d\theta$$

Now it is a much easier integral to solve. The answer will be long though, so I'd suggest putting in bounds so that we don't have to substitute back.

Using Double Angle Example

$$\int_0^1 \frac{x\sqrt{\sqrt{1-x}+\sqrt{1+x}}}{\sqrt{1-x^2}}dx$$

[My High School Integration Bee Journal]

Let $x = \sin(2\theta)$. Then $dx = 2\cos(2\theta)$. This substitution gets rid of $\sqrt{1-x}$ or $\sqrt{1+x}$ because $1 \pm \sin(2\theta) = (\sin\theta \pm \cos\theta)^2$.

$$= \int_0^{\frac{\pi}{4}} \sin(2\theta)\sqrt{|\sin\theta - \cos\theta| + \sin\theta + \cos\theta}2d\theta$$

$$= \int_0^{\frac{\pi}{4}} 4\sqrt{2}\cos^{\frac{3}{2}}\theta\sin\theta d\theta = \frac{2\sqrt{2}}{5}(4 - 2^{\frac{3}{4}})$$

Loophole Example

$$\int e^{\cos^{-1}(x)}dx$$

[MIT Integration Bee 2006]

Let $\theta = \cos^{-1}(x)$. Then $\cos\theta = x$ and $-\sin\theta = dx$.

$$\int -e^{\theta}\sin\theta d\theta$$

Identity Cancellation Example

$$\int_0^1 \cos^{-1}(2x^2-1)dx$$

Let $x = \cos\theta$.

$$= \int_0^{\frac{\pi}{2}} \cos^{-1}(2\cos^2\theta - 1)\sin\theta d\theta = \int_0^{\frac{\pi}{2}} -2\theta\sin\theta d\theta$$

Note: It is true that $\cos^{-1}(\cos(2\theta)) = 2\theta$ for $\theta \in [0, \frac{\pi}{2}]$.

A Very Satisfying Example

$$\int_0^1 \frac{32}{\sqrt{x-x^2}}\left(\tan^{-1}\left(\sqrt{\frac{x}{1-x}}\right)\right)^3 dx$$

[UC Berkeley Integration Bee 2021]

Notice that if I test certain trig functions to plug into x, letting $x = \sin^2\theta$ will definitely simplify it better because it cancels the radicals and the arctangent.

$$= \int_0^{\frac{\pi}{2}} \frac{32}{\sin\theta\cos\theta}(\theta)^3\sin(2\theta)d\theta = \int_0^{\frac{\pi}{2}} 64\theta^3 d\theta = \boxed{\pi^4}$$

1.17 Inverse Functions

Inverse Trig Identities

$$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$$

$$\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

Note: $\tan^{-1}(\frac{1}{x}) = \cot^{-1}(x)$

Inverse Trig Example

$$\int_0^1 \frac{dx}{\frac{\sin^{-1}(x)}{\cos^{-1}(x)} + 1} = \int_0^1 \frac{\cos^{-1}(x)}{\frac{\pi}{2}} dx = \frac{2}{\pi}$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

Symmetry Example

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \cos^{-1}(\tan(x))dx$$

[Silver]

$$= \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \cos^{-1}(\tan(x)) + \sin^{-1}(\tan(x)) - \sin^{-1}(\tan(x))dx = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi}{2} + 0\ dx$$

Note that $\sin^{-1}(\tan(x))$ is an odd function. So this integral equals to $\frac{\pi^2}{3}$.

Sum of Inverse Functions

$$\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$$

Clearly, we would rather find certain functions and bounds such that we can combine this into one integral.

A Simple Example

$$\int_0^1 \left[\sin^3\left(\frac{\pi x}{2}\right) + \frac{2}{\pi}\sin^{-1}(\sqrt[3]{x})\right] dx$$

You can clearly see that they are inverses of each other. Hence, the answer to this integral would be 1. But let's be careful not to make too many integrals like this since competitors can just easily guess.

A Guess-Proof Example

$$\int_0^1 2^{\sqrt{x}}\ln^2(2)+\ln^2(1+x)dx$$

[Carnegie Mellon Integration Bee 2023]

Now it's not so obvious. Most would guess 1, but there's an evil hidden detail about this integral. The inverse of $\ln^2(x+1)$ is $e^{\sqrt{x}}-1$. Notice that $2^{\sqrt{x}}\ln^2(2)=e^{\ln(2)\sqrt{x}}\ln^2(2)$. Let $u=x\ln^2(2)$ for only the first term.

$$\int_0^{\ln^2(2)} e^{\sqrt{u}}du+\int_0^1 \ln^2(1+x)dx$$

Wait, we're not done. We need that -1 in the first term.

$$\int_0^{\ln^2(2)}(e^{\sqrt{u}}-1)+1du+\int_0^1 \ln^2(1+x)dx=\ln^2(2)+\ln^2(2)(1)-0=\boxed{2\ln^2(2)}.$$

Common Substitution Example

$$\int_0^2 \sqrt{1+x^3}+\sqrt[3]{x^2+2x}\ dx$$

$$=\int_0^2\sqrt{1+x^3}dx+\int_0^2\sqrt[3]{(x+1)^2-1}dx$$

$$=\int_0^2\sqrt{1+x^3}dx+\int_1^3\sqrt[3]{u^2-1}du=2(3)-0=\boxed{6}$$

This is the most common way to hide inverse functions, just by u-substitutions to manipulate the bounds.

Making a sum of inverse integral is not easy to do. You're gonna need to play around with functions on a graph or on WolframAlpha until you have an idea.

1.18 King's/Queen's Rule

King's Rule Concept

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx \xrightarrow{u=a+b-x} \int_a^b \frac{f(a+b-u)}{f(u)+f(a+b-u)}du = I$$

Variables are just variables.

$$I + I = 2I = \int_a^b \frac{f(x)+f(a+b-x)}{f(x)+f(a+b-x)}dx = \int_a^b 1dx = (b-a)$$

Hence $I = \frac{b-a}{2}$.

A Simple Example

$$\int_3^4 \frac{\sqrt{x}}{\sqrt{7-x}+\sqrt{x}}dx$$

Let $u = 7 - x$.

$$\int_3^4 \frac{\sqrt{7-u}}{\sqrt{u}+\sqrt{7-u}}du$$

Add it to the first integral.

$$2I = \int_3^4 \frac{\sqrt{x}}{\sqrt{7-x}+\sqrt{x}}dx + \int_3^4 \frac{\sqrt{7-x}}{\sqrt{x}+\sqrt{7-x}}dx = \int_3^4 1dx$$

We get $2I = 1 \to I = \frac{1}{2}$.

Logarithm Example

$$\int_0^1 \frac{\ln(x+1)}{\ln(2+x-x^2)}dx$$

$$= \int_0^1 \frac{\ln(x+1)}{\ln(x+1)+\ln(2-x)}dx \xrightarrow{u=1-x} \int_0^1 \frac{\ln(2-x)}{\ln(2-x)+\ln(x+1)}dx$$

$$2I = \int_0^1 1dx \Rightarrow I = \frac{1}{2}.$$

Flipping the Logarithm Example

$$\int_0^2 \frac{1+\ln(\frac{x}{2-x})}{x^2+(2-x)^2}dx$$

[Silver]

Let $u = 2-x$, then:

$$\int_0^2 \frac{1+\ln(\frac{2-u}{u})}{(2-u)^2+u^2}du$$

Combine the first integral with our u-sub integral, the log will cancel out:

$$\int_0^2 \frac{2}{2(x^2-2x+2)}dx = \int_0^2 \frac{dx}{(x-1)^2+1} = \boxed{\frac{\pi}{4}}.$$

King's Advantage

$$\int_0^1 \frac{x}{e^x+e^{1-x}}dx$$

[Silver]

If you let $u = 1-x$ and add the integral with our original, it'll cancel out the numerator:

$$\int_0^1 \frac{x}{e^x+e^{1-x}} + \frac{1-x}{e^{1-x}+e^x}dx = \int_0^1 \frac{dx}{e^x+e^{1-x}}$$

It is now an elementary integral we can solve!!

Trig Identity Example

$$\int_0^1 \sin(x)\cos(1-x)dx$$

$$2I = \int_0^1 \sin(x)\cos(1-x) + \cos(x)\sin(1-x)dx = \int_0^1 \sin(1)dx$$

Inverse Arctangent Identity Example

$$\int_0^1 \tan^{-1}\left(\frac{x}{1-x}\right)dx$$

$$2I = \int_0^1 \tan^{-1}\left(\frac{x}{1-x}\right) + \tan^{-1}\left(\frac{1-x}{x}\right)dx$$

$$I = \frac{1}{2}\int_0^1 \frac{\pi}{2}dx = \boxed{\frac{\pi}{4}}$$

Queen's Rule Concept

If you let $u = \frac{\pi}{2} - x$, then

$$\int_0^{\frac{\pi}{2}} f(\sin(x))dx = \int_0^{\frac{\pi}{2}} f(\cos(u))du$$

Important Trig Identities:

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x) \qquad \cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$

$$\sin(\pi - x) = \sin(x) \qquad \cos(\pi - x) = -\cos(x)$$

Simple Queen's Rule Example

$$\int_0^{\frac{\pi}{2}} \frac{\sin^3(x)}{\sin^3(x) + \cos^3(x)} dx$$

Let $u = \frac{\pi}{2} - x$:

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^3(x)}{\sin^3(x) + \cos^3(x)} + \frac{\cos^3(x)}{\sin^3(x) + \cos^3(x)} dx = \int_0^{\frac{\pi}{2}} 1 dx$$

$(\pi - x)$ Example

$$\int_0^{\pi} \frac{x\sin(x)}{1 + \cos^2(x)} dx$$

Let $u = \pi - x$:

$$2I = \int_0^{\pi} \frac{x\sin(x)}{1 + \cos^2(x)} + \frac{(\pi - x)\sin(x)}{1 + \cos^2(x)} dx = \int_0^{\pi} \frac{\pi\sin(x)}{1 + \cos^2(x)} dx$$

Trig-Sub Example

$$\int_0^{\infty} \frac{dx}{1 + x + x^2 + x^3}$$

Let $x = \tan\theta$:

$$\int_0^{\infty} \frac{dx}{(x+1)(x^2+1)} = \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \tan\theta} = \int_0^{\frac{\pi}{2}} \frac{\cos\theta}{\cos\theta + \sin\theta} d\theta$$

Let $u = \frac{\pi}{2} - \theta$ and you apply the same concept.

Tangent Example

$$\int_0^{\frac{\pi}{2}} \frac{1}{\ln(\tan(x))} + \frac{1}{1-\tan(x)} dx$$

[JEE Main]

Let $u = \frac{\pi}{2} - x$:

$$\int_0^{\frac{\pi}{2}} \frac{1}{\ln(\cot(x))} + \frac{1}{1-\cot(x)} dx$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{1}{\ln(\tan(x))} - \frac{1}{\ln(\tan(x))} + \frac{1}{1-\tan(x)} - \frac{\tan(x)}{1-\tan(x)} dx$$

$$= \int_0^{\frac{\pi}{2}} 1 dx \Rightarrow I = \frac{\pi}{4}$$

Note: $\tan(\frac{\pi}{2} - x) = \cot(x), \qquad \cot(\frac{\pi}{2} - x) = \tan(x)$

Trig Identities Craze

$$\int_0^{\frac{\pi}{2}} (\sin(\sin^2(x)) + \cos(\cos^2(x)))^2 dx$$

[Silver]

$$\int_0^{\frac{\pi}{2}} \sin^2(\sin^2(x)) + \cos^2(\cos^2(x)) + 2\sin(\sin^2(x))\cos(\cos^2(x)) dx$$

Let $u = \frac{\pi}{2} - x$:

$$\int_0^{\frac{\pi}{2}} \sin^2(\cos^2(u)) + \cos^2(\sin^2(u)) + 2\sin(\cos^2(u))\cos(\sin^2(u)) du$$

$$2I = \int_0^{\frac{\pi}{2}} 1 + 1 + 2\sin(\sin^2(x) + \cos^2(x)) dx$$

$$I = \boxed{(1 + \sin(1))\frac{\pi}{2}}$$

1.19 Gaussian Integrals

Gaussian Integral Formula

$$\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$$

Substitution Examples

$$\int_0^\infty 2^{-x^2}dx$$

$$= \int_0^\infty e^{-\ln(2)x^2}dx \xrightarrow{u=x\sqrt{\ln(2)}} \frac{1}{\sqrt{\ln(2)}}\int_0^\infty e^{-u^2}du = \boxed{\frac{1}{2}\sqrt{\frac{\pi}{\ln(2)}}}$$

$$\int_{-\infty}^\infty e^{-x^2+2x}dx$$

$$= e\int_{-\infty}^\infty e^{-(x+1)^2}dx = \boxed{e\sqrt{\pi}}$$

Symmetry Example

$$\int_{-\infty}^\infty e^{-x^2}(\sin(x)+\cos(x))^2dx$$

$$= \int_{-\infty}^\infty e^{-x^2}(1+\sin(2x))dx = \int_0^\infty 2e^{-x^2}dx + 0 = \boxed{\sqrt{\pi}}$$

Disguised Example

$$\int_0^\infty \frac{x^2+1+e^{x^2}}{e^{x^2+1}(x^2+1)}dx$$

$$= \frac{1}{e}\int_0^\infty e^{-x^2} + \frac{1}{x^2+1}dx = \boxed{\frac{\pi+\sqrt{\pi}}{2e}}$$

Trig Example

$$\int_0^{\frac{\pi}{2}} e^{-\sec^2(x)} \sec^2(x) dx$$

[Stanford Math Tournament Integration Bee 2022]

$$= \frac{1}{e} \int_0^{\frac{\pi}{2}} e^{-\tan^2(x)} \sec^2(x) dx = \frac{1}{e} \int_0^{\infty} e^{-u^2} du = \boxed{\frac{\sqrt{\pi}}{2e}}$$

Integration by Parts Example

$$\int_0^{\infty} x^2 e^{-x^2} dx$$

We derive x and integrate xe^{-x^2}.

$$\left[-\frac{x}{2} e^{-x^2} \right]_0^{\infty} + \frac{1}{2} \int_0^{\infty} e^{-x^2} dx = \boxed{\frac{\sqrt{\pi}}{4}}$$

Note:

$$\lim_{x \to \infty} \frac{x}{2} e^{-x^2} = \lim_{x \to \infty} \frac{x}{2e^{x^2}} = 0$$

A Beautiful U-Substitution Integral

$$\int_0^{\infty} e^{-x^{\frac{2}{3}}} x^{-\frac{2}{3}} dx$$

[Silver]

Let $u = x^{\frac{1}{3}}$, then $du = \frac{1}{3} x^{-\frac{2}{3}} dx$.

$$\int_0^{\infty} 3e^{-u^2} du = \boxed{\frac{3\sqrt{\pi}}{2}}$$

1.20 Special Wildcard

Usually, the last problem is the hardest. This type of integral helps us tiebreak some competitors. Obviously, it still must be appropriate for the qualifying exam, but it can be anything unusual, creative, and use future techniques after this chapter. Here are examples from other qualifying exams that past further techniques than this chapter.

1. $$\int_0^\infty \frac{\ln(\sqrt{x+1})}{x\sqrt{x}}dx$$

2. $$\int_{-1}^\infty e^{-x^2}(5x^4-2x^6)dx$$

3. $$\int_0^{2\pi} \frac{dx}{\sin^4(x)+\cos^4(x)}$$

4. $$\int_0^\infty \frac{\ln(x)(\ln(x)+1)}{x^{2x-1}}dx$$

5. $$\int_0^\infty \frac{\tan^{-1}(x)}{x(\ln^2(x)+1)}dx$$

6. $$\int \frac{3\sin(x)+4\cos(x)}{4\sin(x)+3\cos(x)}dx$$

7. $$\int_0^1 \frac{\ln(1+x+x^2)}{x}dx$$

You'll soon learn how to solve these after you finish reading this whole guide lmao.

Chapter 2

Regular Round Topics

2.1 Modified Textbook Problems

I sometimes use calculus textbook for easy problems, skeletons, or for some inspiration. Let's do an example of how we modified textbook problems.

Textbook Problem 1

$$\int \sqrt{\frac{1-x}{1+x}}dx$$

I chose this problem because I like the idea of conjugating the radicals in a way where it becomes a u-sub and an arcsine function.

$$= \int \frac{1-x}{\sqrt{1-x^2}}dx = \sin^{-1}(x) + \sqrt{1-x^2} + C$$

Can we pull this trick off with a different integral? Maybe arcsecant? But that would be using $\frac{1}{x\sqrt{x^2-1}}$, which is a little harder to work with. How about other conjugating functions? I thought about $\frac{1}{\sqrt{x+1}-\sqrt{x-1}} = \frac{\sqrt{x+1}+\sqrt{x-1}}{2}$. Hmmm, what if:

$$\int \frac{\sqrt{x+1}+\sqrt{x-1}}{\sqrt{x-1}}dx = \int \frac{dx}{\sqrt{x^2-1}-x+1}$$

Ooooh, this looks tricky and nice. I'll use it!!

Textbook Problem 2

$$\int \frac{dx}{1+e^x}$$

This is a very common integral that keeps showing up in a lot of integration bees. Has anyone ever tried squaring it?? Let's see how the integral is.

$$\int \frac{dx}{(1+e^x)^2} = \int \frac{e^{-2x}}{(1+e^{-x})^2}dx = -\int \frac{u}{(1+u)^2}du$$

Hmmm, what if we generalize it?

$$\int \frac{dx}{(1+e^x)^n} = \int \frac{e^{-nx}}{(1+e^{-x})^n}dx = \int \frac{u^{n-1}}{(1+u)^n}du$$

Dang, not elementary, although this does look like something we can use in advanced competitions (beta function trick). Unfortunately, we won't do beta functions here in this integration bee, but we'll keep the first idea.

Textbook Problem 3

$$\int e^{e^x+x}dx$$

Another easy common integral that pops up a lot in integration bee. Let's modify it fancier. Notice the u-sub skeleton:

$$\int e^{e^x}e^x dx = \int e^u du$$

We can turn it into an integration by parts:

$$\int u^2 e^u du = \int e^{e^x}e^{3x}dx = \int e^{e^x+3x}dx$$

Ooooh, that dx form gave me an integral idea:

$$\int e^{(e^x-1)(e^x+1)}e^{2x}dx = \int e^{e^{2x}+2x-1}dx = \frac{1}{2e}\int e^u du$$

Oooooh, we could hide the substitution even more by making the u-sub skeleton into integration by parts:

$$\int e^{(e^x-1)(e^x+1)}e^{4x}dx = \frac{1}{2e}\int ue^u du$$

Looks good!!!

2.2 Sneaky Algebra Manipulation

Factoring Below 1 Degree

$$\int_0^1 \frac{x-4}{\sqrt{x}-2}dx$$

This could be easily solved by factoring $x-4=(\sqrt{x}-2)(\sqrt{x}+2)$.

Absorption

$$\int (x^2+x)^2(x^2-x+1)^2dx$$

This looks like a tedious integral, but...

$$\int x^2(x+1)^2(x^2-x+1)^2dx = \int x^2(x^3+1)^2dx$$

Substitution

$$\int_3^4 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

Let $u = x-3$:

$$\int_0^1 (u+2)(u+1)u(u-1)(u-2)du = \int_0^1 u(u^2-1)(u^2-4)du$$

Let $w = u^2-1$:

$$\frac{1}{2}\int_{-1}^0 w(w-3)dw = \boxed{\frac{5}{12}}$$

Reverse Absorption

$$\int (x^{2024}+x)^{2022}dx$$

This looks disgusting, however:

$$\int x^{2022}(x^{2023}+1)^{2022}dx$$

Now it's easier to see that this is just a u-substitution. Try this one:

$$\int \sqrt{e^x + e^{\frac{3}{2}x}}dx$$

Exchanged Absorption

$$\int (x+1)^3(2x^4+x^3)dx$$

$$\int (x+1)^3x^3(2x+1)dx = \int (x^2+x)^3(2x+1)dx$$

Factoring Below Degree 1 Type 2

$$\int \frac{dx}{(x+1)\sqrt{x}+2x}$$

$$\int \frac{dx}{(x+1+2\sqrt{x})\sqrt{x}} = \int \frac{dx}{(1+\sqrt{x})^2\sqrt{x}}$$

The way you form the integral truly matters. It can always throw off competitors; either to the wrong direction or not.

2.3 Basic Forced U-Substitutions

Forced u-substitutions means to completely perform u-sub without exchanging a proper dx. For example, if I need to do $u = \sqrt{x}$ but I don't have $du = \frac{1}{2\sqrt{x}}dx$, then you must solve for x by having $u^2 = x$, then derive both sides to get $2udu = dx$. After that, you can start subbing the integral.

1. $$\int e^{\sqrt{x}}dx$$

2. $$\int \sin(\ln(x))dx$$

3. $$\int_0^1 (1+\sqrt{x})^8$$

4. $$\int \frac{dx}{\sqrt{\sqrt{x}+1}}$$

5. $$\int (\sin^{-1}(x))^2 dx$$

6. $$\int x\sqrt[3]{2x-1}\,dx$$

7. $$\int_0^8 \frac{dx}{\sqrt{1+\sqrt{1+x}}}$$

8. $$\int \tan^{-1}(\sqrt{x})dx$$

9. $$\int \ln^3(x)dx$$

10. $$\int \frac{dx}{(1+\sqrt{x})^2}$$

11. $$\int_0^{\ln(10)} \frac{e^x\sqrt{e^x-1}}{e^x+8}dx$$

12. $$\int \frac{dx}{\sqrt{2021^x-1}}$$

13. $$\int \frac{dx}{\sqrt[3]{x+1}-1}$$

14. $$\int \frac{dx}{2\sqrt{x+3}+x}$$

15. $$\int 3^{\sqrt{2x+1}}dx$$

16. $$\int_{\ln(2)}^{\infty} \frac{\ln(1-e^{-x})}{e^x-1}dx$$

17. $$\int x\ln^5(x)dx$$

18. $$\int \sqrt{1-\sqrt{x}}dx$$

2.4 Gamma Function Integral

Factorial Integral Formula

$$n! = \int_0^\infty x^n e^{-x} dx$$

Now that you know the formula or the definition of the gamma function, here are some good examples that you can be creative with:

1. $$\int_0^\infty x^5 e^{-x} dx$$

2. $$\int_0^1 \ln^6(x) dx$$

3. $$\int_{-\infty}^0 e^x (x^4 + x^3 + x^2 + x) dx$$

4. $$\int_0^\infty x^5 e^{-x^2} dx$$

5. $$\int_0^\infty e^{-x}(x+1)^4 dx$$

6. $$\int_0^\infty \sqrt{x} e^{-\sqrt{x}} dx$$

7. $$\int_0^\infty e^{-x^{\frac{1}{3}}} x^{-\frac{1}{3}} dx$$

8. $$\int_0^\infty \frac{x}{e^{\sqrt{x}}} dx$$

9. $$\int_{-\infty}^\infty e^{3x - e^x} dx$$

10. $$\int_1^\infty \left(\frac{\ln(x)}{x}\right)^4 dx$$

11. $$\int_0^\infty \frac{1}{x^7 e^{\frac{1}{x}}} dx$$

12. $$\int_0^{\frac{\pi}{2}} \sec^6(x) e^{-\tan(x)} dx$$

2.5 Integration by Parts

Here are some good examples of integration by parts for regular rounds. Again, it's just a little trickier but also not too difficult.

1. $$\int x\ln(x+1)dx$$

2. $$\int e^x\sin(x)\cos(x)dx$$

3. $$\int \frac{\ln^2(x)}{x^3}dx$$

4. $$\int \frac{x}{1+\sin(x)}dx$$

5. $$\int \sec^3(x)dx$$

6. $$\int \frac{\ln(x+1)}{x\sqrt{x}}dx$$

7. $$\int \frac{\tan^{-1}(x)}{x^2}dx$$

8. $$\int e^x\sin^2(x)dx$$

9. $$\int \frac{x\sin^{-1}(x)}{\sqrt{1-x^2}}dx$$

10. $$\int \frac{xe^x}{\sqrt{e^x-1}}dx$$

11. $$\int \frac{\sin(\ln(x))}{x^3}dx$$

12. $$\int x^2\sec^{-1}(x)dx$$

13. $$\int \sec(x)dx$$

14. $$\int \frac{x\ln(x)}{\sqrt{x^2-1}}dx$$

15. $$\int \sin(x)\ln(\sin(x))dx$$

16. $$\int xe^x\sin(x)dx$$

17. $$\int \frac{\ln(1+x^4)}{x^3}dx$$

18. $$\int \ln(\tan(x))\cos(2x)dx$$

2.6 Trig-Identities Manipulation

Here, we use trig-identities to make a little trickier integrals. Here are some good examples for this regular round.

1. $$\int \sqrt{1+\cos(x)}dx$$

2. $$\int \frac{\sec(x)}{\sec(x)-\tan(x)}dx$$

3. $$\int \frac{dx}{\sin^2(x)\cos^2(x)}$$

4. $$\int \sin^3(x)+\cos^3(x)dx$$

5. $$\int \sin\left(x+\frac{\pi}{4}\right)\sin\left(x-\frac{\pi}{4}\right)dx$$

6. $$\int \frac{\cos(x)}{\cos(2x)-1}dx$$

7. $$\int \frac{\cos(2x)}{\sin(x)+\cos(x)}dx$$

8. $$\int \cos(x)\sqrt{\cos(2x)}dx$$

9. $$\int \frac{dx}{1+8\cos^2(x)}$$

10. $$\int \sec^2(x)\sqrt{\tan(x)\sqrt{\tan(x)\sqrt{\tan(x)\sin(x)}}}dx$$

11. $$\int \frac{2\tan(x)}{1-\tan^2(x)}dx$$

12. $$\int \frac{dx}{\sqrt{3}\sin(x)+\cos(x)}$$

13. $$\int \frac{\tan(x)}{\tan(x)+\cot(x)}dx$$

14. $$\int_0^{\frac{\pi}{4}} \ln(2\cos^2(x)-1)\cos(x)dx$$

15. $$\int \sin(\sin(x)\cos(x))\cos(2x)dx$$

16. $$\int \sec(x)\csc(x)dx$$

2.7 Intermediate Trig-Substitution

It's just like the qualifying exam chapter, except that the integrals are a bit trickier and we have more time for each integral. Here are some good examples of intermediate trig-substitution for this regular round.

1.
$$\int \frac{x}{(1-x)\sqrt{1-x^2}}dx$$

2.
$$\int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{\frac{3}{2}}}dx$$

3.
$$\int_{-1}^{1} \sin^{-1}(\sqrt{1-x^2})dx$$

4.
$$\int_0^1 \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)dx$$

5.
$$\int_0^\infty \frac{\sqrt{1+\sqrt{1+x^2}}}{1+x^2}dx$$

6.
$$\int_0^1 \frac{1-x}{\sin^{-1}(x)\sqrt{1-x^2}+1-x^2}dx$$

7.
$$\int_{\sqrt{2}}^\infty \frac{dx}{x^2(x^2-1)^{\frac{3}{2}}}$$

8.
$$\int_0^\infty \frac{\tan^{-1}(x)}{(\sqrt{1+x^2}+x)\sqrt{1+x^2}}dx$$

9.
$$\int_1^{\sqrt{2}} e^{\sec^{-1}(x)}\sqrt{x^2-1}dx$$

10.
$$\int_0^1 \frac{\sin^{-1}(\sqrt{x})}{\sqrt{1-x}}dx$$

11.
$$\int_0^1 \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$$

12.
$$\int_1^4 \tan^{-1}\left(\sqrt{\sqrt{x}-1}\right)dx$$

13.
$$\int_0^1 \frac{dx}{(1+x^2)\sqrt{1-x^2}}$$

2.8 Infinite Functions

Algebraic Functions

$$\int \frac{dx}{1+\frac{1}{1+\frac{1}{1+\ldots}}}$$

$$y = \frac{1}{1+\frac{1}{1+\ldots}} \Rightarrow y = \frac{1}{1+y} \Rightarrow y^2+y-1=0 \Rightarrow y = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi}$$

Remember that $y > 0$, hence

$$\int \frac{dx}{1+\frac{1}{1+\frac{1}{1+\ldots}}} = \int \frac{dx}{\varphi}$$

Geometry Series Convergence

$$\int_0^{2023} \frac{10dx}{(1+x^4)(1+x^8)(1+x^{16})\ldots}$$

[UC Berkeley Integration Bee 2023]

$$\int_0^{2023} \frac{10(1+x)(1+x^2)dx}{1+x+x^2+x^3+x^4+\ldots}$$

For $0 < x < 1$, the geometric series applies. For $1 < x$, the polynomial goes to infinity, but since it's the denominator, the whole integral goes to zero.

$$= \int_0^1 10(1-x^4)dx = \boxed{8}$$

BE VERY AWARE ABOUT THE DOMAIN!!

Force U-Subbing the Infinance

$$\int_0^{\frac{\pi}{2}+1} \sin(x-\sin(x-\sin(x-\ldots)))dx$$

[MIT Integration Bee 2023 - Regular Round]

Let $u = \sin(x-u)$. Then $\sin^{-1}(u) + u = x \to \left(\frac{1}{\sqrt{1-u^2}}+1\right)du = dx$.

$$\int_0^1 u + \frac{u}{\sqrt{1-u^2}}du = \boxed{\frac{3}{2}}$$

Taylor Series

$$\int \sqrt{x\sqrt[3]{x\sqrt[4]{x\sqrt[5]{x\ldots}}}}dx$$

$$\int x^{\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\cdots}dx = \int x^{\sum_{n=2}^{\infty}\frac{1}{n!}}dx = \int x^{e-2}dx$$

Function Inside a Taylor Series

$$\int \frac{e^{3x}+\frac{e^{4x}+\frac{e^{5x}+\ldots}{4}}{3}}{2}dx$$

$$\int \frac{e^{3x}}{2!}+\frac{e^{4x}}{3!}+\frac{e^{5x}}{4!}+\ldots dx = \int e^{x}\sum_{n=2}^{\infty}\frac{e^{nx}}{n!}dx = \int (e^{e^x}-e^x-1)e^x dx$$

Iterative Bound

$$\int_1^{\left(\int_1^{\left(\int_1^{\cdots} xdx\right)} xdx\right)} xdx$$

[Cambridge University Integration Bee]

$$y = \int_1^y xdx \Rightarrow y = \frac{y^2-1}{2} \Rightarrow y^2-2y-1=0$$

Since $y > 0$, the integral equals to $y = \boxed{1+\sqrt{2}}$.

2.9 Sneaky Symmetry

U-Substitution

$$\int_2^4 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

Let $u = x - 3$:

$$\int_{-1}^{1} u(u^2-1)(u^2-4)du = \boxed{0}$$

Removing a Term

$$\int_{\frac{1}{2}}^{2} \frac{\ln^2(x)(\sin^{-1}(\ln(x))+1)}{x}dx$$

Let $u = \ln(x)$:

$$\int_{-\ln(2)}^{\ln(2)} u^2(\sin^{-1}(u)+1)du = 2\int_0^{\ln(2)} u^2 du$$

Super easy, yet intimidating or distracting for competitors. Here's another good one for you to try:

$$\int_{-1}^{1} \left(\frac{x^5+1}{x^2+1}\right)(\tan^{-1}(x))^2 dx$$

A Very Dangerous Type of Symmetry Trick

$$\int_0^2 x^3\cos(x^3-3x^2+4x-2)dx$$

[Silver]

We will never ever use this integral in UC Berkeley Integration Bee, so I suggest a simpler integral, but for now I'm just showing you this dangerous symmetry concept.
Let $u = x - 1$ to purposely make the bound symmetrical.

$$\int_{-1}^{1} (u+1)^3\cos(u^3+u)du = \int_0^1 2(3u^2+1)\cos(u^3+u)du = \boxed{2\sin(2)}$$

So what happened here is that we hid a nice polynomial u^3+u by letting $u = x-1$, making it uglier and unpredictable. The $(u+1)^3$ was expressed and u^3+3u is an odd function cancelling out of the integral. This leaves us our needed derivative $3u^2+1$ which is why we had u^3+u inside cosine.

2.10 Sneaky Product Rule

Secant Hiding Spot

$$\int \left[\sec^3(x) + \sec(x)\tan^2(x)\right] dx$$

This will definitely scare the hell out of speed integrators, especially beginner calculus mathletes. Even though each term is famous for having such an ugly answer and a terrible integration by parts process, adding both of these integrals gives a very sneaky product rule function.

$$\int \sec(x)\sec^2(x) + (\sec(x)\tan(x))\tan(x)dx$$

And now you see that the answer is just $\boxed{\sec(x)\tan(x) + C}$.

Exponential Product Rule

$$\int \frac{e^x(2x+1)}{\sqrt{x}}dx$$

$$= \int e^x\left(2\sqrt{x} + \frac{1}{\sqrt{x}}\right) = \boxed{2\sqrt{x}e^x + C}$$

Most beginners may not be able to see this. They'll probably just do integration by parts. You can also trick them with this as well:

$$\int e^x(\sin(x) - \cos(x))dx$$

They'll accidentally think its not product rule because of the minus sign.

Hiding the Exponential Product Rule

$$\int \sin(\ln(x)) + \cos(\ln(x))dx$$

Let $u = \ln(x)$:

$$\int e^u(\sin(u) + \cos(u))du$$

xs and Logarithms

$$\int \ln(\ln(x)) + \frac{1}{\ln(x)} dx$$

$$= \int 1 \cdot \ln(\ln(x)) + x\left(\frac{1}{x\ln(x)}\right) dx = \boxed{x\ln(\ln(x)) + C}$$

Radical Product Rule

$$\int \sqrt{\frac{\ln(x)}{x}} + \frac{1}{\sqrt{x\ln(x)}} dx$$

$$= \int \sqrt{\ln(x)} \cdot \frac{1}{\sqrt{x}} + \sqrt{x} \cdot \frac{1}{x\sqrt{\ln(x)}} = \boxed{2\sqrt{x\ln|x|} + C}$$

Multi-Product Rule

$$\int_0^{2\pi} \cos(x)(x^3 + 6x) dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

$$= \int_0^{2\pi} \cos(x)(x^3 + 6x) + \sin(x)(3x^2 - 3x^2) dx$$

$$= \int_0^{2\pi} (x^3\cos(x) + 3x^2\sin(x)) + (6x\cos(x) - 3x^2\sin(x)) dx$$

$$= \left[x^3\sin(x) + 3x^2\cos(x)\right]_0^{2\pi} = \boxed{12\pi^2}$$

An Imposter Integral

$$\int_{-1}^{\infty} e^{-x^2}(5x^4 - 2x^6) dx$$

[Silver]

No, this is not a Gaussian integral.

$$= \int_{-1}^{\infty} 5x^4 e^{-x^2} + x^5\left(-2xe^{-x^2}\right) dx$$

$$= \left[x^5 e^{-x^2}\right]_{-1}^{\infty} = \boxed{\frac{1}{e}}$$

Chapter 3

Intermediate Topics

3.1 Power Substitution

Getting Rid of Radicals

$$\int \frac{\sqrt{1-\sqrt{x}}}{x}dx$$

Let $u = \sqrt{1-\sqrt{x}}$, then $(1-u^2)^2 = x \to -4(1-u^2)u = dx$.

$$\int \frac{-4u^2(1-u^2)}{(1-u^2)^2}du = 4\int \frac{u^2}{u^2-1}du$$

Power Subbing

$$\int_0^1 \frac{dx}{1+\sqrt[3]{x}}$$

Let $u = 1+\sqrt[3]{x} \to (u-1)^3 = x$, then $3(u-1)^2du = dx$.

$$\int_1^2 \frac{3(u-1)^2}{u}du = 3\int_1^2 \left(u-2+\frac{1}{u}\right)du$$

Fulfilling All Radicals

$$\int \frac{dx}{\sqrt{x}+\sqrt[3]{x}}$$

Let $u^6 = x$ so that all radicals are cancelled.

$$\int \frac{6u^5du}{u^3+u^2} = \int \frac{6u^3du}{u+1}du = 6\int u^2-u+1-\frac{1}{1+u}du$$

3.2 Sneaky U-Sub Tricks

Inverse Functions

Note that $W(x)$ is the inverse function of xe^x.

$$\int_0^e e^{W(x)}dx$$

[Silver]

Let $u = W(x)$. Then $ue^u = x \to e^u(u+1)du = dx$.

$$\int_0^1 e^{2u}(u+1)du$$

Trig Manipulation

$$\int \frac{\sec(x) - \tan(x)}{\sqrt{\sin(x)}}dx$$

$$= \int \frac{1-\sin(x)}{\cos(x)\sqrt{\sin(x)}}dx = \int \frac{\cos(x)dx}{(1+\sin(x))\sqrt{\sin(x)}} = \int \frac{du}{(1+u)\sqrt{u}}$$

Trig Manipulation 2

$$\int_0^{\frac{\pi}{4}} \frac{\tan(x)}{\sqrt{\cos(2x)}}dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)\sqrt{2\cos^2(x)-1}}dx = \int_{\frac{\sqrt{2}}{2}}^1 \frac{du}{u\sqrt{2u^2-1}} = \boxed{\frac{\pi}{4}}$$

Norwegian Substitution

$$\int_0^\infty \frac{6^x}{4^x+9^x}dx$$

[Integral Kokeboken]

$$= \int_0^\infty \frac{\left(\frac{6}{4}\right)^x}{1+\left(\frac{9}{4}\right)^x}dx = \int_0^\infty \frac{\left(\frac{3}{2}\right)^x}{1+\left(\frac{3}{2}\right)^{2x}}dx \xrightarrow{u=\left(\frac{3}{2}\right)^x} \frac{1}{\ln(\frac{3}{2})}\int_1^\infty \frac{du}{1+u^2}$$

Blind Substitution

$$\int \frac{dx}{(2x+1)\sqrt{x^2+x}}$$

Blind substitutions are very hard to come up with. Pretty much there's a hidden manipulation inside the u-sub mechanism. Notice that if I let $u = \sqrt{x^2+x}$, observe that

$$\begin{aligned} u^2 &= x^2 + x \\ 4u^2 + 1 &= 4x^2 + 4x + 1 \\ \sqrt{4u^2+1} &= (2x+1). \end{aligned}$$

Also,

$$\begin{aligned} du &= \frac{2x+1}{2\sqrt{x^2+x}}dx \\ du &= \frac{\sqrt{4u^2+1}}{2u}dx \\ \frac{2u}{\sqrt{4u^2+1}}du &= dx. \end{aligned}$$

So now we have,

$$\int \frac{1}{u\sqrt{4u^2+1}} \cdot \frac{2u}{\sqrt{4u^2+1}}du = \int \frac{2}{4u^2+1}du = \boxed{\tan^{-1}(2\sqrt{x^2+x}) + C}$$

WOOOOOOAHHH, definitely did not see that coming. It's just a simple arctangent integral. But wait a minute...WolframAlpha has another answer: $\boxed{2\tan^{-1}\left(\frac{x}{x+1}\right) + C}$.

To check if both answers are valid, take the derivative of the difference of those answers on WolframAlpha or Mathematica. If the derivative equals to zero, they are both valid. If not, there is a domain issue. In that case, just add in bounds to the integrals, easy fix!!

Blind Substitution 2

$$\int \left(1 + \frac{1}{x}\right)\sqrt{xe^x}dx$$

Let $u = \sqrt{xe^x} \to u^2 = xe^x$, then

$$\begin{aligned} 2udu &= e^x(x+1)dx \\ \frac{2u}{u^2}du &= \frac{e^x(x+1)}{xe^x}dx \\ \frac{2}{u}du &= \left(1 + \frac{1}{x}\right)dx \end{aligned}$$

British Substitution

$$\int_0^1 \left(\frac{x-1}{x+1}\right)^3 \frac{dx}{x+1}$$

[Oxford University]

Let $u = \frac{x-1}{x+1} = 1 - \frac{2}{x+1}$. Then $x+1 = \frac{2}{1-u} \to dx = \frac{2}{(1-u)^2}du$.

$$\int_{-1}^0 u^3 \left(\frac{1-u}{2}\right)\left(\frac{2}{(1-u)^2}\right) du = \int_{-1}^0 \frac{u^3}{1-u} du$$

British Substitution 2

$$\int_1^2 \sqrt{\frac{2-x}{x-1}} dx$$

[National University of Singapore]

Let $u = \sqrt{\frac{2-x}{x-1}}$. Then $u^2 = \frac{1}{x-1} - 1 \to \frac{1}{u^2+1} + 1 = x$. Also, $dx = -\frac{2u}{(u^2+1)^2}du$.

$$\int_0^\infty u\left(\frac{2u}{(u^2+1)^2}\right) du = 2\int_0^{\frac{\pi}{2}} \sin^2\theta d\theta = \boxed{\frac{\pi}{2}}$$

Last integral was made by trig-subbing $u = \tan\theta$.

Needing More Power

$$\int_1^\infty \frac{dx}{x\sqrt{x^5-1}}$$

$$= \int_1^\infty \frac{x^4}{x^5\sqrt{x^5-1}} dx = \frac{1}{5}\int_0^\infty \frac{du}{(u+1)\sqrt{u}} = \boxed{\frac{\pi}{5}}$$

Hiding the Rational Exponent

$$\int \frac{\sqrt{x}}{1+x^3} dx$$

$$= \int \frac{\sqrt{x}}{1+(x^{\frac{3}{2}})^2} \xrightarrow{u = x^{\frac{3}{2}}} \frac{2}{3}\int \frac{du}{1+u^2} = \boxed{\frac{2}{3}\tan^{-1}(x^{\frac{3}{2}}) + C}$$

Caveman Trick

$$\int \frac{dx}{x^4 + x}$$

This always tricks beginners. It looks like partial fractions, but...

$$\int \frac{dx}{x(x^3+1)} = \int \frac{dx}{x^4(1+x^{-3})}$$

As simple as it looks, this is not very easy to see. Most of us aren't comfortable with forcing variables to have a negative exponent. But in this case, it is super helpful to do so for a simple u-substitution.

Let $u = 1 + x^{-3}$. Then $du = -3x^{-4}dx$.

$$= -\frac{1}{3}\int \frac{du}{u} = \boxed{-\frac{1}{3}\ln|1+x^{-3}| + C}$$

Again, as simple as it is, it's not very easy to see. If you were thrown with a bunch of random integrals, you wouldn't be able to tell. If this is a bit too hard to see, giving more power also works!!

$$\int \frac{dx}{x(1+x^3)} = \int \frac{x^2}{x^3(1+x^3)}dx = \frac{1}{3}\int \frac{1}{u} - \frac{1}{u+1}du$$
$$= \frac{1}{3}\left[\ln|x^3| - \ln|x^3+1|\right] + C = -\frac{1}{3}\ln\left|\frac{x^3+1}{x^3}\right| + C$$

Multi Logged

$$\int \frac{\ln(x)\ln(\sqrt{2}x)\ln(2x)}{x}dx$$

[UC Berkeley Integration Bee 2022 - Tiebreaker]

Let $u = \ln(x)\ln(2x)$. Then, $du = \frac{\ln(2x^2)}{x}dx = \frac{2\ln(\sqrt{2}x)}{x}dx$.

LOGS EVERYWHERE

$$\int x(\ln^2(x)+1)(\ln(x)+1)^2dx$$

[Silver]

Letting $u = \ln(x)$ is a death sentence.
Let $u = x(\ln^2(x)+1)$. You're welcome lol
Sneaky Note: The derivative $xf(\ln(x))$ will give you a function full of $\ln(x)$s.

3.3 Wallis' Trick

Wallis' Trick

For all $n \in \mathbb{N}$,

$$\int_0^{\frac{\pi}{2}} \sin^n(x)dx = \int_0^{\frac{\pi}{2}} \cos^n(x)dx = \begin{cases} \frac{\pi}{2} \times \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \ldots \times \frac{n-1}{n} & \text{if } n \text{ is even} \\ \frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} \times \frac{8}{9} \times \ldots \times \frac{n-1}{n} & \text{if } n \text{ is odd} \end{cases}$$

Basic Example

$$\int_0^{\frac{\pi}{2}} \cos^4(x)dx$$

By Wallis' Trick,

$$\int_0^{\frac{\pi}{2}} \cos^4(x)dx = \frac{\pi}{2} \times \frac{1}{2} \times \frac{3}{4} = \boxed{\frac{3\pi}{16}}$$

Manipulation Example

$$\int_0^{\frac{\pi}{2}} \sin^2(x)\cos^2(x)dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{4}\sin^2(2x)dx = \frac{1}{8}\int_0^{\pi} \sin^2(u)du$$

Let $w = \frac{\pi}{2} - u$ to turn it into Wallis' Trick integral:

$$\frac{1}{4}\int_0^{\frac{\pi}{2}} \cos^2(u)du = \boxed{\frac{\pi}{16}}$$

Helpful Wallis

$$\int_0^{\frac{\pi}{2}} (1+\sin(x))^3dx$$

$$\int_0^{\frac{\pi}{2}} (\sin^3(x) + 3\sin^2(x) + 3\sin(x) + 1)dx$$

$$= \frac{2}{3} + \frac{3\pi}{4} + 3 + \frac{\pi}{2} = \boxed{\frac{11}{3} + \frac{5\pi}{4}}$$

3.4 Advanced Trig-Substitution

Getting Rid of the Inverse Trig Functions

$$\int_0^1 \sin^{-1}\left(\sqrt{\frac{x}{x+1}}\right) dx$$

Let $x = \tan^2\theta$. Then $dx = 2\tan\theta\sec^2\theta d\theta$.

$$\int_0^{\frac{\pi}{4}} \sin^{-1}\left(\frac{\tan\theta}{\sec\theta}\right) 2\tan\theta\sec^2\theta d\theta = \int_0^{\frac{\pi}{4}} \theta \cdot 2\tan\theta\sec^2\theta d\theta$$

A Little Trig Manipulation

$$\int_1^{\sqrt{2}} \frac{\sqrt{x^2-1}}{x^2+x} dx$$

Let $x = \sec\theta$.

$$\int_0^{\frac{\pi}{4}} \frac{\sec\theta\tan^2\theta}{\sec\theta(\sec\theta+1)} d\theta = \int_0^{\frac{\pi}{4}} (\sec\theta - 1) d\theta = \boxed{\ln(1+\sqrt{2}) - \frac{\pi}{4}}$$

An Intimidating Friendly Integral

$$\int_0^1 \frac{32}{\sqrt{x-x^2}} \left(\tan^{-1}\left(\sqrt{\frac{x}{1-x}}\right)\right)^3 dx$$

[UC Berkeley Integration Bee 2021]

Let $x = \sin^2\theta$ to get rid of the inverse tangent.

$$\int_0^{\frac{\pi}{2}} \frac{32\theta^3}{\sin\theta\cos\theta} \sin(2\theta) d\theta = \boxed{\pi^4}$$

Another Arcsine Example

$$\int_0^1 \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

Let $x = \tan\theta$.

$$\int_0^{\frac{\pi}{4}} \sin^{-1}\left(\frac{2\tan\theta}{\sec^2\theta}\right) d\theta = \int_0^{\frac{\pi}{4}} 2\theta d\theta = \boxed{\frac{\pi^2}{16}}$$

An Iterating Cosine Trig-Sub

$$\int_0^2 \sqrt{2+\sqrt{2+\sqrt{2+x}}}\,dx$$

This is also one of those integrals we will never put in this integration bee. I'm only showing you this for the sake of the concept. As weird as this looks, there's a cool ugly trick for this. Let $x = 2\cos(2\theta)$. Note that $1+\cos(2x) = 2\cos^2(x)$.

$$\int_0^{\frac{\pi}{4}} \sqrt{2+\sqrt{2+\sqrt{2+2\cos(2\theta)}}}4\sin(2\theta)d\theta = \int_0^{\frac{\pi}{4}} \sqrt{2+\sqrt{2+2\cos\theta}}4\sin(2\theta)d\theta$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{2+2\cos\left(\frac{\theta}{2}\right)}4\sin(2\theta)d\theta = \int_0^{\frac{\pi}{4}} 8\cos\left(\frac{\theta}{4}\right)\sin(2\theta)d\theta$$

$$4\int_0^{\frac{\pi}{4}} \sin\left(\frac{9}{4}\theta\right)+\sin\left(\frac{7}{4}\theta\right)d\theta = \boxed{\frac{32}{63}\left(8-\sin\left(\frac{\pi}{16}\right)\right)}$$

The double angle trig-identity and the constant worked together and caused a chain reaction; cancelling multiple radicals throughout the integral. Although, the constants are a mess.

Subbing Double Angle Cosine

$$\int_0^1 \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx$$

Let $x = \cos(2\theta)$.

$$\int_0^{\frac{\pi}{4}} \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)2\sin(2\theta)d\theta = \int_0^{\frac{\pi}{4}} 2\theta\sin(2\theta)d\theta = \boxed{\frac{1}{2}}$$

Logarithm Example

$$\int_0^1 \ln(\sqrt{1-x}+\sqrt{1+x})dx$$

Let $x = \sin(2\theta)$. Note that $1\pm\sin(2x) = (\sin(x)\pm\cos(x))^2$

$$\int_0^{\frac{\pi}{4}} \ln(|\sin\theta-\cos\theta|+\sin\theta+\cos\theta)2\cos(2\theta)d\theta = \int_0^{\frac{\pi}{4}} 2\ln(2\cos\theta)\cos(2\theta)d\theta$$

After integration by parts, you should get $\boxed{\frac{\pi}{4}+\frac{1}{2}\ln(2)-\frac{1}{2}}$

Famous Indian Trig-Sub Monster!!!

$$\int \frac{\sqrt[3]{x+\sqrt{2-x^2}}\,\sqrt[6]{1-x\sqrt{2-x^2}}}{\sqrt[3]{1-x^2}}dx$$

[JEE Main]

It's one of those beautiful integrals where it's horrendous on the outside but very nice on the inside. Let $x=\sqrt{2}\sin\theta$. Note that $1-2\sin^2\theta=\cos(2\theta)$.

$$=\int \frac{\sqrt[3]{\sqrt{2}\sin\theta+\sqrt{2}\cos\theta}\,\sqrt[6]{1-2\sin\theta\cos\theta}}{\sqrt[3]{\cos(2\theta)}}\sqrt{2}\cos\theta d\theta$$

Recall that $1-\sin(2\theta)=(\sin\theta-\cos\theta)^2$. We use $\cos\theta-\sin\theta$ to cancel out $\cos(2\theta)$.

$$=\int \frac{\sqrt[6]{2}\sqrt[3]{\sin\theta+\cos\theta}\sqrt[3]{\cos\theta-\sin\theta}}{\sqrt[3]{\cos(2\theta)}}\sqrt{2}\cos\theta d\theta=\int 2^{\frac{2}{3}}\cos\theta d\theta$$

$$=2^{\frac{2}{3}}\sin\theta+C=2^{\frac{2}{3}}\left(\frac{x}{\sqrt{2}}\right)+C=\boxed{2^{\frac{1}{6}}x+C}$$

Yes, that whole function was just a constant. Very wild.

Constants Assisting Us

$$\int_{-1}^{1}\frac{\sqrt{1+x}}{\sqrt{2}+\sqrt{1-x}}dx$$

Let $x=\cos(2\theta)$.

$$\int_0^{\frac{\pi}{2}}\frac{\sqrt{2}\cos\theta}{\sqrt{2}+\sqrt{2}\sin\theta}2\sin(2\theta)d\theta=\int_0^{\frac{\pi}{2}}\frac{4\sin\theta\cos^2\theta}{1+\sin\theta}d\theta$$

$$\int_0^{\frac{\pi}{2}}4\sin\theta(1-\sin\theta)d\theta=\boxed{4-\pi}$$

Another Use for Cosine

$$\int_{\frac{\sqrt{3}}{2}}^{1}\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{2x^2-1}\right)dx$$

Let $x=\cos\theta$. Note that $2\cos^2(x)-1=\cos(2x)$. Then,

$$\int_0^{\frac{\pi}{6}}\tan^{-1}\left(\frac{\sin(2\theta)}{\cos(2\theta)}\right)\sin\theta d\theta=\int_0^{\frac{\pi}{6}}2\theta\sin\theta d\theta=\boxed{1-\frac{\pi}{2\sqrt{3}}}$$

A Triple Angle Identity

$$\int_0^{\sqrt{3}} \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right) dx$$

[Cambridge University]

Okay...probably not use the identity for tangent, but maybe for sine or cosine. Let $x = \tan\theta$. It is NOT well known that:

$$\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \tan(3\theta)$$

HOWEVER!!! Please realize that we are about to do $\tan^{-1}(\tan(3\theta)) = 3\theta$.
Be very careful that our integral WILL NOT BE:

$$\int_0^{\frac{\pi}{3}} 3\theta\sec^2\theta d\theta$$

Notice our top bound $\theta = \frac{\pi}{3}$ because of $x = \sqrt{3} \to \theta = \tan^{-1}(\sqrt{3})$.
It is true that in general, $-\frac{\pi}{2} \leq \tan^{-1}(x) \leq \frac{\pi}{2}$, but $\tan^{-1}(\tan(3 \cdot \frac{\pi}{3})) = 3\left(\frac{\pi}{3}\right) = \pi > \frac{\pi}{2}$.
This is a domain problem, meaning we have to split our bounds.
Since we need $3\theta = \frac{\pi}{2} \to \theta = \frac{\pi}{6}$, we must split our bounds at $\frac{\pi}{6}$.
Since $\tan^{-1}\left(\tan\left(3 \cdot \frac{\pi}{3}\right)\right)$ is actually 0, which is a π difference than what we want, then $\tan^{-1}(\tan(3\theta)) = 3\theta - \pi$ when we go beyond the bound interval of $\frac{\pi}{6}$.
Hence, our true integral is

$$\int_0^{\frac{\pi}{6}} 3\theta\sec^2\theta d\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (3\theta - \pi)\sec^2\theta d\theta = \boxed{\frac{\pi}{\sqrt{3}} - \ln(8)}.$$

Now without being too complicated like this evil integral, try to use the triple angle identity for sine and cosine, and turn it into a trig-sub integral.

3.5 Awkward & Advanced Integration by Parts

Not Every Log is Derived

$$\int \ln(x)\ln(1-\ln(x))dx$$

Integrating $\ln(x)$ gives us $x\ln(x) - x = x(\ln(x)-1)$ which will nicely cancel out with the derivative.

Nonelementary Intimidation

$$\int xe^x \ln(1-x)dx$$

This scares advanced mathletes because this looks like an impossible Taylor Series trick due to the integral having no bounds. Integrating xe^x gives us $xe^x - e^x = e^x(x-1)$ which will nicely cancel out with the derivative.

A Hidden Arctangent Cancellation

$$\int \frac{3x^2-1}{2x\sqrt{x}}\tan^{-1}(x)dx$$

[Cambridge University]

Now this integral is super difficult to see. I honestly couldn't figure out the trick to this until WolframAlpha showed me that it's just integration by parts.
Observe that

$$\frac{3x^2-1}{2x\sqrt{x}} = \frac{3\sqrt{x}}{2} - \frac{1}{2x\sqrt{x}} \Rightarrow \int \frac{3\sqrt{x}}{2} - \frac{1}{2x\sqrt{x}}dx = \frac{x^2+1}{\sqrt{x}} + C$$

Notice that with x^2+1, we cancel it out nicely when we derive arctangent. Thus,

$$\int \frac{3x^2-1}{2x\sqrt{x}}\tan^{-1}(x)dx = \boxed{\left(\frac{x^2+1}{\sqrt{x}}\right)\tan^{-1}(x) - 2\sqrt{x} + C}$$

The reason why I didn't want to do integration by parts was because it didn't look pretty to perform, but I wouldn't know.

Utilizing Trig-Manipulation

$$\int_0^\pi \ln(1+\sin(x))\sin(x)dx$$

[Silver]

$$[-\cos(x)\ln(1+\sin(x))]_0^\pi + \int_0^\pi (1-\sin(x))dx = \boxed{\pi - 2}$$

3.6 Intermediate Partial Fractions

Two Methods Example

$$\int \frac{1}{(x^2+6x+13)(x^2+6x+10)}dx$$

The first method I will show you is dummying. Pretty much, you will treat any nonlinear functions as linear for the sake of partial fractions. Notice that if I let $\alpha = x^2+6x$, then

$$\frac{1}{(x^2+6x+13)(x^2+6x+10)} = \frac{1}{(\alpha+13)(\alpha+10)}$$

Now we can just perform basic partial fractions.

$$\frac{-\frac{1}{3}}{\alpha+13} + \frac{\frac{1}{3}}{\alpha+10} = \frac{1}{3(x^2+6x+10)} - \frac{1}{3(x^2+6x+13)}$$

Now this is just an easier integral full of completing squares and arctangents:

$$\int \frac{1}{(x^2+6x+13)(x^2+6x+10)}dx = \frac{1}{3}\int \frac{1}{x^2+6x+10} - \frac{1}{x^2+6x+13}dx$$

$$\frac{1}{3}\int \frac{1}{(x+3)^2+1} - \frac{1}{(x+3)^2+4}dx = \boxed{\frac{1}{3}\tan^{-1}(x+3) - \frac{1}{6}\tan^{-1}\left(\frac{x+3}{2}\right) + C}$$

The second method is a bit more advanced. In this case, you utilize constants and perform algebraic manipulation. Because we have x^2+6x in both parenthesis, we will perform a zero substitution with it. Watch very closely:

$$\frac{1}{3}\int \frac{13-10}{(x^2+6x+13)(x^2+6x+10)}dx = \frac{1}{3}\int \frac{(x^2+6x+13)-(x^2+6x+10)}{(x^2+6x+13)(x^2+6x+10)}dx$$

$$= \frac{1}{3}\int \frac{1}{x^2+6x+10} - \frac{1}{x^2+6x+13}dx$$

After that, the rest of the process is the same as the previous method.
Other problems like below, you just have to use Heaviside method (;-;).

$$\int \frac{dx}{1+x+x^2+x^3} \qquad \int \frac{dx}{(1+x)^2(4+x)}$$

3.7 Basic Periodicity

Periodicity Concept

For any integers m, n,

$$\int_0^{2\pi} \sin(mx)\sin(nx)dx = 0$$

$$\int_0^{2\pi} \sin(mx)\cos(nx)dx = 0$$

$$\int_0^{2\pi} \cos(mx)\cos(nx)dx = 0$$

Note: For the first and third case, $m \neq n$ otherwise it will be a square. If the answers are going to be in terms of sine functions, then the bounds will make it equal to zero. For cosines, you must be aware of the coefficients inside each cosines.

Consider the integral:

$$\int_0^{\pi} \sin(23x)\cos(19x)dx$$

By trig identities, we know that this function is in terms of sine functions because of the trig-identity: $\sin(A+B) = \sin(A)\cos(B) + \sin(B)\cos(A)$. Coefficients are nasty, so let's just be general. p and q are some integer related to 23 and 19 whatever the result is:

$$\int_0^{\pi} \frac{\sin(px)+\sin(qx)}{2}dx = \left[-\frac{1}{2p}\cos(px) - \frac{1}{2q}\cos(qx)\right]_0^{\pi}$$

Uh oh, the answers are in terms of cosine, this is kind of bad because you actually have to know what the values of p and q are. This is because $\cos(0) = \cos(2n\pi) = 1$ and $\cos(\pi) = \cos((2n+1)\pi) = -1$ for $n \in \mathbb{Z}$. These answers will not cancel out to zero.

If we consider the integral:

$$\int_0^{\pi} \sin(23x)\sin(19x)dx$$

then by trig-identities, this function is in terms of cosine functions.

$$\int_0^{\pi} \frac{\cos(px)+\cos(qx)}{2}dx = \left[\frac{1}{2p}\sin(px) + \frac{1}{2q}\sin(qx)\right]_0^{\pi}$$

By periodicity, no matter what integers p, q are, it all equals to 0.

Periodicity Example

$$\int_0^{2\pi} (\cos(2020x) - 2020\sin(x))(\cos(x) - \sin(x))dx$$

[UC Berkeley Integration Bee 2020]

By inspection, $\cos(2020x)\cos(x)$ is going to be in terms of cosines, which will give answers of sine functions. So this term will become zero. $\cos(2020x)\sin(x)$ will also become zero. And $2020\sin(x)\cos(x)$ also becomes zero as well. $\sin^2(x)$ will not become zero, thus:

$$\int_0^{2\pi} (\cos(2020x) - 2020\sin(x))(\cos(x) - \sin(x))dx = \int_0^{2\pi} 2020\sin^2(x)dx$$

The reason why square's don't become zero is because they create additional constants: $\sin^2(x) = \frac{1}{2} - \frac{1}{2}\cos(2x)$. Hence,

$$= \int_0^{2\pi} 1010 - 1010\cos(2x)dx = \int_0^{2\pi} 1010 = \boxed{2020\pi}$$

KEEP ON A LOOKOUT FOR SQUARES!!!!

An Invasion of Periodicity

$$\int_0^{2\pi} (\sin(x) + \sin(2x) + \sin(3x))^2 dx$$

Obviously, I'm not gonna express this, so I'll take advantage of periodicity. Notice that each term is going to be $\sin(ax)\sin(bx)$, which becomes in terms of cosines and then integrated into functions of sines; giving us zero due to the bounds. However, we do have squares: $\sin^2(x), \sin^2(2x)$, and $\sin^2(3x)$. Hence,

$$\int_0^{2\pi} (\sin(x) + \sin(2x) + \sin(3x))^2 dx = \int_0^{2\pi} \sin^2(x) + \sin^2(2x) + \sin^2(3x)dx$$

$$\int_0^{2\pi} \frac{3}{2} - \frac{1}{2}\cos(2x) - \frac{1}{2}\cos(4x) - \frac{1}{2}\cos(6x)dx = \boxed{3\pi}$$

Since those cosines will be integrated into sines, those also become zero.

3.8 Speed Bashing

We now begin the art of making speed bashing integrals. These integrals are mainly for tiebreaking competitors. Technically, we use speed bashing in all rounds as tiebreakers.

All Standard Methods in One Integral

$$\int_0^1 \tan^{-1}(\sqrt{1-x^2})dx$$

[Cambridge University]

It is very hard to create and find a good integral that utilizes all standard methods, but this is a good example. First, trig-substitution:

$$\int_0^1 \tan^{-1}(\sqrt{1-x^2})dx = \int_0^{\frac{\pi}{2}} \tan^{-1}(\cos\theta)\cos\theta d\theta$$

Second, integration by parts:

$$= [\sin\theta\tan^{-1}(\cos\theta)]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{\sin^2\theta}{1+\cos^2\theta}d\theta$$

Third, trigonometric manipulation:

$$= 0 + \int_0^{\frac{\pi}{2}} \frac{\tan^2\theta}{\sec^2\theta+1}d\theta = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta-1}{\tan^2\theta+2}d\theta = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta}{\tan^2\theta+2} - \frac{\sec^2\theta}{(\tan^2\theta+2)(\tan^2\theta+1)}d\theta$$

Fourth, u-substitution into partial fractions:

$$= \int_0^{\infty} \frac{1}{u^2+2} - \frac{1}{(u^2+2)(u^2+1)}du = \int_0^{\infty} \frac{u^2}{(u^2+2)(u^2+1)}du$$

$$\int_0^{\infty} \frac{\frac{-2}{-1}}{u^2+2} + \frac{\frac{-1}{1}}{u^2+1}du = \int_0^{\infty} \frac{2}{u^2+2} - \frac{1}{u^2+1}du$$

Fifth, improper integral evaluation:

$$= \left[\sqrt{2}\tan^{-1}\left(\frac{u}{\sqrt{2}}\right) - \tan^{-1}(u)\right]_0^{\infty} = \boxed{\frac{\pi}{2}(\sqrt{2}-1)}$$

Can you find a faster solution for this integral???

Sprinting Example

$$\int x^5 e^x dx$$

By utilizing pattern recognition,

$$= e^x(x^5 - 5x^4 + 20x^3 - 60x^2 + 120x - 120) + C$$

Symmetry Example

$$\int_{-1}^{1} \left(x + \sqrt{1-x^2}\right)^3 dx$$

By Symmetry and letting $x = \sin\theta$,

$$= 2\int_0^1 3x^2\sqrt{1-x^2} + (\sqrt{1-x^2})^3 dx = \int_0^{\frac{\pi}{2}} 6\sin^2\theta\cos^2\theta + 2\cos^4\theta d\theta = \boxed{1 + \frac{3\pi}{8}}$$

The following integral can be solved using Wallis' Trick. Psssh...factor $\cos^2\theta$! ;)

Integration by Parts Example

$$\int x(e^x + \sin(x) + 1)dx$$

The x can just be integrated. Instead of doing IBP separately, you could just integrate $(e^x + \sin(x))$ together to not waste time.

Trig-Sprinting

$$\int \sin^3(x) + \cos^3(x) + \tan^3(x)dx$$

$$= \int \sin(x)(1-\cos^2(x)) + \cos(x)(1-\sin^2(x)) + \tan(x)\sec^2(x) - \tan(x)dx$$

Algebra Sprinting

$$\int \frac{1}{x^2+1}\left(\frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2\right)dx$$

$$\int \frac{1}{x^2+1}\left(\left(\frac{1}{x^2}+1\right) + \left(x + \frac{1}{x}\right) + x^2\right)dx = \int \frac{1}{x^2} + \frac{1}{x} + 1 - \frac{1}{x^2+1}dx$$

Here are some speed bashing integral examples I've made over the past years:

1. $$\int_0^{\ln^2(4)} \sqrt{\frac{e^{\sqrt{e^{\sqrt{x}}+\sqrt{x}}}}{x}}dx$$

[UC Berkeley Integration Bee 2023]

2. $$\int_{-1}^{1} 15(1+\sqrt[3]{x})^3 dx$$

3. $$\int_0^{\infty} e^{-x^2}(x+1)^3 dx$$

4. $$\int_1^{\infty} \frac{dx}{x(x^{\ln(4)}+1)}dx$$

5. $$\int_{-\infty}^{\infty} \frac{dx}{x^2+\pi x+\pi^2}$$

6. $$\int_0^{\infty} \frac{\ln(\sqrt{x+1})}{x\sqrt{x}}dx$$

[Stanford Math Tournament Integration Bee 2022]

7. $$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin(x)+\cos(x)+\tan(x)}{\sec^2(x)}dx$$

8. $$\int_{-1}^{1} \sqrt{x+\sqrt{x^2-1}}+\sqrt{x-\sqrt{x^2-1}}dx$$

9. $$\int_1^4 \sec^{-1}\left(\sqrt{\sqrt{x}}\right) dx$$

10. $$\int \frac{x(x+1)}{(x-1)x+1}dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

11. $$\int_0^{\infty} \frac{dx}{e^x+\frac{1}{e^x+1}}dx$$

12. $$\int_0^{\infty} \frac{dx}{\sqrt{1+\sqrt{e^x}}}$$

13. $$\int_0^9 e^{\sqrt{1+\sqrt{x}}}dx$$

14. $$\int \frac{\tan(2x)}{\tan^2(x)}dx$$

Chapter 4

A Little Advanced Topics

4.1 Undertow Technique

Beginner Example

$$\int \frac{2x^3-1}{x^4+x}dx$$

[MIT Integration Bee 2006]

This confuses a lot of beginners because the derivatives don't match and it looks awkward. The worst thing most students do is performing partial fractions. But it does look very sus though because the numerator does look like the derivative of the denominator. Watch happens when we keep dividing top and bottom by x:

$$\int \frac{2x^3-1}{x^4+x}dx = \int \frac{2x^2-\frac{1}{x}}{x^3+1}dx = \int \frac{2x-\frac{1}{x^2}}{x^2+\frac{1}{x}}dx$$

This is very difficult to see if you haven't been speed integrating a lot, but it is common to know for speed integration that:

$$\frac{d}{dx}\left[\frac{1}{x}\right] = -\frac{1}{x^2}.$$

A good thing to keep in mind is that reciprocal power rule alternates the sign. In this case, we have our answer: $\boxed{\ln\left|x^2+\frac{1}{x}\right|+C}$

Here's a very difficult version:

$$\int \frac{4x^5-1}{(x^5+x+1)^2}dx$$

[UC Berkeley Integration Bee 2020]

Nonlinear Functions Example

$$\int \frac{\sin(x)+\cos(x)}{e^x+\cos(x)}dx$$

[JEE Main]

$$= \int \frac{\sin(x)+\cos(x)}{e^x} \cdot \frac{dx}{1+\frac{\cos(x)}{e^x}}$$

Let $u = 1 + e^{-x}\cos(x)$.

A Sneaky Exponential Example

$$\int \frac{x^2+e^x}{x^2+xe^x}dx$$

[Silver]

$$\int \frac{x^2e^{-x}+1}{x^2e^{-x}+x}dx = \int \frac{e^{-x}+\frac{1}{x^2}}{e^{-x}+\frac{1}{x}}dx$$

This is super difficult to see, but let $u = e^{-x} + \frac{1}{x}$.

Logarithm Example

$$\int \frac{1-2\ln(x)}{x(x^2+\ln(x))}dx$$

This integral actually requires experimentation in order to figure out the answer to this. Let's factor an x out of the denominator.

$$\int \frac{1-2\ln(x)}{x^2(x+\frac{\ln(x)}{x})}dx$$

If we let $u = x + \frac{\ln(x)}{x}$, then $du = 1 + \frac{1-\ln(x)}{x^2}dx$. This substitution does not work unfortunately, so we must divide x again.

$$\int \frac{1-2\ln(x)}{x^3(1+\frac{\ln(x)}{x^2})}dx$$

If we let $u = 1 + \frac{\ln(x)}{x^2}$, then $du = \frac{x-2x\ln(x)}{x^4}dx = \frac{1-2\ln(x)}{x^3}dx$. Jackpot!!!

$$\int \frac{du}{u} = \boxed{\ln\left|1+\frac{\ln|x|}{x^2}\right| + C}$$

4.2 Square-Square Technique

Square-Square Concept

$$\left(x+\frac{1}{x}\right)^2 = x^2+\frac{1}{x^2}+2$$
$$\left(x-\frac{1}{x}\right)^2 = x^2+\frac{1}{x^2}-2$$
$$\Rightarrow \left(x+\frac{1}{x}\right)^2-2 = \left(x-\frac{1}{x}\right)^2+2 = x^2+\frac{1}{x^2}$$

$$\frac{d}{dx}\left[x+\frac{1}{x}\right] = 1-\frac{1}{x^2}$$
$$\frac{d}{dx}\left[x-\frac{1}{x}\right] = 1+\frac{1}{x^2}$$

Beginner Example

$$\int \frac{x^2+1}{x^4-x^2+1}dx$$

Everytime you see a bunch of squares, a good thing to come in mind is to utilize the square-square technique. Divide everything by x^2. This will make us complete a square using the square-square concept.

$$\int \frac{1+\frac{1}{x^2}}{x^2-1+\frac{1}{x^2}}dx$$

To determine which way to complete the square, we follow our du or the derivative on the numerator. Because we have $1+\frac{1}{x^2}$, we want to do $u = x-\frac{1}{x}$, so we complete the square in that form.

$$\int \frac{1+\frac{1}{x^2}}{(x^2+\frac{1}{x^2}-2)+2-1}dx = \int \frac{1+\frac{1}{x^2}}{(x-\frac{1}{x})^2+1}dx$$

Hence, performing $u = x-\frac{1}{x}$ gives us $\boxed{\tan^{-1}\left(x-\frac{1}{x}\right)+C}$.

Trig-Sub Example

$$\int \frac{x^2-1}{x^2+1} \cdot \frac{dx}{\sqrt{1+x^4}}$$

[Calculus - Spivak]

$$\int \frac{(x^2-1)}{x(x+\frac{1}{x})x\sqrt{x^2+\frac{1}{x^2}}}dx = \int \frac{1-\frac{1}{x^2}}{(x+\frac{1}{x})\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\int \frac{1-\frac{1}{x^2}}{(x+\frac{1}{x})\sqrt{(x+\frac{1}{x})^2-2}}dx = \int \frac{du}{u\sqrt{u^2-2}}$$

Zero Substitution Example

$$\int_0^\infty \frac{dx}{(x+\frac{1}{x})^2}$$

$$\int_0^\infty \frac{dx}{x^2+2+\frac{1}{x^2}} = \frac{1}{2}\int_0^\infty \frac{2}{x^2+\frac{1}{x^2}+2}dx = \frac{1}{2}\int_0^\infty \frac{1+1+\frac{1}{x^2}-\frac{1}{x^2}}{x^2+\frac{1}{x^2}+2}$$

$$\frac{1}{2}\int_0^\infty \frac{1-\frac{1}{x^2}}{(x+\frac{1}{x})^2}+\frac{1+\frac{1}{x^2}}{(x-\frac{1}{x})^2+4}dx = \frac{1}{2}\left[-\frac{1}{x+\frac{1}{x}}+\frac{1}{2}\tan^{-1}\left(\frac{x-\frac{1}{x}}{2}\right)\right]_0^\infty = \boxed{\frac{\pi}{4}}$$

Trig Example

$$\int_0^{\frac{\pi}{2}} \frac{dx}{\sin^4(x)+\cos^4(x)}$$

Let $u = \tan(x)$.

$$\int_0^{\frac{\pi}{2}} \frac{\sec^2(x)(1+\tan^2(x))}{1+\tan^4(x)}dx = \int_0^\infty \frac{1+u^2}{1+u^4}du$$

$$\int_0^\infty \frac{1+\frac{1}{u^2}}{u^2+\frac{1}{u^2}}du = \int_0^\infty \frac{1+\frac{1}{u^2}}{(u-\frac{1}{u})^2+2}du = \boxed{\frac{\pi}{\sqrt{2}}}$$

4.3 $1 + \sin(2x)$ Trick

$1 + \sin(2x)$ Concept

Very Similar to the Square-Square Technique

$$1 + \sin(2x) = (\sin(x) + \cos(x))^2$$

$$1 - \sin(2x) = (\sin(x) - \cos(x))^2$$

Beginner Example

$$\int \frac{\cos(x) - \sin(x)}{1 + \sqrt{1 + \sin(2x)}} dx$$

$$\int \frac{\cos(x) - \sin(x)}{1 + \sin(x) + \cos(x)} dx = \boxed{\ln|1 + \sin(x) + \cos(x)| + C}$$

Trig-Sub Example

$$\int_{-1}^{1} \frac{\sqrt{1 - x^2} - x}{\sqrt{1 - x^2}(1 + x\sqrt{1 - x^2})} dx$$

Let $x = \sin\theta$:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\theta - \sin\theta}{1 + \sin\theta\cos\theta} d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2(\cos\theta - \sin\theta)}{2 + 2\sin\theta\cos\theta} d\theta$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2(\cos\theta - \sin\theta)}{1 + (\sin\theta + \cos\theta)^2} d\theta = 2\int_{-1}^{1} \frac{du}{u^2 + 1} = \boxed{\pi}$$

Completing the Square

$$\int \frac{\cos(x) - \sin(x)}{\sqrt{\sin(2x)}} dx$$

$$\int \frac{\cos(x) - \sin(x)}{\sqrt{(\sin(x) + \cos(x))^2 - 1}} dx = \int \frac{du}{\sqrt{u^2 - 1}}$$

Trig Manipulation

$$\int \sqrt{\tan(x)} + \sqrt{\cot(x)}dx$$

$$\int \frac{\sin(x)+\cos(x)}{\sqrt{\sin(x)\cos(x)}}dx = \sqrt{2}\int \frac{\sin(x)+\cos(x)}{\sqrt{1-(\sin(x)-\cos(x))^2}}dx$$
$$= \sqrt{2}\int \frac{du}{\sqrt{1-u^2}} = \boxed{\sqrt{2}\sin^{-1}(\sin(x)-\cos(x))+C}$$

Zero Substitution Example

$$\int \frac{dx}{\sec(x)+\sin(x)}$$

[JEE Main]

$$\int \frac{\cos(x)}{1+\sin(x)\cos(x)}dx = \int \frac{2\cos(x)}{2+\sin(2x)}dx = \int \frac{\cos(x)+\cos(x)+\sin(x)-\sin(x)}{2+\sin(2x)}dx$$
$$= \int \frac{\cos(x)-\sin(x)}{1+(\sin(x)+\cos(x))^2} + \frac{\sin(x)+\cos(x)}{3-(\sin(x)-\cos(x))^2}dx$$
$$= \int \frac{du}{1+u^2} + \int \frac{dw}{3-w^2}$$

A Satisfying Example

$$\int_0^{\frac{\pi}{4}} \frac{\sqrt{\cot(x)}-\sqrt{\tan(x)}}{1+\sin(2x)}dx$$

$$\int_0^{\frac{\pi}{4}} \frac{\cos(x)-\sin(x)}{\sqrt{\sin(x)\cos(x)}(\sin(x)+\cos(x))^2}dx$$

Let $u = \sin(x)+\cos(x)$. Then $\frac{u^2-1}{2} = \sin(x)\cos(x)$.

$$\int_1^{\sqrt{2}} \frac{\sqrt{2}}{u^2\sqrt{u^2-1}}du = \int_0^{\frac{\pi}{4}} \sqrt{2}\cos\theta d\theta = \boxed{1}$$

4.4 Advanced King's/Queen's Rule

Disguised Example

$$\int_0^2 \sqrt{x^2-x+1}-\sqrt{x^2-3x+3}dx$$

Let $u=2-x$:

$$\int_0^2 \sqrt{(2-u)^2-(2-u)+1}-\sqrt{(2-u)^2-3(2-u)+3}du$$

$$=\int_0^2 \sqrt{u^2-3u+3}-\sqrt{u^2-u+1}du=-I$$

If the original integral $I=-I$, then $I=\boxed{0}$.

Exponential Example

$$\int_0^1 \frac{x}{x+(x-x^2)^x}dx$$

[Silver]

$$=\int_0^1 \frac{x}{x+x^x(1-x)^x}dx=\int_0^1 \frac{x^{1-x}}{x^{1-x}+(1-x)^x}dx$$

Taking Advantage of King's Rule

$$\int_0^1 \ln(x-x^2)\left(\ln^2\left(\frac{x}{1-x}\right)+\ln(x)\ln(1-x)\right)dx$$

[Silver]

After some algebra simplification:

$$\int_0^1 \ln^3(x)+\ln^3(1-x)dx=2\int_0^1 \ln^3(x)dx=\boxed{-12}$$

The last integral can be solved using Gamma function.

Trig Example

$$\int_0^1 \frac{\tan(x)}{1-\tan(x)\tan(1-x)}dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

Let $u = 1 - x$:

$$= \int_0^1 \frac{\tan(1-x)}{1-\tan(x)\tan(1-x)}dx$$

$$2I = \int_0^1 \tan(x+1-x)dx \Rightarrow I = \boxed{\frac{\tan(1)}{2}}$$

A Crazy Example!!

Let $f(x) = \frac{4^x}{2+4^x}$. Then compute,

$$\int_{\frac{1}{4}}^{\frac{3}{4}} f(f(x))dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

This is actually very hard to see. Notice that $f(x) = \frac{4^x}{2+4^x} = \frac{2^x}{2^{1-x}+2^x}$.
Because of this formation, it is sneaky to see that $f(x) + f(1-x) = 1$.
Also notice that $f(\frac{1}{4} + \frac{3}{4} - x) = f(1-x)$. So we can definitely use King's Rule!!
By $u = 1 - x$,

$$\int_a^b f(f(x))dx = \int_a^b f(f(1-x))dx$$

Since $f(1-x) = 1 - f(x)$,

$$\int_a^b f(f(1-x))dx = \int_a^b f(1-f(x))dx = \int_a^b 1 - f(f(x))dx$$

So now

$$2I = \int_{\frac{1}{4}}^{\frac{3}{4}} 1dx \Rightarrow I = \boxed{\frac{1}{4}}.$$

MINDBLOWING!!! I don't think I can ever make another integral like this though ;-;
Definitely one of the most unique King's Rule integral I've ever made.

A Very Famous Queen's Rule

$$\int_0^{\frac{\pi}{2}} \ln(\sin(x))dx$$

Like most beginners, you can easily mess up the process if you're not careful enough. Usually, I'd just memorize the answer to this integral. Let $u = \frac{\pi}{2} - x$. Note that x and u are just variables.

$$\int_0^{\frac{\pi}{2}} \ln(\cos(x))dx \Rightarrow 2I = \int_0^{\frac{\pi}{2}} \ln(\sin(x)\cos(x))dx$$

$$I = \frac{1}{2}\int_0^{\frac{\pi}{2}} \ln\left(\frac{1}{2}\sin(2x)\right)dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} -\ln(2) + \ln(\sin(2x))dx$$

$$= -\frac{\pi\ln(2)}{4} + \frac{1}{4}\int_0^{\pi} \ln(\sin(u))du$$

Observe closely that

$$\int_0^{\pi} \ln(\sin(u))du = \int_0^{\frac{\pi}{2}} \ln(\sin(u))du + \int_{\frac{\pi}{2}}^{\pi} \ln(\sin(u))du$$

So we have

$$I = -\frac{\pi\ln(2)}{4} + \frac{1}{4}I + \frac{1}{4}\int_{\frac{\pi}{2}}^{\pi} \ln(\sin(x))dx$$

Then $u = \frac{\pi}{2} - x$ gives,

$$\int_{\frac{\pi}{2}}^{\pi} \ln(\sin(x))dx = \int_{-\frac{\pi}{2}}^{0} \ln(\cos(u))du = \int_0^{\frac{\pi}{2}} \ln(\cos(u))du = I$$

Hence,

$$\begin{aligned} I &= -\frac{\pi\ln(2)}{4} + \frac{1}{2}I \\ \frac{1}{2}I &= -\frac{\pi\ln(2)}{4} \\ I &= \boxed{-\frac{\pi}{2}\ln(2)} \end{aligned}$$

That was a lot to take in. As you can see, you can easily get lost in the process. And a lot of integrals keep using this famous integral in different forms. Because of that, I just remember the answer of this integral.

Trig-Manipulation

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos(x)}}{(\sqrt{\sin(x)}+\sqrt{\cos(x)})^5} dx$$

[JEE Main]

Let $u = \frac{\pi}{2} - x$:

$$= \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin(x)}}{(\sqrt{\sin(x)}+\sqrt{\cos(x)})^5} dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{dx}{(\sqrt{\sin(x)}+\sqrt{\cos(x)})^4}$$

$$\frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{(\sqrt{\tan(x)}+1)^4} dx = \frac{1}{2}\int_0^{\infty} \frac{du}{(\sqrt{u}+1)^4} = \boxed{\frac{1}{6}}$$

Absolute Value Example

$$\int_0^{\frac{\pi}{2}} e^{|x-\frac{\pi}{4}|}\sin^2(x) dx$$

[Silver]

Let $u = \frac{\pi}{2} - x$:

$$= \int_0^{\frac{\pi}{2}} e^{|x-\frac{\pi}{4}|}\cos^2(x) dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} e^{|x-\frac{\pi}{4}|} dx = \boxed{e^{\frac{\pi}{4}} - 1}$$

Just King's Rule

$$\int_0^{\frac{\pi}{6}} \frac{x}{\cos(x)\cos(\frac{\pi}{6}-x)} dx$$

[Cambridge University Integration Bee]

Let $u = \frac{\pi}{6} - x$:

$$\frac{\pi}{12}\int_0^{\frac{\pi}{6}} \frac{dx}{\cos(x)\cos(\frac{\pi}{6}-x)} = \frac{\pi}{6}\int_0^{\frac{\pi}{6}} \frac{dx}{\cos(x)(\sqrt{3}\cos(x)+\sin(x))}$$

$$\frac{\pi}{6}\int_0^{\frac{\pi}{6}} \frac{\sec^2(x)}{(\sqrt{3}+\tan(x))} dx = \boxed{\frac{\pi}{6}\ln\left(\frac{4}{3}\right)}$$

The Evil Queen's Rule

$$\int_0^{\frac{\pi}{4}} \ln(1+\tan(x))dx$$

This is super hard to see, yet the trig-manipulation is so satisfying!!!
Let $u = \frac{\pi}{4} - x$. Note that:

$$\tan\left(\frac{\pi}{4} - x\right) = \frac{1-\tan(x)}{1+\tan(x)}$$

Then,

$$\int_0^{\frac{\pi}{4}} \ln\left(1+\frac{1-\tan(x)}{1+\tan(x)}\right)dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1+\tan(x)}\right)dx$$

$$2I = \int_0^{\frac{\pi}{4}} \ln(2)dx \Rightarrow I = \boxed{\frac{\pi}{8}\ln(2)}$$

Who would've thought that algebra manipulation would cause the logarithms to cancel out so nicely.

Trigonometric Algebra

$$\int_0^{\frac{\pi}{2}} \frac{\sin^3(x)}{1-\sin(x)\cos(x)}dx$$

[Silver]

$$\int_0^{\frac{\pi}{2}} \frac{\sin^3(x)+\cos^3(x)}{2(1-\sin(x)\cos(x))}dx = \int_0^{\frac{\pi}{2}} \frac{(s+c)(s^2-sc+c^2)}{2(1-sc)}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin(x)+\cos(x)}{2}dx = \boxed{1}.$$

4.5 Weierstrass Substitution

Weierstrass Concept

Let $u = \tan\left(\frac{x}{2}\right)$, then:

$$dx = \frac{2}{1+u^2}du$$

$$\sin(x) = \frac{2u}{1+u^2}$$

$$\cos(x) = \frac{1-u^2}{1+u^2}$$

The purpose of this technique is to turn ugly trig integrals into rational polynomials. Please use this method sparringly or you'll end up making the integral worse.

Beginner Example

$$\int_0^{2\pi} \frac{dx}{2+\cos(x)}$$

Notice we can't exactly do this substitution because $\tan(\frac{2\pi}{2})$ will go beyond the domain. However, because this function is periodic at 2π, we can actually translate the function since the bounds add up to 2π:

$$\int_0^{2\pi} \frac{dx}{2+\cos(x)} = \int_{-\pi}^{\pi} \frac{dx}{2+\cos(x)}$$

Let $u = \tan(\frac{x}{2})$:

$$\int_{-\infty}^{\infty} \frac{1}{2+(\frac{1-u^2}{1+u^2})} \cdot \frac{2du}{1+u^2} = \int_{-\infty}^{\infty} \frac{2du}{2+2u^2+1-u^2} = \int_{-\infty}^{\infty} \frac{2du}{u^2+3} = \boxed{\frac{2\pi}{\sqrt{3}}}$$

BE VERY CAREFUL ABOUT THE DOMAIN. Sometimes, tangent functions are dangerous to mess around with if you're not careful enough.

A Log Example

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin(x)+\cos(x)}$$

Let $u = \tan(\frac{x}{2})$:

$$= \int_0^1 \frac{du}{1+(\frac{2u}{1+u^2})+(\frac{1-u^2}{1+u^2})} \cdot \frac{2}{1+u^2}$$

$$= \int_0^1 \frac{2du}{1+u^2+2u+1-u^2} = \int_0^1 \frac{du}{u+1} = \boxed{\ln(2)}$$

4.6 Reverse Quotient Rule

Reverse quotient rule works the same way as reverse product rule. However, it's so much harder to solve than reverse product rules. Things can get super ugly and sometimes, you have to find the right function that fits the quotient rule.

A Simple Example

$$\int \frac{xe^{2x}}{(2x+1)^2}dx$$

We know that by quotient rule, we can assume $g(x) = (2x+1)$. Then,

$$f'(x)(2x+1) - f(x)(2) = xe^{2x}$$

What if $f(x) = e^{2x}$? Then,

$$2e^{2x}(2x+1) - 2e^{2x} = 4xe^{2x}$$

Aha, we just need that 4. Hence, our answer for the integral is $\boxed{\frac{e^{2x}}{4(2x+1)} + C}$.

As you can see, it's not an easy process as the reverse product rule. You actually have to guess and check for functions that best fits the function in the integral.

Logarithm Example

$$\int \frac{\ln(x)}{(1+\ln(x))^2}dx$$

Let $g(x) = 1 + \ln(x)$, then $f'(x)(1+\ln(x)) - f(x)\frac{1}{x} = \ln(x)$.
If $f(x) = x$, then $(1+\ln(x)) - 1 = \ln(x)$, which matches the numerator.

Trig Example

$$\int \frac{\sin^2(x)}{(x\cos(x) - \sin(x))^2}dx$$

Assume $g(x) = x\cos(x) - \sin(x)$, then

$$f'(x)(x\cos(x) - \sin(x)) - f(x)(-x\sin(x)) = \sin^2(x)$$

The first thing I saw was wanting to cancel out $f'(x)x\cos(x)$ with $f(x)x\sin(x)$.
This works by letting $f(x) = \cos(x)$.

This also gives $\sin^2(x)$, so our answer is $\boxed{\frac{\cos(x)}{x\cos(x) - \sin(x)} + C}$.

A Sinister Example

$$\int \left(\frac{\tan^{-1}(x)}{x-\tan^{-1}(x)}\right)^2 dx$$

[JEE Main]

This is probably the most evil integral to ever give in an integration bee.

$$\int \left(\frac{\tan^{-1}(x)}{x-\tan^{-1}(x)}\right)^2 dx = \int \left(\frac{x}{x-\tan^{-1}(x)} - 1\right)^2 dx$$

$$= \int \frac{x^2 - 2x(x-\tan^{-1}(x))}{(x-\tan^{-1}(x))^2} + 1dx = \int \frac{2x\tan^{-1}(x) - x^2}{(x-\tan^{-1}(x))^2} + 1dx$$

Let $g(x) = x - \tan^{-1}(x)$. Then

$$f'(x)(x-\tan^{-1}(x)) - f(x)\left(1 - \frac{1}{x^2+1}\right) = 2x\tan^{-1}(x) - x^2$$

I need that x^2+1 to cancel out. So let $f(x) = x^2+1$.

$$2x(x-\tan^{-1}(x)) - x^2 = x^2 - 2x\tan^{-1}(x)$$

How evil, trying to leave out a negative.

So our answer is (DON'T FORGET THE x!!!) $\boxed{x - \frac{x^2+1}{x-\tan^{-1}(x)} + C}$

When I first tried to solve this integral, I literally tried doing quotient rule with the $\tan^{-1}(x)$ on top. It was so terrible, I didn't know I was suppose to perform zero-substitution first and then reverse quotient rule.

Chapter 5

Advanced Topics

5.1 Semi-Integration by Parts

Beginner Example

$$\int e^{x^2}(2x^2+1)dx$$

Sometimes it's super difficult to see the reverse product rule. So it's better not to waste time trying to think which function is which. Instead, whenever we see each term being nonelementary, we perform semi-integration by parts. Instead of integrating by parts with the whole integral, we will only do it to the first term $2x^2e^{x^2}$.

$$\int 2x^2e^{x^2}dx + \int e^{x^2}dx = xe^{x^2} - \int e^{x^2}dx + \int e^{x^2}dx$$

The nonelementary integrals satisfyingly cancels out, so that gives us our answer $\boxed{xe^{x^2}+C}$.

A Double Example

$$\int e^{\sin(x)} \cdot \frac{x\cos^3(x)-\sin(x)}{\cos^2(x)}dx$$

[Calculus - Spivak]

Let's simplify the integral further:

$$\int e^{\sin(x)}(x\cos(x)-\sec(x)\tan(x))dx$$

We will do IBP on each terms differently.

$$xe^{\sin(x)} - \int e^{\sin(x)}dx - e^{\sin(x)}\sec(x) + \int e^{\sin(x)}dx$$

The nonelementary integrals cancels out, giving us $\boxed{e^{\sin(x)}(x-\sec(x))+C}$.

5.2 Advanced Trig Integrals

A Little Scary Example

$$\int \frac{dx}{9\sin^2(x) + 4\cos^2(x)}$$

$$\int \frac{\sec^2(x)}{9\tan^2(x) + 4} dx = \int \frac{du}{9u^2 + 4}$$

A Constant Changes Everything

$$\int \frac{dx}{\sqrt{3}\sin(x) + \cos(x)}$$

[JEE Main]

$$\frac{1}{2} \int \frac{dx}{\frac{\sqrt{3}}{2}\sin(x) + \frac{1}{2}\cos(x)} = \frac{1}{2} \int \sec\left(x - \frac{\pi}{3}\right) dx$$

Linear Algebra???

$$\int \frac{\sin(x) + 2\cos(x)}{3\sin(x) + 4\cos(x)} dx$$

[JEE Main]

We want to turn this function into two different rational functions. One function where the numerator is the same as the denominator and the other function where the numerator is the derivative of the denominator. In that way, we have two easy integrals: one term just being a constant and the other being a simple u-sub logarithm. But in order to do that, we must find the appropriate coefficients:

$$\alpha(3\sin(x) + 4\cos(x)) + \beta(3\cos(x) - 4\sin(x)) = \sin(x) + 2\cos(x)$$

This gives us a system of equations by separating in terms of sines and cosines:

$$\left.\begin{aligned} 3\alpha - 4\beta = 1 \\ 4\alpha + 3\beta = 2 \end{aligned}\right\} \Rightarrow \alpha = \frac{11}{25}, \quad \beta = \frac{2}{25}$$

Now that we have our coefficients, our integral is now easier:

$$\int \frac{11}{25} + \frac{2}{25}\left(\frac{3\cos(x) - 4\sin(x)}{3\sin(x) + 4\cos(x)}\right) dx = \boxed{\frac{11}{25}x + \frac{2}{25}\ln|3\sin(x) + 4\cos(x)| + C}$$

A Difficult Trig-Manipulation

$$\int \frac{dx}{\sin(x+2)\sin(x+3)}$$

[JEE Main]

$$\frac{1}{\sin(1)} \int \frac{\sin((x+3)-(x+2))}{\sin(x+2)\sin(x+3)} dx$$
$$= \frac{1}{\sin(1)} \int \frac{\sin(x+3)\cos(x+2) - \cos(x+3)\sin(x+2)}{\sin(x+2)\sin(x+3)} dx$$
$$= \frac{1}{\sin(1)} \int \cot(x+2) - \cot(x+3) dx$$

Double Angle Identities Chaos

$$\int \cos^8(x) - \sin^8(x) dx$$

Let $c = \cos(x)$ and $s = \sin(x)$. Then,

$$\begin{aligned} c^8 - s^8 &= (c^4 + s^4)(c^4 - s^4) \\ &= (c^4 + s^4)\cos(2x) \\ &= (c^2(1-s^2) + s^2(1-c^2))\cos(2x) \\ &= (1 - 2s^2c^2)\cos(2x) \\ &= \left(1 - \frac{1}{2}\sin^2(2x)\right)\cos(2x). \end{aligned}$$

So now our integral is just

$$\int \left(1 - \frac{1}{2}\sin^2(2x)\right)\cos(2x) dx$$

This simplified a lot nicer than I expected.

A Chain of Tangents

$$\int \tan(x)\tan(2x)\tan(3x)dx$$

[JEE Main]

This is also super difficult to see, especially when you're not comfortable with tangent trig-identities. Observe that,

$$\begin{aligned}
\tan(3x) &= \frac{\tan(x)+\tan(2x)}{1-\tan(x)\tan(2x)} \\
\tan(3x)(1-\tan(x)\tan(2x)) &= \tan(x)+\tan(2x) \\
\tan(3x)-\tan(2x)-\tan(x) &= \tan(x)\tan(2x)\tan(3x).
\end{aligned}$$

So our integral is

$$\int \tan(3x)-\tan(2x)-\tan(x)dx$$

Again!! Waaay nicer than I expected. Although, manipulating trig-identities like this is super difficult to get comfortable with. I would consider sticking with easier identities for this integration bee.

A Chain of Sines

$$\int_0^{\frac{\pi}{2}} \sin(x)\sin(2x)\sin(3x)dx$$

[MIT Integration Bee 2010]

Note that $\sin(x)\sin(3x) = \frac{1}{2}(\cos(2x)-\cos(4x))$. Then,

$$\frac{1}{2}\int_0^{\frac{\pi}{2}} \sin(2x)(\cos(2x)-\cos(4x))dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin(2x)(\cos(2x)-2\cos^2(2x)+1)dx$$

Let $u=\cos(2x)$ and apply symmetry:

$$= \frac{1}{4}\int_{-1}^{1}(1+u-2u^2)du = \frac{1}{2}\int_0^1(1-2u^2)du = \boxed{\frac{1}{6}}$$

Sneakiest Trinomial Conjugation

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\sec(x)+\tan(x)+1}dx$$

[JEE Main]

It is NOT well known that

$$(\sec(x)+\tan(x)+1)(1+\tan(x)-\sec(x)) = 2\tan(x)$$

We will use this to conjugate this nasty integral:

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\sec(x)+\tan(x)+1}\cdot\frac{1+\tan(x)-\sec(x)}{1+\tan(x)-\sec(x)}dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin(x)+\sin(x)\tan(x)-\tan(x)}{2\tan(x)}dx = \frac{1}{2}\int_0^{\frac{\pi}{4}} \cos(x)+\sin(x)-1dx$$

Look how beautiful that turned out. Just one sneaky trig-manipulation literally turned a monster integral into a very easy integral. Obviously, we will not use this integral in our integration bee. Consider making an easier function or a trig-identity that is more known and easier to see.

Just Simplification

$$\int \ln\left(\frac{\sin(2x)}{1+\cos(2x)}\right)\sec^2(x)dx$$

$$\int \ln\left(\frac{2\sin(x)\cos(x)}{2\cos^2(x)}\right)\sec^2(x)dx = \int \ln(\tan(x))\sec^2(x)dx$$

5.3 Advanced Gaussian Integral

Gaussian Integral in Log Form

$$\int_0^1 \sqrt{\ln\left(\frac{1}{x}\right)}dx$$

Let $u = \sqrt{\ln(\frac{1}{x})}$. Then $e^{-u^2} = x \to -2ue^{-u^2}du = dx$.

$$\int_0^\infty 2u^2e^{-u^2}du = \boxed{\frac{\sqrt{\pi}}{2}}$$

A Very Cool Gaussian Log

$$\int_0^\infty e^{-\ln^2(x)}dx$$

[UC Berkeley Integration Bee 2023]

Let $u = \ln(x)$:

$$\int_{-\infty}^\infty e^{-u^2+u}du = \int_{-\infty}^\infty e^{-(u^2-u+\frac{1}{4})+\frac{1}{4}}du = e^{\frac{1}{4}}\int_{-\infty}^\infty e^{-(u-\frac{1}{2})^2}du = \boxed{\sqrt{\pi\sqrt{e}}}$$

More Log Integrals

$$\int_{\frac{1}{e}}^\infty \frac{\sqrt{1+\ln(x)}}{x^2}dx$$

[Silver]

Let $u = \sqrt{1+\ln(x)}$, then $x = e^{u^2-1}$.

$$\int_0^\infty \frac{u\cdot 2ue^{u^2-1}}{e^{2u^2-2}}du = \int_0^\infty 2u^2e^{-u^2+1}du = \boxed{\frac{e\sqrt{\pi}}{2}}$$

This Converges???

$$\int_0^\infty 1-e^{-\frac{1}{x^2}}dx$$

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{1-e^{-u^2}}{u^2}du = \left[\frac{1-e^{-u^2}}{u}\right]_0^\infty + 2\int_0^\infty e^{-u^2}du = \boxed{\sqrt{\pi}}$$

Radical Form of Gaussian Integral

$$\int_1^\infty \frac{dx}{e^x\sqrt{x-1}}$$

[UC Berkeley Integration Bee 2023]

Let $u=\sqrt{x-1}$:

$$\int_0^\infty \frac{2udu}{e^{u^2+1}u}du = \frac{2}{e}\int_0^\infty e^{-u^2}du = \boxed{\frac{\sqrt{\pi}}{e}}$$

A Beautiful Gaussian Integral

$$\int_0^\infty \left(\frac{1}{\sqrt{x}}\right)^{\ln(x)} dx$$

$$= \int_0^\infty e^{\ln\left(\frac{1}{\sqrt{x}}\right)\ln(x)}dx = \int_0^\infty e^{-\frac{1}{2}\ln^2(x)}dx = \boxed{\sqrt{2\pi e}}$$

A Crazy Lambert W Function Integral

$$\int_0^\infty W\left(\frac{1}{x^2}\right)dx$$

Recall that $W(x)$ is the inverse function of xe^x. Let $u=W\left(\frac{1}{x^2}\right)$. Then,

$$\begin{aligned} ue^u &= \frac{1}{x^2} \\ e^u(1+u)du &= -\frac{2}{x^3}dx \\ -\frac{u+1}{2u\sqrt{ue^u}} &= dx \end{aligned}$$

That was a lot of manipulation, so now we have

$$\int_0^\infty \frac{\sqrt{u}+\frac{1}{\sqrt{u}}}{2e^{\frac{u}{2}}}du = \int_0^\infty \frac{u+1}{2\sqrt{u}e^{\frac{u}{2}}}du = \int_0^\infty \frac{2w+1}{\sqrt{2w}e^w}dw$$

$$\int_0^\infty \sqrt{2}(2t^2+1)e^{-t^2}dt = \boxed{\sqrt{2\pi}}$$

5.4 Dirichlet Integral

Dirichlet Integral Formula

$$\int_0^\infty \frac{\sin(x)}{x}dx = \frac{\pi}{2}$$

Substitution Example

$$\int_0^\infty \frac{\sin(x^3)}{x}dx$$

Let $u = x^3$:

$$\int_0^\infty \frac{\sin(x^3)x^2}{x^3}dx = \int_0^\infty \frac{\sin(u)}{3u}du = \boxed{\frac{\pi}{6}}$$

Trig-Identity Example

$$\int_0^\infty \frac{\sin(5x)\cos(3x)}{x}dx$$

$$= \int_0^\infty \frac{\sin(8x)+\sin(2x)}{2x}dx = \boxed{\frac{\pi}{2}}$$

IBP Example

$$\int_0^\infty \frac{1-\cos(x)}{x^2}dx$$

Derive $1-\cos(x)$ and integrate $\frac{1}{x^2}$:

$$= \left[\frac{\cos(x)-1}{x}\right]_0^\infty + \int_0^\infty \frac{\sin(x)}{x}dx = \boxed{\frac{\pi}{2}}$$

Exponential Example

$$\int_{-\infty}^\infty \sin(e^x)dx$$

Let $u = e^x$:

$$\int_0^\infty \frac{\sin(u)}{u}du = \boxed{\frac{\pi}{2}}$$

Trig U-sub Example

$$\int_0^{\frac{\pi}{2}} \frac{\sin(\tan(x))}{\sin(2x)} dx$$

Let $u = \tan(x)$:

$$\int_0^{\infty} \frac{\sin(u)}{2u} du = \boxed{\frac{\pi}{4}}$$

IBP Example 2

$$\int_0^{\infty} \frac{\sin^3(x)}{x^3} dx$$

Derive $\sin^3(x)$ and $\frac{1}{x^3}$ twice.

$$= \left[\frac{3\sin^2(x)\cos(x)}{2x} - \frac{\sin^3(x)}{2x^2}\right]_0^{\infty} + \int_0^{\infty} \frac{3\sin(2x)}{2x} - \frac{3\sin^3(x)}{2x} dx$$

$$= 0 + \int_0^{\infty} \frac{3}{2}\left(\frac{\sin(2x)}{x}\right) - \frac{3}{2}\left(\frac{\sin(x)(\frac{1}{2} - \frac{1}{2}\cos(2x))}{x}\right) dx$$

$$= \frac{3\pi}{4} - \frac{3\pi}{8} + \frac{3}{4}\int_0^{\infty} \frac{\sin(3x) - \sin(4x)}{x} dx = \boxed{\frac{3\pi}{8}}$$

Intimidating Example

$$\int_0^{\infty} \frac{(x-1)\sin(x) + x\cos(x)}{x^2} dx$$

$$= \int_0^{\infty} \frac{\sin(x)}{x} + \frac{x\cos(x) - \sin(x)}{x^2} dx = \boxed{\frac{\pi}{2} - 1}$$

Chapter 6

Very Advanced Topics

6.1 Inverse Jailbreaking

Inverse Jailbreaking Concept

We learned that

$$\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$$

Let's say we come across a disgusting definite integral. Is the inverse of the function easier to integrate??? If so, we can perform:

$$\int_a^b f(x)dx = bf(b) - af(a) - \int_{f(a)}^{f(b)} f^{-1}(x)dx$$

Beginner Example

$$\int_1^2 \csc^{-1}(\sqrt{x})dx$$

[Caltech Homework]

I don't wanna deal with inverse cosecants, so let's integrate the inverse instead.
If $y = \csc^{-1}(\sqrt{x})$, then $f^{-1}(x) = \csc^2(x)$. If the bounds were from 1 to 2, then our new bounds are from $\csc^{-1}(1) = \frac{\pi}{2}$ to $\csc^{-1}(\sqrt{2}) = \frac{\pi}{4}$.

$$\int_1^2 \csc^{-1}(x)dx = 2 \cdot \frac{\pi}{4} - \frac{\pi}{2} - \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \csc^2(x)dx = \boxed{1}$$

Thank you symmetry!!

$$\int_0^8 \tan^{-1}(\sqrt[3]{x}-1)dx$$

[Silver]

The inverse function of that is $(\tan(x)+1)^3$. Our new integral is now:

$$= 8\left(\frac{\pi}{4}\right) - 0 - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\tan(x)+1)^3 dx = 2\pi - 2\int_0^{\frac{\pi}{4}} 3\tan^2(x)+1\; dx$$

No leftovers.

$$\int_0^1 \tan^{-1}\left(\sqrt[3]{\frac{1-x}{x}}\right)dx$$

[Silver]

The inverse function for this integral is $\frac{1}{1+\tan^3(x)}$. Notice that

$$\int_0^1 \tan^{-1}\left(\sqrt[3]{\frac{1-x}{x}}\right)dx + \int_{\frac{\pi}{2}}^0 \frac{dx}{1+\tan^3(x)} = 1(0) - 0\left(\frac{\pi}{2}\right)$$

This means that

$$\int_0^1 \tan^{-1}\left(\sqrt[3]{\frac{1-x}{x}}\right)dx = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan^3(x)} = \boxed{\frac{\pi}{4}}$$

A Familiar Integral

$$\int_0^{\ln(2)} \tan^{-1}(e^x-1)dx$$

[Silver]

The inverse function of this is $\ln(1+\tan(x))$. So we have:

$$= \frac{\pi}{4}\ln(2) - \int_0^{\frac{\pi}{4}} \ln(1+\tan(x))$$

If you remember from a past section of sneaky queen's rule, the following answer becomes $\boxed{\frac{\pi}{8}\ln(2)}$.

Infinite Function Example

$$\int_0^6 \sqrt[3]{x+\sqrt[3]{x+\sqrt[3]{x+...}}}dx$$

[Harvard-MIT Math Tournament Integration Bee 2021 - Finals]

If we try to simplify the function algebraically, we get

$$y = \sqrt[3]{x+\sqrt[3]{x+...}}$$
$$y = \sqrt[3]{x+y}$$
$$y^3 - y = x.$$

I stopped here because solving for y is too much of a pain, but notice that I just found the inverse of the function. It's just a simple polynomial. So we'll use inverse jailbreak.
If the lower bound is 0, then our new lower bound is 0......right???
Oh no....

$$y^3 - y = y(y-1)(y+1) = 0$$

There's more than one solution for y, but which one is the right output???

Well, if our upper bound is 6, then our new upper bound is 2 (by inspection of the equation).
Realize that $x = y^3 - y$ is a sleeping cubic function. Let's test some numbers.
At $y =$ -1 to 0, x travels forward ($y = -\frac{3}{4}$) then back ($y = -\frac{1}{4}$). Not okay for integration.
At $y = 0$ to 1, x just travels back then forward, also not okay for integration.
Then $y = 1$ is the output we choose because x travels in only one direction which we can actually integrate through the interval [1,2]. Thus,

$$\int_0^6 \sqrt[3]{x+\sqrt[3]{x+\sqrt[3]{x+...}}}dx = 6(2) - 0 - \int_1^2 (y^3 - y)dy$$

$$= 12 - \frac{9}{4} = \boxed{\frac{39}{4}}$$

This is definitely the most dangerous inverse jailbreak integrals I've ever solved.

Logarithmic Example

$$\int_{\frac{\sqrt{3}}{4}}^{\frac{1}{2}} \ln\left(\frac{\sqrt{1-4x^2}+1}{2x}\right) dx$$

[Silver]

I have no patience to do IBP on this, and I'm not risking any small mistakes. Amazingly, the inverse function of this is $\frac{e^x}{1+e^{2x}}$. So we have

$$= -\frac{\sqrt{3}}{8}\ln(3) - \int_{\ln(\sqrt{3})}^{0} \frac{e^x}{1+e^{2x}}dx = \boxed{\frac{\pi}{12} - \frac{\sqrt{3}}{8}\ln(3)}$$

Retreat from Trig-Sub

$$\int_0^1 \sin^{-1}\left(\sqrt{1-\sqrt{x}}\right) dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

If you try $x = \sin^4\theta$ or $x = \cos^4\theta$, you're gonna have to deal with integration by parts. Instead, the inverse function is just $\cos^4(x)$. This also gave us no leftovers!!

$$\int_0^1 \sin^{-1}\left(\sqrt{1-\sqrt{x}}\right) dx = \int_0^{\frac{\pi}{2}} \cos^4(x)dx$$

Wallis' Trick!!

A Very Satisfying Polynomial Inverse Example

$$\int_0^7 \left(\sqrt[3]{\sqrt{x^2+1}+x} - \sqrt[3]{\sqrt{x^2+1}-x}\right) dx$$

[Silver]

This is clearly ugly to work with. But watch very closely...

$$\begin{aligned}
y &= \sqrt[3]{\sqrt{x^2+1}+x} - \sqrt[3]{\sqrt{x^2+1}-x} \\
y^3 &= (\sqrt{x^2+1}+x) - 3(1)\sqrt[3]{\sqrt{x^2+1}+x} + 3(1)\sqrt[3]{\sqrt{x^2+1}-x} - (\sqrt{x^2+1}-x) \\
y^3 &= 2x - 3y \\
\frac{y^3+3y}{2} &= x.
\end{aligned}$$

Beautiful. If you lose track very easily, you can do what most AIME people do, let $\alpha = \sqrt[3]{\sqrt{x^2+1}+x}$ and let $\beta = \sqrt[3]{\sqrt{x^2+1}-x}$. Then simplify in terms of those variables. So now this integral is:

$$= 14 - \int_0^2 \frac{x^3+3x}{2}dx = \boxed{9}$$

How unimaginable. This answer is just way too clean, but it turns out to be true. Don't believe me? Go ahead and WolframAlpha it. The amount of times I WolframAlpha this integral just to check if this was actually true 0.0

6.2 Taylor Series

Taylor Series Concept

Let $xg(x)$ be an ugly function and

$$g(x) = \sum_{n=0}^{\infty} f(xn).$$

Then,

$$\int_a^b xg(x)dx = \sum_{n=0}^{\infty} \int_a^b xf(nx)dx.$$

We can now integrate with an easier function and then compute the infinite sum using taylor series or other sums.

A Beginner Example

$$\int_0^1 \frac{\ln(1-x)}{x} dx$$

Turn $\ln(1-x)$ into Taylor series:

$$= \int_0^1 \sum_{n=1}^{\infty} \frac{-x^n}{nx} dx = -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 x^{n-1} dx = -\sum_{n=1}^{\infty} \frac{1}{n^2} = \boxed{-\frac{\pi^2}{6}}$$

Sneaky Geometric Series

$$\int_0^1 \frac{\ln(x)}{1-x^2} dx$$

Turn $\frac{1}{1-x^2}$ into a Geo series:

$$= \int_0^1 \sum_{n=0}^{\infty} x^{2n} \ln(x) dx = \sum_{n=0}^{\infty} \int_0^1 x^{2n} \ln(x) dx = \sum_{n=0}^{\infty} -\frac{1}{(2n+1)^2} = \boxed{-\frac{\pi^2}{8}}$$

Famous Taylor Series Log Integral

$$\int_0^1 \ln(x)\ln(1-x)dx$$

Turn $\ln(1-x)$ into a Taylor Series:

$$= \int_0^1 \ln(x)\sum_{n=1}^{\infty} -\frac{x^n}{n}dx = -\sum_{n=1}^{\infty}\frac{1}{n}\int_0^1 x^n \ln(x)dx = \sum_{n=1}^{\infty}\frac{1}{n(n+1)^2}$$

$$= \sum_{n=1}^{\infty}\frac{1}{n(n+1)} - \frac{1}{(n+1)^2} = \boxed{2-\frac{\pi^2}{6}}$$

Exponential Case

$$\int_0^{\infty} \ln(1+e^{-x})dx$$

$$= \int_0^{\infty}\sum_{n=1}^{\infty}(-1)^n\frac{(e^{-x})^n}{n}dx = \sum_{n=1}^{\infty}\frac{(-1)^n}{n}\int_0^{\infty}(e^{-x})^n dx = \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = \boxed{-\frac{\pi^2}{12}}$$

Exponential Geo Series

$$\int_0^{\infty}\frac{x}{1+e^x}dx$$

We can not use e^x otherwise the bound to infinity will not converge.
So we stick with e^{-x}.

$$= \int_0^{\infty}\frac{xe^{-x}}{e^{-x}+1}dx = \int_0^{\infty} x\sum_{n=1}^{\infty}(-1)^n e^{-nx}dx = \sum_{n=1}^{\infty}(-1)^n\int_0^{\infty} xe^{-nx}dx$$

$$= \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = \boxed{-\frac{\pi^2}{12}}$$

Taylor Series Modification

$$\int_0^1 \frac{x\ln(x)}{(1+x)^2}dx$$

We can also modify our Taylor Series!!
Observe that differentiating both sides gives us

$$\begin{aligned}
\frac{1}{1+x} &= \sum_{n=0}^{\infty}(-1)^n x^n \\
\frac{-1}{(1+x)^2} &= \sum_{n=0}^{\infty} n x^{n-1}(-1)^n \\
\frac{-x}{(1+x)^2} &= \sum_{n=0}^{\infty} n(-1)^n x^n \\
\frac{x}{(1+x)^2} &= -\sum_{n=1}^{\infty} n(-1)^n x^n.
\end{aligned}$$

After all that simplification, we can now substitute it in our integral:

$$\int_0^1 \frac{x\ln(x)}{(1+x)^2}dx = \int_0^1 -\sum_{n=1}^{\infty} n(-1)^n x^n \ln(x)dx = -\sum_{n=1}^{\infty} n(-1)^n \int_0^1 x^n \ln(x)dx$$

$$= \sum_{n=1}^{\infty} \frac{n(-1)^n}{(n+1)^2} = -\sum_{n=2}^{\infty} \frac{(n-1)(-1)^n}{n^2} = \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2} - \frac{(-1)^n}{n} = \boxed{\ln(2) - \frac{\pi^2}{12}}$$

Double Taylor Series!

$$\int_0^{\infty} \ln\left(\frac{e^x+1}{e^x-1}\right)dx$$

$$\begin{aligned}
&= \int_0^{\infty} \ln\left(\frac{1+e^{-x}}{1-e^{-x}}\right)dx = \int_0^{\infty} \ln(1+e^{-x}) - \ln(1-e^{-x})dx \\
&= \int_0^{\infty} \sum_{n=1}^{\infty}(-1)^n \frac{(e^{-x})^n}{n}dx + \int_0^{\infty} \sum_{n=1}^{\infty} \frac{(e^{-x})^n}{n}dx \\
&= \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} + \sum_{n=1}^{\infty} \frac{1}{n^2} = \boxed{\frac{\pi^2}{4}}
\end{aligned}$$

Algebra Manipulation

$$\int_0^1 \frac{\ln(1+x+x^2)}{x}dx$$

We can separate it into to Taylor Series since $1+x+x^2 = \frac{1-x^3}{1-x}$.
By log rules, we can get

$$= \int_0^1 \frac{\ln(1-x^3)-\ln(1-x)}{x}dx$$

$$= \int_0^1 \sum_{n=1}^{\infty} \frac{-x^{3n}}{nx}dx + \int_0^1 \sum_{n=1}^{\infty} \frac{x^n}{nx}dx = \boxed{\frac{\pi^2}{9}}$$

U-Sub Example

$$\int_0^{\frac{\pi}{2}} \frac{\ln(\sec(x))}{\tan(x)}dx$$

Let $u = \ln(\sec(x))$:

$$= \int_0^{\infty} \frac{u}{\tan^2(x)}du = \int_0^{\infty} \frac{u}{e^{2u}-1}du = \int_0^{\infty} \frac{ue^{-2u}}{1-e^{-2u}}du$$

$$= \int_0^{\infty} \sum_{n=1}^{\infty} u(e^{-2u})^n du = \sum_{n=1}^{\infty} \int_0^{\infty} ue^{-2nu}du = \sum_{n=1}^{\infty} \frac{1}{4n^2} = \boxed{\frac{\pi^2}{24}}$$

I have never seen an integral where it actually uses a "double summation" of Taylor Series. Experimentation and creativity are the most powerful tools of problem writing!!

6.3 Complexifying the Integral

Beginner Concept

We have the following formulas:

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\frac{a^n - b^n}{a - b} = a^{n-1} + a^{n-2}b + a^{n-3}b^2 + ... + ab^{n-2} + b^{n-1}$$

Trust me, you'll be surprised how many times you need this.

A Basic Example

$$\int_0^{2\pi} \frac{\sin(10x)}{\sin(x)} dx$$

Let's not try to use trig-identities. Although, you could probably do periodicity, but let's complexify this integral anyways.

$$= \int_0^{2\pi} \frac{e^{10ix} - e^{-10ix}}{e^{ix} - e^{-ix}} dx$$

Let $a = e^{ix}$ and $b = e^{-ix}$. Then,

$$\frac{a^{10} - b^{10}}{a - b} = a^9 + a^8b + a^7b^2 + ... + ab^8 + b^9$$

Note that $ab = 1$, hence

$$= e^{9ix} + e^{7ix} + e^{5ix} + ... + e^{-5ix} + e^{-7ix} + e^{-9ix}$$

$$= \int_0^{2\pi} (2\cos(9x) + 2\cos(7x) + 2\cos(5x) + 2\cos(3x) + 2\cos(x)) dx$$

At this point, it is really nice to use periodicity, and it is safe to say that this integral is just $\boxed{0}$. If you do this a lot, you'll start to get a sense of the algebraic pattern very quickly.

The Significance of Periodicity

$$\int_0^{2\pi} \left(\frac{\sin(3x)}{\sin(x)}\right)^3 dx$$

[Harvard-MIT Math Tournament Integration Bee 2019]

$$= \int_0^{2\pi} \left(\frac{e^{3ix} - e^{-3ix}}{e^{ix} - e^{-ix}}\right)^3 dx = \int_0^{2\pi} (e^{2ix} + 1 + e^{-2ix})^3 dx = \int_0^{2\pi} (2\cos(2x) + 1)^3 dx$$

Let $u = 2x$ to simplify periodicity:

$$= \int_0^{4\pi} \frac{1}{2}(8c^3 + 12c^2 + 6c + 1)du = \int_0^{4\pi} [2c(1 + \cos(2x)) + 3(1 + \cos(2x)) + \frac{1}{2}]du$$

$$= \int_0^{4\pi} 3 + \frac{1}{2} du = \boxed{14\pi}$$

There was a lot of disappearing terms and cancellations thanks to periodicity.

In fact, for a very helpful tip, for every $n \in \mathbb{N}$:

$$\int_0^{2\pi} \sin^{2n+1}(x)dx = \int_0^{2\pi} \cos^{2n+1}(x)dx = \boxed{0}$$

No Wallis' Trick???

$$\int \cos^6(x)dx$$

No Wallis' Trick??? Oh naww, I'm complexifying this integral immediately!!

$$= \int \left(\frac{e^{ix} + e^{-ix}}{2}\right)^6 dx = \frac{1}{2^6} \int (e^{6ix} + 6e^{4ix} + 15e^{2ix} + 20 + ...)dx$$

$$= \frac{1}{2^6} \int (2\cos(6x) + 12\cos(4x) + 30\cos(2x) + 20)dx$$

YAY, no double angle trig-identity algebra spamming!!

Advanced Concept

If you remember Euler's complex formula, it is very well known that

$$e^{ix} = \cos(x) + i\sin(x).$$

Then,

$$\int \cos(x)f(x)dx = \Re\left(\int e^{ix}f(x)dx\right)$$

$$\int \sin(x)f(x)dx = \Im\left(\int e^{ix}f(x)dx\right)$$

A Physics Example

$$\int_{-\infty}^{\infty} \cos(x^2)dx$$

$$= \Re\left(\int_{-\infty}^{\infty} e^{ix^2}dx\right) = \Re\left(\frac{1+i}{\sqrt{2}}\int_{-\infty}^{\infty} e^{-x^2}dx\right) = \boxed{\sqrt{\frac{\pi}{2}}}$$

Complex Algebra Chaos!!!

$$\int_{0}^{\infty} e^{-x^2}\sin(x^2)dx$$

$$= \Im\left(\int_{0}^{\infty} e^{-x^2(1-i)}dx\right) = \Im\left(\frac{1}{\sqrt{1-i}}\int_{0}^{\infty} e^{-u^2}du\right)$$

$$= \Im\left(\frac{\sqrt{\pi}}{2}\left[\frac{\sqrt{1+i}}{\sqrt{2}}\right]\right) = \Im\left(\frac{\sqrt{\pi}}{2\sqrt[4]{2}}e^{\frac{\pi}{8}i}\right) = \boxed{\frac{\sqrt{\pi}}{2\sqrt[4]{2}}\sin\left(\frac{\pi}{8}\right)}$$

Let's try to steer away from integrals that requires a lot of complex algebra knowledge.

No Boundaries???

$$\int e^{\cos(x)}\cos(x+\sin(x))dx$$

[MIT Integration Bee 2023]

$$= \Re\left(\int e^{\cos(x)}e^{i(x+\sin(x))}dx\right) = \Re\left(\int e^{e^{ix}}e^{ix}dx\right) = \Re\left(\int -ie^{u}du\right) = \Re\left(-ie^{e^{ix}}\right)$$

$$= \Re\left(-ie^{\cos(x)+i\sin(x)}\right) = \Re\left(-ie^{\cos(x)}(\cos(\sin(x)) + i\sin(\sin(x)))\right) = \boxed{e^{\cos(x)}\sin(\sin(x)) + C}$$

6.4 Ninja Substitution

The following technique derives from King's rule, except a little easier to work with because the substitution is just $u = -x$.

Exponential Case

$$\int_{-1}^{1} \frac{x^2}{1+e^x} dx$$

Let $u = -x$:

$$\int_{-1}^{1} \frac{x^2}{1+e^{-x}} dx = \int_{-1}^{1} \frac{x^2 e^x}{1+e^x} dx$$

$$2I = \int_{-1}^{1} \frac{x^2(1+e^x)}{1+e^x} dx = \int_{0}^{1} 2x^2 dx \Rightarrow I = \boxed{\frac{1}{3}}$$

Arctangent Case

$$\int_{-1}^{1} \tan^{-1}(e^x) dx$$

Let $u = -x$:

$$\int_{-1}^{1} \tan^{-1}(e^{-x}) dx$$

$$2I = \int_{-1}^{1} \tan^{-1}(e^x) + \tan^{-1}(e^{-x}) dx = \int_{-1}^{1} \frac{\pi}{2} dx \Rightarrow I = \boxed{\frac{\pi}{2}}$$

A Null Case

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(\sec(x) - \tan(x)) dx$$

Let $u = -x$:

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(\sec(x) + \tan(x)) dx$$

$$2I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(1) dx \Rightarrow I = \boxed{0}$$

Logarithmic Case

$$\int_{-1}^{1} x\ln(1+e^x)dx$$

Let $u=-x$:

$$= \int_{-1}^{1} -x\ln(1+e^{-x})dx = \int_{-1}^{1} x\ln\left(\frac{e^x}{e^x+1}\right)dx$$

$$2I = \int_{-1}^{1} x\ln(e^x)dx \Rightarrow I = \boxed{\frac{1}{3}}$$

Radical Case

$$\int_{-\sqrt{3}}^{\sqrt{3}} \frac{\tan^{-1}(x)}{x+\sqrt{x^2+1}}dx$$

[UC Berkeley Integration Bee 2023]

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \tan^{-1}(x)(\sqrt{x^2+1}-x)dx$$

Let $u=-x$:

$$= \int_{-\sqrt{3}}^{\sqrt{3}} -\tan^{-1}(x)(\sqrt{x^2+1}+x)dx$$

$$2I = \int_{-\sqrt{3}}^{\sqrt{3}} -2x\tan^{-1}(x)dx$$

A Cool Gaussian Integral

$$\int_{0}^{\infty} \frac{\tan^{-1}(x)}{x^{\ln(x)+1}}dx$$

[Cambridge University Integration Bee]

Let $u=\ln(x)$:

$$\int_{0}^{\infty} \frac{\tan^{-1}(x)}{xe^{\ln^2(x)}}dx = \int_{-\infty}^{\infty} \tan^{-1}(e^u)e^{-u^2}du$$

Let $w=-u$:

$$\frac{1}{2}\int_{-\infty}^{\infty} \frac{\pi}{2}e^{-u^2}du = \boxed{\frac{\pi\sqrt{\pi}}{4}}$$

6.5 Inversion

A Logarithmic Case

$$\int_0^\infty \frac{\ln(2x)}{1+x^2}dx$$

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{\ln(\frac{2}{u})}{1+\frac{1}{u^2}} \cdot \frac{du}{u^2} = \int_0^\infty \frac{\ln(2)-\ln(u)}{u^2+1}du$$

$$2I = \int_0^\infty \frac{\ln(4)}{u^2+1}du \Rightarrow I = \boxed{\frac{\pi}{2}\ln(2)}$$

A Logarithmic Case 2

$$\int_0^\infty \frac{\ln(x)}{x^2+2x+4}dx$$

Let $u = \frac{4}{x}$:

$$\int_0^\infty \frac{\ln(\frac{4}{u})\cdot 4du}{(\frac{16}{u^2}+\frac{8}{u}+4)u^2} = \int_0^\infty \frac{\ln(4)-\ln(u)}{u^2+2u+4}du$$

$$2I = \int_0^\infty \frac{\ln(4)}{x^2+2x+4}dx = \int_0^\infty \frac{\ln(4)}{(x+1)^2+3}dx \Rightarrow I = \boxed{\frac{\pi\ln(2)}{3\sqrt{3}}}$$

Arctangent Case

$$\int_0^\infty \frac{\tan^{-1}(x)}{(x+1)\sqrt{x}}dx$$

Let $u = \frac{1}{x}$:

$$= \int_0^\infty \frac{\tan^{-1}(\frac{1}{u})}{(\frac{1}{u}+1)\sqrt{\frac{1}{u}}}\frac{du}{u^2} = \int_0^\infty \frac{\tan^{-1}(\frac{1}{u})}{(u+1)\sqrt{u}}du$$

$$2I = \int_0^\infty \frac{\tan^{-1}(x)+\tan^{-1}(\frac{1}{x})}{(x+1)\sqrt{x}}dx \Rightarrow I = \boxed{\frac{\pi^2}{4}}$$

Hybrid Example

$$\int_0^\infty \frac{\tan^{-1}(x)}{x(\ln^2(x)+1)}dx$$

Let $u = \frac{1}{x}$:

$$= \int_0^\infty \frac{\tan^{-1}(\frac{1}{x})}{x(\ln^2(x)+1)}dx = \frac{1}{2}\int_0^\infty \frac{\frac{\pi}{2}}{x(\ln^2(x)+1)}dx = \boxed{\frac{\pi^2}{2}}$$

Exponential Example

$$\int_0^\infty \frac{dx}{(x^2+1)(1+e^{x-\frac{1}{x}})}$$

Let $u = \frac{1}{x}$:

$$= \int_0^\infty \frac{dx}{(x^2+1)(1+e^{\frac{1}{x}-x})} = \int_0^\infty \frac{e^{x-\frac{1}{x}}}{(x^2+1)(1+e^{x-\frac{1}{x}})}dx$$

$$= \frac{1}{2}\int_0^\infty \frac{dx}{(x^2+1)} = \boxed{\frac{\pi}{4}}$$

Needing a Proper Bound

$$\int_1^\infty \frac{\ln(x)}{x(x-1)}dx$$

Let $u = \frac{1}{x}$:

$$= \int_0^1 \frac{\ln(\frac{1}{u})}{u(1-u)}dx = \int_0^1 \frac{\ln(u)}{u(u-1)}du$$

Now that the bounds are from 0 to 1, we can legally perform Taylor Series!!

Double Log

$$\int_0^\infty \frac{\ln(1-x+x^2)}{(1+x^2)\ln(x)}dx$$

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{\ln(1-\frac{1}{x}+\frac{1}{x^2})}{(1+x^2)\ln(\frac{1}{x})}dx = \int_0^\infty \frac{\ln(x^2-x+1)-2\ln(x)}{(1+x^2)\ln(\frac{1}{x})}dx$$

$$= \int_0^\infty \frac{2\ln(x)}{(1+x^2)\ln(x)} - \frac{\ln(x^2-x+1)}{(1+x^2)\ln(x)}dx = \frac{1}{2}\int_0^\infty \frac{2}{(1+x^2)}dx = \boxed{\frac{\pi}{2}}$$

Dirichlet Example

$$\int_0^1 \frac{\sin(x)+\sin(\frac{1}{x})}{x}dx$$

Let $u=\frac{1}{x}$:

$$= \int_1^\infty \frac{\sin(x)+\sin(\frac{1}{x})}{x}dx$$

$$2I = \int_0^1 \frac{\sin(x)+\sin(\frac{1}{x})}{x}dx + \int_1^\infty \frac{\sin(x)+\sin(\frac{1}{x})}{x}dx = \int_0^\infty \frac{\sin(x)+\sin(\frac{1}{x})}{x}dx$$

$$I = \frac{1}{2}\int_0^\infty \frac{\sin(x)}{x} + \frac{\sin(x)}{x}dx = \boxed{\frac{\pi}{2}}$$

Advanced Dirichlet Example

$$\int_{-\infty}^\infty \frac{\sin(x+\frac{1}{x})\cos(x-\frac{1}{x})}{x+\frac{1}{x}}dx$$

This is an even function, so we use symmetry. Then we can use $u=\frac{1}{x}$:

$$= \int_0^\infty \frac{2\sin(u+\frac{1}{u})\cos(u-\frac{1}{u})}{(u+\frac{1}{u})u^2}du = \int_0^\infty \frac{\sin(2u)+\sin(\frac{2}{u})}{u(u^2+1)}du$$

Separate terms and do inversion again only for the right term:

$$= \int_0^\infty \frac{\sin(2x)}{x(x^2+1)} + \frac{x^2\sin(2x)}{x(x^2+1)}dx = \int_0^\infty \frac{\sin(2x)}{x}dx = \boxed{\frac{\pi}{2}}$$

Self-Substitution

$$\int_0^\infty xe^{-(x^2-\frac{1}{x^2})^2}dx$$

[UC Berkeley Integration Bee 2023]

Let $u=\frac{1}{x}$:

$$= \int_0^\infty \frac{1}{x^3}e^{-(x^2-\frac{1}{x^2})^2}dx$$

$$2I = \int_0^\infty e^{-(x^2-\frac{1}{x^2})^2}\left(x+\frac{1}{x^3}\right)dx = \frac{1}{2}\int_{-\infty}^\infty e^{-u^2}du \Rightarrow I = \boxed{\frac{\sqrt{\pi}}{4}}$$

A Super Cool Famous Inversion Integral

$$\int_0^\infty \frac{(x-1)}{\sqrt{2^x-1}\ln(2^x-1)}dx$$

[JEE Main]

Now this integral is actually very tough to do. It's also one of those integrals where you can easily lose track of certain things. Let $u = \sqrt{2^x-1}$. Then,

$$= \int_0^\infty \frac{(\log_2(u^2+1)-1)}{u\ln(u^2)\ln(2)} \cdot \frac{2udu}{u^2+1} = \frac{1}{\ln^2(2)}\int_0^\infty \frac{\ln(u^2+1)-\ln(2)}{(u^2+1)\ln(u)}du$$

Let $u = \frac{1}{x}$:

$$= \int_0^\infty \frac{\ln(\frac{1}{x^2}+1)-\ln(2)}{\ln^2(2)(x^2+1)\ln(\frac{1}{x})}dx = \int_0^\infty \frac{\ln(x^2+1)-2\ln(x)-\ln(2)}{-\ln^2(2)(x^2+1)\ln(x)}dx$$

$$= \int_0^\infty -\frac{1}{\ln^2(2)}\left(\frac{\ln(x^2+1)-\ln(2)}{(x^2+1)\ln(x)} - \frac{2}{x^2+1}\right)dx = \int_0^\infty \frac{2}{\ln^2(2)(x^2+1)} - \frac{1}{\ln^2(2)}\left(\frac{\log_2(x^2+1)}{(x^2+1)\ln(x)}\right)dx$$

Notice in my first line of the equation, the last integral form is the same as that second term I have now, just different signs.

$$I = \int_0^\infty \frac{2}{\ln^2(2)(x^2+1)}dx - I$$

$$2I = \int_0^\infty \frac{2}{\ln^2(2)(x^2+1)}dx$$

$$I = \boxed{\frac{\pi}{2\ln^2(2)}}$$

Notice how quickly and ugly it got at the very start. It is a lot to process, and it looked like you were going nowhere at first glance. Thankfully all the ugliness cancels out nicely, giving us a clean answer.

6.6 Frullani Integrals

Frullani Formula

$$\int_0^\infty \frac{f(ax)-f(bx)}{x}dx = (f(\infty)-f(0))\ln\left(\frac{a}{b}\right)$$

Beginner Example

$$\int_0^\infty \frac{\tan^{-1}(x)-\tan^{-1}(\pi x)}{x}dx$$

By Frullani Formula,

$$= \left(\lim_{x\to\infty}\tan^{-1}(x) - \lim_{x\to 0}\tan^{-1}(x)\right)\ln\left(\frac{1}{\pi}\right) = \boxed{-\frac{\pi}{2}\ln(\pi)}$$

U-Sub Example

$$\int_0^\infty \sin\left(\frac{1}{x}\right) - \frac{1}{\pi}\sin\left(\frac{\pi}{x}\right)dx$$

[CMIMC Integration Bee 2020]

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{\sin(u)}{u^2} - \frac{\sin(\pi u)}{\pi u^2}du = \left(\lim_{x\to\infty}\frac{\sin(x)}{x} - \lim_{x\to 0}\frac{\sin(x)}{x}\right)\ln\left(\frac{1}{\pi}\right) = \boxed{\ln(\pi)}$$

Algebraic Manipulation

$$\int_0^\infty \frac{(e^{3x}-e^x)}{x(e^x+1)(e^{3x}+1)}dx$$

[MIT Integration Bee 2006]

$$= \int_0^\infty \frac{1}{x}\left(\frac{1}{e^x+1} - \frac{1}{e^{3x}+1}\right)dx = \boxed{\frac{\ln(3)}{2}}$$

Disguised Frullani???

$$\int_0^\infty \frac{8}{x^4+8x} - \frac{1}{x^4+x} dx$$

You could do this with partial fractions or caveman, but....

$$\int_0^\infty \frac{1}{x}\left(\frac{1}{\frac{1}{8}x^3+1} - \frac{1}{x^3+1}\right) dx = \boxed{\ln(2)}$$

Frullani Intimidation

$$\int_0^\infty \left(\frac{1}{x^{x+1}}\right)\left(1 - \frac{e^{-xe}}{(x^x)^{e-1}}\right) dx$$

[Silver]

$$= \int_0^\infty \frac{\frac{1}{x^x} - \frac{1}{(ex)^{ex}}}{x} dx = \left(\lim_{x\to\infty}\frac{1}{x^x} - \lim_{x\to 0}\frac{1}{x^x}\right)\ln\left(\frac{1}{e}\right) = \boxed{1}$$

A Challenging Frullani

$$\int_0^\infty \sin\left(\frac{1}{x}\right)\left(\cos\left(\frac{1}{x}\right) - \sec\left(\frac{1}{2x}\right)\right) dx$$

[Silver]

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{\sin(u)(\cos(u) - \sec(\frac{u}{2}))}{u^2} du = \int_0^\infty \frac{1}{u^2}\left(\sin(u)\cos(u) - \frac{\sin(u)}{\cos(\frac{u}{2})}\right) du$$

$$= \int_0^\infty \frac{1}{u}\left(\frac{\sin(2u)}{2u} - \frac{\sin(\frac{u}{2})}{(\frac{u}{2})}\right) du = \boxed{-\ln(4)}$$

Another Challenging Example

$$\int_0^\infty \frac{\cos(2x) - 4\cos(x) + 3}{x^3} dx$$

[Silver]

$$= \int_0^\infty \frac{(\cos(2x) - 1) + 4(1 - \cos(x))}{x^3} dx$$

$$= \int_0^\infty \frac{1}{x}\left[4\left(\frac{1-\cos(x)}{x^2}\right) - 4\left(\frac{1-\cos(2x)}{(2x)^2}\right)\right] dx = \boxed{2\ln(2)}$$

6.7 Feynman Technique

Feynman Trick Concept

Here's the easiest way to sort of explain the purpose of this trick. We have a weird monstrous integral:

$$\int_a^b \frac{f(3g(x))}{g(x)}dx$$

What we do is let a constant be a variable and derive it based from that variable. In this case, let $n = 3$:

$$I(n) = \int_a^b \frac{f(ng(x))}{g(x)}dx \Rightarrow I'(n) = \frac{\partial}{\partial n}\int_a^b \frac{f(ng(x))}{g(x)}dx = \int_a^b f'(ng(x))dx$$

So what we did here is we simplified the integral by taking advantage of its derivative, cancelling out the denominator. After integrating, you should get an answer in terms of n. When you get that, integrate the n and substitute back $n = 3$.

Beginner Feynman Example

$$\int_0^1 \frac{x^{10}-1}{\ln(x)}dx$$

I don't blame you, when I was learning this technique, I was a bit uncomfortable because you can place n wherever you want. However, the main purpose of this trick is just to make the integral easier. One common case is utilizing:

$$\frac{\partial}{\partial n}[x^n] = x^n \ln(x)$$

Multivariables is hard to work with mentally, so experimenting is crucial. Notice with this derivative, I can cancel out $\ln(x)$ at the bottom in the integral. Cool! I'll do that.
Let $n = 10$, then:

$$I(n) = \int_0^1 \frac{x^n - 1}{\ln(x)}dx \Rightarrow I'(n) = \int_0^1 x^n dx$$

So pretty much, we made an integral function, and then derive it in terms of n.
Now we have an easier function!!! After finding our integral function in terms of n, we integrate it back to our original integral function.

$$I'(n) = \int_0^1 x^n dx = \frac{1}{n+1} \Rightarrow I(n) = \ln(n+1)$$

We had $n = 10$, so we just plug it back in.

$$I(10) = \int_0^1 \frac{x^{10}-1}{\ln(x)}dx = \boxed{\ln(11)}$$

A Deceiving Example

$$\int_0^{\pi} \ln(2-\cos(x))dx$$

Alright, so where do we place our n??? If we try $n=2$, then

$$\frac{\partial}{\partial n}\left[\ln(n-\cos(x))\right] = \frac{1}{n-\cos(x)}$$

REMEMBER THAT FUNCTIONS of X are CONSTANTS!!!
So it would NOT be $\frac{\sin(x)}{n-\cos(x)}$ in case you're wondering. Other than that, we can definitely integrate with this derivative, so we'll take it!!

$$I(n) = \int_0^{\pi} \ln(n-\cos(x))dx \Rightarrow I'(n) = \int_0^{\pi} \frac{dx}{n-\cos(x)}$$

Let $u = \tan(\frac{x}{2})$:

$$I'(n) = \int_0^{\infty} \frac{2du}{n+nu^2-1+u^2} = \int_0^{\infty} \frac{2du}{u^2(n+1)+n-1} = \frac{2}{n+1}\int_0^{\infty} \frac{du}{u^2+\frac{n-1}{n+1}} = \frac{\pi}{\sqrt{n^2-1}}$$

Then integrating this gives us $I(n) = \pi\cosh^{-1}(n)$ or $\pi\ln(n+\sqrt{n^2-1})$.
So our answer is $I(2) = \pi\ln(2+\sqrt{3})$......right???
WRONG!!! IT IS NOT!!! You must be very careful because we actually missed something.

The truth is....when we integrate $I'(n)$, we still have to add $+C$. So we actually have $\pi\ln(n+\sqrt{n^2-1})+C$. But how do we find C??? You must plug in another value of n and integrate the original with it. Unfortunately, the only value we can possibly do is $n=1$, so....

$$I(1) = \int_0^{\pi} \ln(1-\cos(x))dx = \int_0^{\pi} \ln\left(2\sin^2\left(\frac{x}{2}\right)\right)dx = -\pi\ln(2)$$

$$\Rightarrow I(1) = -\pi\ln(2) = \pi\ln(1+\sqrt{1^2-1})+C$$

Ahhhhh, so $C = -\pi\ln(2)$, not zero. Hence, our answer is actually

$$I(2) = \boxed{\pi\ln(2+\sqrt{3})-\pi\ln(2)}$$

In the previous problem (the beginner example), $C=0$ which most commonly happens, so we were able to get away with it luckily. If you let $n=0$, then $I(0) = \int_0^1 \frac{x^0-1}{\ln(x)}dx = 0$. Because of this, $0 = \ln(0+1)+C \to C=0$.

The Purpose of n Placement

$$\int_0^{\pi} \sec(x)\ln(1+\cos(x))dx$$

So do we let $n = 1$??? Well let's see.

$$I(n) = \int_0^{\pi} \frac{\ln(n+\cos(x))}{\cos(x)}dx \Rightarrow I'(n) = \int_0^{\pi} \frac{dx}{\cos(x)(n+\cos(x))}$$

That is not an easy integral, so we'll retreat. As you can see, placing n isn't about substituting a constant, it's about positioning it in a way where we can cancel out functions like constants. That way, we have an easier integral.

We'll technically still let $n = 1$, but if we place it differently

$$I(n) = \int_0^{\pi} \frac{\ln(1+n\cos(x))}{\cos(x)}dx \Rightarrow I'(n) = \int_0^{\pi} \frac{\cos(x)}{\cos(x)(1+n\cos(x))}dx = \int_0^{\pi} \frac{dx}{1+n\cos(x)}$$

Let $u = \tan(\frac{x}{2})$:

$$= \int_0^{\infty} \frac{2du}{1+u^2+n-nu^2} = \int_0^{\infty} \frac{2du}{u^2(1-n)+1+n} = \frac{2}{1-n}\int_0^{\infty} \frac{du}{u^2+\frac{1+n}{1-n}} = \frac{\pi}{\sqrt{1-n^2}}$$

$$I'(n) = \frac{\pi}{\sqrt{1-n^2}} \Rightarrow I(n) = \pi\sin^{-1}(n) + C$$

We can easily find C by letting $n = 0$ in our original integral.

$$I(0) = \int_0^{\pi} \frac{\ln(1+0)}{\cos(x)}dx = 0 = \pi\sin^{-1}(0) + C \Rightarrow C = 0$$

Cool, we're safe. So then our answer is

$$I(1) = \pi\sin^{-1}(1) = \boxed{\frac{\pi^2}{2}}$$

Arctangent Example

$$\int_0^{\infty} \frac{\tan^{-1}(2x)}{x(x^2+1)}dx$$

Let $n = 2$, then

$$I(n) = \int_0^{\infty} \frac{\tan^{-1}(nx)}{x(x^2+1)}dx \Rightarrow I'(n) = \int_0^{\infty} \frac{dx}{(x^2+1)(n^2x^2+1)} = \int_0^{\infty} \frac{\frac{1}{\frac{1}{n^2}-1}}{x^2+1} - \frac{\frac{1}{1-\frac{1}{n^2}}}{x^2+\frac{1}{n^2}}dx$$

$$= \frac{1}{n^2-1}\int_0^{\infty} \frac{1}{x^2+\frac{1}{n^2}} - \frac{1}{x^2+1}dx = \frac{1}{n^2-1}\left(\frac{\pi}{2}(n-1)\right) = \frac{\pi}{2}\left(\frac{1}{n+1}\right)$$

$$\Rightarrow I(n) = \frac{\pi}{2}\ln(n+1) + C \Rightarrow I(0) = 0 = \frac{\pi}{2}\ln(1) + C \rightarrow C = 0$$

Hence our answer is $I(2) = \boxed{\frac{\pi}{2}\ln(3)}$.

Arctangent Example 2

$$\int_0^{\frac{\pi}{2}} \frac{\tan^{-1}(\cos(x))}{\cos(x)} dx$$

Let $n = 1$ and place n in a way where we can cancel out $\cos(x)$ from the bottom.

$$I(n) = \int_0^{\frac{\pi}{2}} \frac{\tan^{-1}(n\cos(x))}{\cos(x)} dx \Rightarrow I'(n) = \int_0^{\frac{\pi}{2}} \frac{dx}{n^2\cos^2(x) + 1}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{n^2 + 1 + \tan^2(x)} dx = \int_0^{\infty} \frac{du}{n^2 + 1 + u^2} = \frac{\pi}{2}\left(\frac{1}{\sqrt{n^2+1}}\right)$$

$$I(n) = \frac{\pi}{2}\ln(n + \sqrt{n^2+1}) + C \Rightarrow I(0) = 0 \to C = 0$$

Hence our answer is $I(1) = \boxed{\frac{\pi}{2}\ln(1+\sqrt{2})}$.

Logarithmic Example

$$\int_0^{\infty} \frac{\ln(x^2+4)}{x^2+1} dx$$

Let $n = 4$, then

$$I(n) = \int_0^{\infty} \frac{\ln(x^2+n)}{x^2+1} dx \Rightarrow I'(n) = \int_0^{\infty} \frac{dx}{(x^2+n)(x^2+1)}$$

$$= \int_0^{\infty} \frac{\frac{1}{n-1}}{x^2+1} + \frac{\frac{1}{1-n}}{x^2+n} dx = \frac{\pi}{2(n-1)}\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right) = \frac{\pi}{2\sqrt{n}(\sqrt{n}+1)}$$

$$I(n) = \pi\ln(\sqrt{n}+1) + C \Rightarrow I(0) = \int_0^{\infty} \frac{2\ln(x)}{x^2+1} dx = 0 \to C = 0$$

That integral can be easily solved by inversion.

So therefore, our answer is $I(4) = \boxed{\pi\ln(3)}$.

How do we make these Feynman Technique integrals???

Like always, we need to build it off from a skeleton. Let's do some examples.
A good skeleton for this trick is a generalized integral formula. For me, I found this integral:

$$I'(n) = \int_1^\infty \frac{dx}{x\sqrt{x^n - 1}} = \frac{\pi}{n}$$

Then integrating it gives us:

$$I(n) = \int_1^\infty \frac{2\sec^{-1}(x^{\frac{n}{2}})}{x\ln(x)} dx = \pi\ln(n) + C$$

Immediately, I see a problem. How are we going to find C???
Realize that $n = 0$ makes the integral zero, but it also makes $\ln(n)$ undefined.
I'm very sorry to break it up with you, but this means that you can not use this integral. Even though the Feynman trick works nicely, the process will not work.
You're going to come across things like this A LOT, and please don't feel bad about it. These integrals aren't easy to make anyways.

We know that

$$I'(n) = \int_0^1 x^n dx = \frac{1}{n+1} \Rightarrow I(n) = \int_0^1 \frac{x^n - 1}{\ln(x)} dx$$

The reason why we have to add -1 constant on top is so that the integral can actually converge. Without it, the whole integral will diverge. What if we use other functions like e^x?

$$I'(n) = \int_0^1 e^{xn} dx = \frac{e^n - 1}{n}$$

Nevermind we can not do that o-o. That function in terms of n is not integrable.
Like I said, a lot of these integrals you'll come across will either diverge or the process will just not work. Even if you think you got it, please check the integral by using Desmos or WolframAlpha to make sure it actually converges. Most of the time in my experience, it will not.

6.8 Modified Famous Integrals

The easiest way to modify famous integrals is by u-substitution. However, you can add more than one integrals together and simplify it or use integration by parts to disguise it. You could even utilize other techniques and mix it with them. Let's do some examples!!

An Engineering Joke

$$\int_0^1 \frac{(1-x)^4x^4}{1+x^2}dx = \frac{22}{7} - \pi$$

Hehe...let $x = \tan\theta$, then $dx = \sec^2\theta d\theta$.

$$\int_0^{\frac{\pi}{4}} (1-\tan\theta)^4 \tan^4\theta d\theta = \int_0^{\frac{\pi}{4}} (1-\tan(x))^4 \tan^4(x) dx = \int_0^{\frac{\pi}{4}} (\tan(x) - \tan^2(x))^4 dx$$

Perfect lol.

Famous Taylor Series Integral

$$\int_0^1 \ln(x)\ln(1-x)dx = 2 - \frac{\pi^2}{6}$$

We can do some algebra stuff with it.

$$\ln^2(x) + 2\ln(x)\ln(1-x) + \ln^2(1-x) = \ln^2(x-x^2)$$

Oh this will work nicely. We can use King's Rule to make the integral of $\ln^2(x) = \ln^2(1-x)$, and then Taylor's Series.

$$\int_0^1 \ln^2(x-x^2)dx = \boxed{8 - \frac{\pi^2}{3}}$$

Now that is a very beautiful answer.

Fresnel Integral

$$\int_{-\infty}^{\infty} \cos(x^2)dx = \int_{-\infty}^{\infty} \sin(x^2)dx = \sqrt{\frac{\pi}{2}}$$

Oooooh. Let's use trig-identities!!! I like to propose this beautiful integral:

$$\int_{-\infty}^{\infty} \sin(x^2 + x\sqrt{3\pi} + \pi)dx$$

You complete the square, use the angle addition trig-identity, and then you get $\boxed{\sqrt{\pi}}$ as your answer!! This is literally why I enjoy making integrals so much. Now go have fun making creative integrals!!!

6.9 Unique Tricks

The truth is, there are infinitely many integration techniques. As ridiculous as it sounds, it's actually true. Do we really know 80% of the integration techniques, or did we only discover 25%? Who know's. But we can still keep finding them.

In this section, I will be sharing some integrals with such unique tricks that I came up with and integrals that I found online.

A Derivative in an Integral???

$$\int_1^e \left(\frac{d^2}{dx^2}\left[x^x\right]\right)\frac{dx}{x^x(1+\ln(x))}$$

[UC Berkeley Integration Bee 2023]

Note that $\frac{d}{dx}[x^x] = x^x(1+\ln(x))$. So that means:

$$= \int_1^e \frac{\frac{d}{dx}[x^x(1+\ln(x))]}{x^x(1+\ln(x))}dx = \int_1^{2e^e}\frac{du}{u} = \boxed{e+\ln(2)}$$

An Inversion Example

$$\int_0^\infty \frac{\ln(x)\ln(\frac{1}{x})+1}{(x^2+1)(\ln(x)+1)}dx$$

[Stanford Math Tournament Integration Bee 2023]

Let $u = \frac{1}{x}$:

$$\int_0^\infty \frac{\ln(x)\ln(\frac{1}{x})+1}{(x^2+1)(\ln(\frac{1}{x})+1)}dx$$

$$= \frac{1}{2}\int_0^\infty \frac{\ln(x)\ln(\frac{1}{x})+1}{(x^2+1)}\left(\frac{1}{\ln(x)+1}+\frac{1}{\ln(\frac{1}{x})+1}\right)dx$$

$$= \frac{1}{2}\int_0^\infty \frac{\ln(x)\ln(\frac{1}{x})+1}{(x^2+1)}\left(\frac{2}{\ln(x)\ln(\frac{1}{x})+1}\right)dx$$

$$= \int_0^\infty \frac{dx}{(x^2+1)} = \boxed{\frac{\pi}{2}}$$

Yuepeng's Ugly Trick

$$\int_0^{\frac{\pi}{3}} \left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)(\sec(x)+\csc(x))dx$$

[Silver]

$$= \int_0^{\frac{\pi}{3}} e^{\ln\left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)}(\sec(x)+\csc(x))dx$$

$$= \int_0^{\frac{\pi}{3}} e^{\ln(\sec(x)+\tan(x))-\ln(\csc(x)+\cot(x))}(\sec(x)+\csc(x))dx$$

Let $u = \ln(\sec(x)+\tan(x)) - \ln(\csc(x)+\cot(x))$.

$$= \int_{-\infty}^{\ln(\frac{2}{\sqrt{3}}+1)} e^u du = \boxed{1+\frac{2}{\sqrt{3}}}$$

Holy Constants

$$\int_0^{\frac{\pi}{2}} \frac{\sin(2x)}{\sqrt{e^{\cos^2(x)}+e^{-\sin^2(x)}}}dx$$

[Silver]

$$= \int_0^{\frac{\pi}{2}} \frac{\sin(2x)e^{\frac{1}{2}\sin^2(x)}}{\sqrt{e+1}}dx = \frac{1}{\sqrt{e+1}}\int_0^{\frac{1}{2}} 2e^u du = \boxed{\frac{2(\sqrt{e}-1)}{\sqrt{e+1}}}$$

Anti Conjugating

$$\int_0^1 \frac{x-\sqrt{x-x^2}}{2x-1}dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

$$= \int_0^1 \frac{x-\sqrt{x-x^2}}{2x-1}\cdot\frac{x+\sqrt{x-x^2}}{x+\sqrt{x-x^2}}dx = \int_0^1 \frac{2x^2-x}{(2x-1)(x+\sqrt{x-x^2})}dx$$

$$= \int_0^1 \frac{x}{x+\sqrt{x-x^2}}dx = \int_0^1 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{1-x}}dx = \boxed{\frac{1}{2}}$$

The last integral can be easily solved by King's Rule.

Self-Supporting U-Substitution

$$\int_0^1 e^{x^2+x} + e^{x+\sqrt{x}} dx$$

[AoPS Community]

Note that:

$$\int_0^1 e^{x+\sqrt{x}} dx = \int_0^1 e^{u^2+u} 2u du$$

Then

$$\int_0^1 e^{x^2+x} + e^{x^2+x} 2x dx = \int_0^1 e^{x^2+x}(2x+1) dx$$

$$= \int_0^2 e^u du = \boxed{e^2 - 1}$$

A Crazy Complexification

$$\int e^{e^x} e^x \left(\cos\left(e^{e^x}\right) + \cos(e^x)\right) dx$$

[Harvard-MIT Math Tournament Integration Bee Mock Training]

$$= \Re\left(\int e^{e^x+x}\left(e^{ie^{e^x}} + e^{ie^x}\right) dx\right) = \Re\left(\int e^{ie^{e^x}+e^x+x} + e^{ie^x+e^x+x} dx\right)$$

Let $u = e^{e^x}$ and $w = e^x$:

$$= \Re\left(\int e^{iu} du + \int e^{(i+1)w} dw\right) = \Re\left(-ie^{iu} + \frac{1-i}{2} e^{(1+i)w}\right)$$

$$= \sin\left(e^{e^x}\right) + \Re\left(\frac{1-i}{2} e^w(\cos(w) + i\sin(w))\right)$$

$$= \boxed{\sin(e^{e^x}) + \frac{1}{2} e^{e^x}(\cos(e^x) + \sin(e^x)) + C}$$

A Random e^x???

$$\int \frac{x+1}{x\sqrt{xe^x - 1}} dx$$

[Silver]

Let $u = \sqrt{xe^x - 1}$:

$$= \int \frac{2u du}{xe^x u} = \int \frac{2du}{u^2+1} = \boxed{2\tan^{-1}(\sqrt{xe^x - 1}) + C}$$

A Beautiful Sum of Inverses

$$\int_0^{\frac{1}{2}} \frac{x}{x+\frac{x}{x+\ldots}} + x^2 + x^3 + x^4 + \ldots \, dx$$

[UC Berkeley Integration Bee 2022]

Notice that

$$y = \frac{x}{x+y} \Rightarrow \frac{y^2}{1-y} = x$$

$$x = \frac{y^2}{1-y} = y^2 + y^3 + y^4 + \ldots$$

Yes....both of those infinite functions are secretly inverses of each other. Cool right!!!!

So our answer is just $\boxed{\frac{1}{4}}$

A Super Evil U-Substitution

$$\int_0^{\frac{\pi}{4}} \left(3\sec(x)\tan(x) + \sec(x)\tan^3(x)\right)^4 dx$$

[Silver]

$$= \int_0^{\frac{\pi}{4}} \sec^4(x)\left(3\tan(x) + \tan^3(x)\right)^4 dx$$

Let $u = 3\tan(x) + \tan^3(x)$

$$= \frac{1}{3}\int_0^4 u^4 du = \boxed{\frac{1024}{15}}$$

This Converges???

$$\int_0^{\infty} \left(1 - x\sin\left(\frac{1}{x}\right)\right) dx$$

$$= \int_0^{\infty} \left(\frac{\frac{1}{x} - \sin\left(\frac{1}{x}\right)}{\frac{1}{x}}\right) dx \xrightarrow{u=\frac{1}{x}} \int_0^{\infty} \frac{u - \sin(u)}{u^3} du$$

Integrate by parts:

$$= -\frac{1}{u} + \frac{\sin(u)}{2u^2} + \frac{\cos(u)}{2u} + \int_0^{\infty} \frac{\sin(u)}{2u} du$$

$$= \left[\frac{\sin(u) - 2u + u\cos(u)}{2u^2}\right]_0^{\infty} + \frac{\pi}{4} = \boxed{\frac{\pi}{4}}$$

The rest equals to zero by Squeeze Theorem!!

Advanced Loophole Logging

$$\int_0^\infty \frac{\ln(x)(\ln(x)+1)}{x^{2x-1}}dx$$

[Silver]

$$= \int_0^\infty x^{1-2x}\ln(x)(\ln(x)+1)dx = \int_0^\infty xe^{-2x\ln(x)}\ln(x)(\ln(x)+1)dx$$

Let $u = x\ln(x)$:

$$= \int_0^\infty ue^{-2u}du = \boxed{\frac{1}{4}}$$

A Sneaky Triggy U-Sub

$$\int \frac{\sin^{2021}(x+1)}{\sin^{2023}(x)}dx$$

[Silver]

Let $u = \frac{\sin(x+1)}{\sin(x)}$, then $du = -\frac{\sin(1)}{\sin^2(x)}dx$.

$$= \frac{-1}{\sin(1)}\int u^{2021}du = \boxed{-\frac{1}{2022\sin(1)}\left(\frac{\sin(x+1)}{\sin(x)}\right)^{2022} + C}$$

Holy Golden Ratio!!

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\tan(x)}{\sqrt{2+\tan^2(x)}}dx$$

[Silver]

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sec(x)\tan(x)}{\sec(x)\sqrt{1+\sec^2(x)}}dx = \int_2^\infty \frac{du}{u\sqrt{1+u^2}} = \int_{\tan^{-1}(2)}^{\frac{\pi}{2}} \csc\theta d\theta$$

$$= \left[-\ln\left|\frac{1+\cos\theta}{\sin\theta}\right|\right]_{\tan^{-1}(2))}^{\frac{\pi}{2}} = \ln\left(\frac{1+\frac{1}{\sqrt{5}}}{\frac{2}{\sqrt{5}}}\right) = \boxed{\ln(\varphi)}$$

Sneakiest Area of a Partial Circle

$$\int_0^{\frac{\pi}{2}} \sqrt{\sin(2x)}\sin(x)dx$$

[JEE Main]

Let $u = \frac{\pi}{2} - x$:

$$= \int_0^{\frac{\pi}{2}} \sqrt{\sin(2x)}\cos(x)dx$$

$$2I = \int_0^{\frac{\pi}{2}} \sqrt{1-(\sin(x)-\cos(x))^2}(\sin(x)+\cos(x))dx$$

$$\Rightarrow I = \frac{1}{2}\int_{-1}^{1} \sqrt{1-u^2}du = \boxed{\frac{\pi}{4}}$$

A Surprising Substitution!!

$$\int_0^{\frac{\pi}{2}} \frac{dx}{(1+\cos(x))^4}$$

[Cambridge University]

Let $u = \frac{1}{1+\cos(x)}$, then $\frac{1}{u^2}du = \sin(x)dx$. Note that $\sin(x) = \sqrt{\frac{2}{u} - \frac{1}{u^2}}$.

$$= \int_{\frac{1}{2}}^{1} \frac{u^4}{u^2} \cdot \frac{du}{\sqrt{\frac{2u-1}{u^2}}} = \int_{\frac{1}{2}}^{1} \frac{u^3}{\sqrt{2u-1}}du$$

Let $w = \sqrt{2u-1}$:

$$= \int_0^1 \frac{1}{8}(w^2+1)^3 dw = \boxed{\frac{12}{35}}$$

Complex Taylor Series???

$$\int_0^\infty \frac{e^{\cos(x)}\sin(\sin(x))}{x}dx$$

It is secretly known that

$$e^{\cos(x)}\sin(\sin(x)) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n!}$$

Hence,

$$= \sum_{n=1}^{\infty} \int_0^\infty \frac{\sin(nx)}{xn!}dx = \sum_{n=1}^{\infty} \frac{1}{n!}\int_0^\infty \frac{\sin(u)}{u}du = \boxed{\frac{\pi}{2}(e-1)}$$

A Trippy Factorial U-Substitution

$$\int \frac{x^3}{1+x+\frac{x^2}{2}+\frac{x^3}{6}}dx$$

[MIT Integration Bee 2022]

$$= 6\int \frac{\frac{x^3}{6}}{1+x+\frac{x^2}{2}+\frac{x^3}{6}}dx = 6\int 1 - \frac{\frac{x^2}{2}+x+1}{1+x+\frac{x^2}{2}+\frac{x^3}{6}}dx$$

Let $u = \frac{x^3}{6}+\frac{x^2}{2}+x+1$:

$$= \boxed{6x - 6\ln\left|\frac{x^3}{6}+\frac{x^2}{2}+x+1\right| + C}$$

That's Not Linear Algebra...

$$\int_0^{\frac{\pi}{2}} \frac{(2\sin(x)+5)}{(5\sin(x)+2)^2}dx$$

[Nepal Integral Bee]

$$= \int_0^{\frac{\pi}{2}} \frac{(2\sec(x)\tan(x)+5\sec^2(x))}{(5\tan(x)+2\sec(x))^2}dx = \int_2^{\infty}\frac{du}{u^2} = \boxed{\frac{1}{2}}$$

Adhesion Trick by U-Substitution???

$$\int_{\frac{1}{\sqrt{3}}}^{\frac{\sqrt{2}}{\sqrt{3}}} (1+x\sin(x-\sqrt{1-x^2}))dx$$

[Nepal Integral Bee]

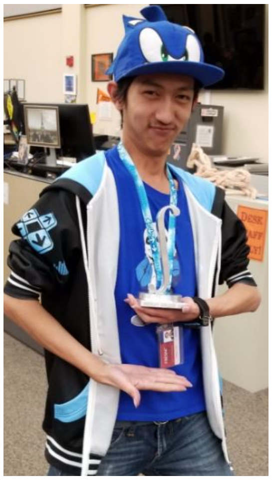
Let $u = \sqrt{1-x^2}$, then $x = \sqrt{1-u^2}$:

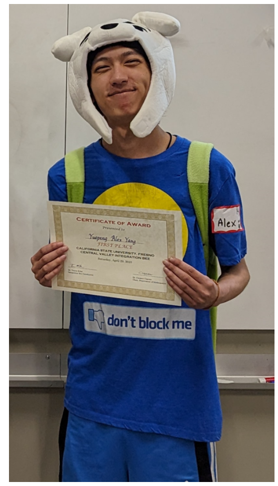
$$= \int_{\sqrt{\frac{2}{3}}}^{\sqrt{\frac{1}{3}}} -(1+\sqrt{1-u^2}\sin(\sqrt{1-u^2}-u))\frac{udu}{\sqrt{1-u^2}}$$

$$2I = \int_{\sqrt{\frac{1}{3}}}^{\sqrt{\frac{2}{3}}} 2 + x\sin(x-\sqrt{1-x^2}) - x\sin(x-\sqrt{1-x^2})dx$$

$$I = \sqrt{\frac{2}{3}} - \sqrt{\frac{1}{3}} = \boxed{\frac{\sqrt{2}-1}{\sqrt{3}}}$$

What About Me???





Wasabi!! (Wassup) I am Yuepeng Alex Yang, mostly known as Silver on the internet. I've been speed integrating since 11th grade and started writing my own integration bee problems in 12th grade. During the time, integration bees were not very popular, and I wished that it was. Since there weren't that many competitions that share their integrals, it was hard to find newer integration bee problems. In addition, I scold any universities who don't share their past integration bee problems because I believe they just reuse them. That's where I decided to make my own integrals and recorded very cool discoveries and unimaginable integration techniques.

I've actually gotten into competitive math since 7th grade and joined Mathorama. However in my 8th grade year, I completely stopped due to bullying and the toxic elitism in my school. I didn't feel comfortable around any "mathletes", so I quit pursuing in math and began majoring in Chemistry. Because of this, I've never competed in an actual Math contest in my life. I remember during high school, my math teacher would always begged me to compete at a Math Circle, but then again, I just wasn't comfortable being in a group of mathletes.

During 12th grade, I debated to compete at an integration bee competition. Because it was mostly undergrads competing, I decided to give it a try. Thankfully, it wasn't too bad and I was okay being there since my friends were with me. Not only that, there were rarely any high school students competing with me; the majority were undergrads. Unfortunately in my first integration bee competition, I got eliminated at semi-finals. I was going too fast which made me become inaccurate with my answers; a complete lesson learned. Sadly, integration bees are annual, so at the end of my 12th grade year, I kept coming up with more cool integration bee problems as well as trained an amazing friend of mine who later became top 16 qualifier for MIT Integration Bee 2022. (So proud of her!!!) During my freshmen undergrad, I started problem writing for many integration bees and math tournaments, as well as winning three of them in my hometown: first from Reedley College and the rest from my own school, CSU Fresno, 2-years in a row both perfectly scored in all rounds. I wanted to win more, but the pandemic hit sadly. Ironically, I switched late being a math major, and thankfully finished in only two years. Now, I pursue in becoming a math professor for a community college or a university, and currently focus on popularizing speed integration. If no one wants to share all the dark forbidden arts of integration techniques, then I will.